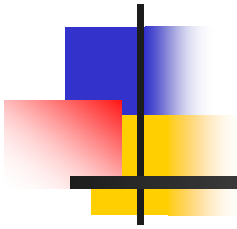


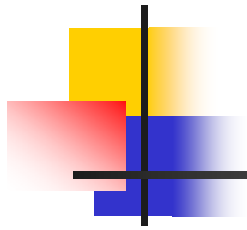
Fatigue and Fracture (Basic Course)

**Fatigue, How and Why
Physics of Fatigue**



Professor Darrell F. Socie
Department of Mechanical Science and Engineering
University of Illinois at Urbana-Champaign

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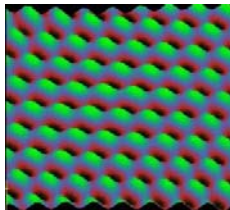


Fatigue, How and Why

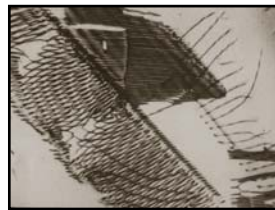
- **Physics of Fatigue**
- Material Properties
- Similitude
- Fatigue Calculator

Size Scale for Studying Fatigue

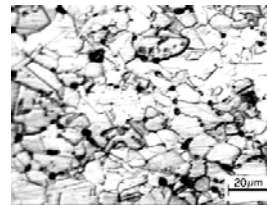
Atoms



Dislocations



Crystals



Specimens



Structures



10^{-10}

10^{-8}

10^{-6}

10^{-4}

10^{-2}

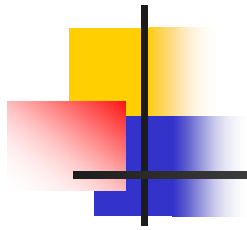
10^0

10^2

Understand the physics on this scale

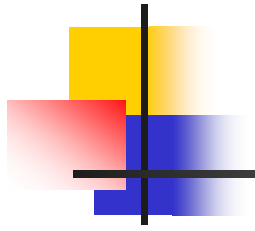
Model the physics on this scale

Use the models on this scale



The Fatigue Process

- Crack nucleation
- Small crack growth in an elastic-plastic stress field
- Macroscopic crack growth in a nominally elastic stress field
- Final fracture



Mechanisms Crack Nucleation

Nucleation in Slip Bands inside Grain

Nucleation at Grain Boundaries

Nucleation at Inclusions

1903 - Ewing and Humfrey



$N = 1,000$



$N = 2,000$



$N = 10,000$

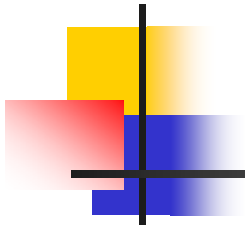


$N = 40,000$

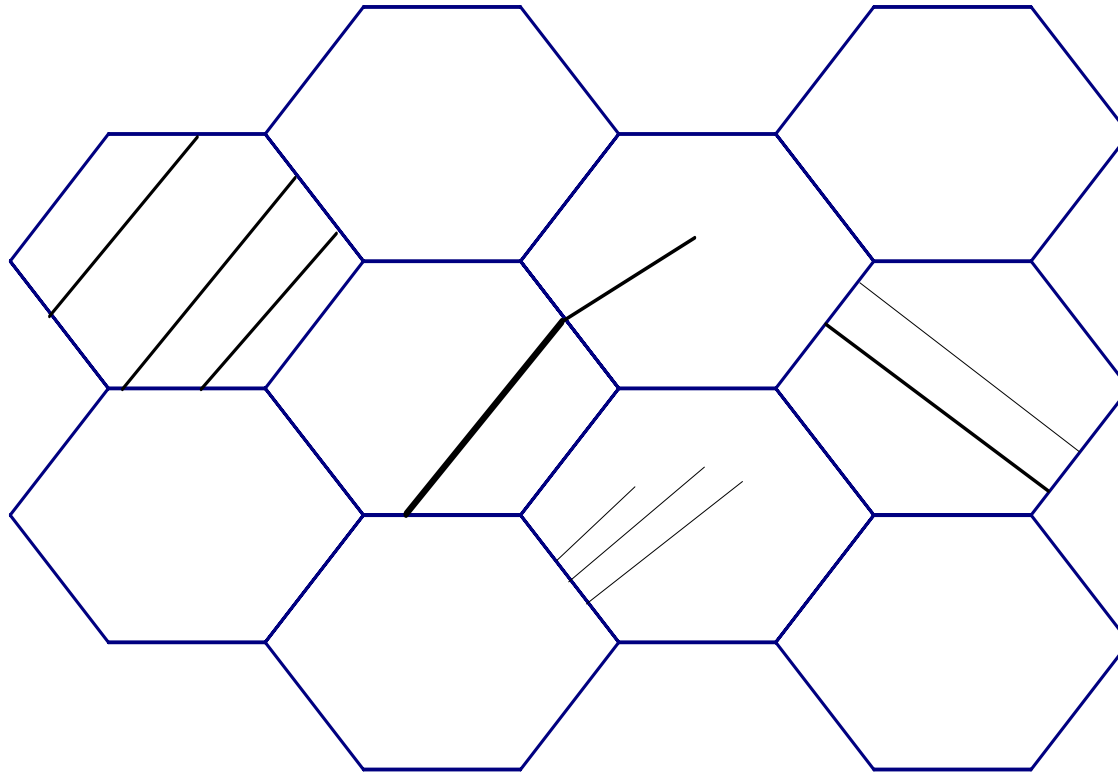
$N_f = 170,000$

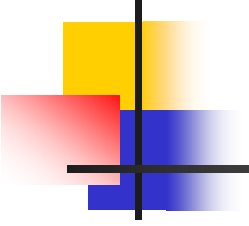
Cyclic deformation leads to the development of slip bands and fatigue cracks

Ewing, J.A. and Humfrey, J.C. "The fracture of metals under repeated alterations of stress", *Philosophical Transactions of the Royal Society*, Vol. A200, 1903, 241-250

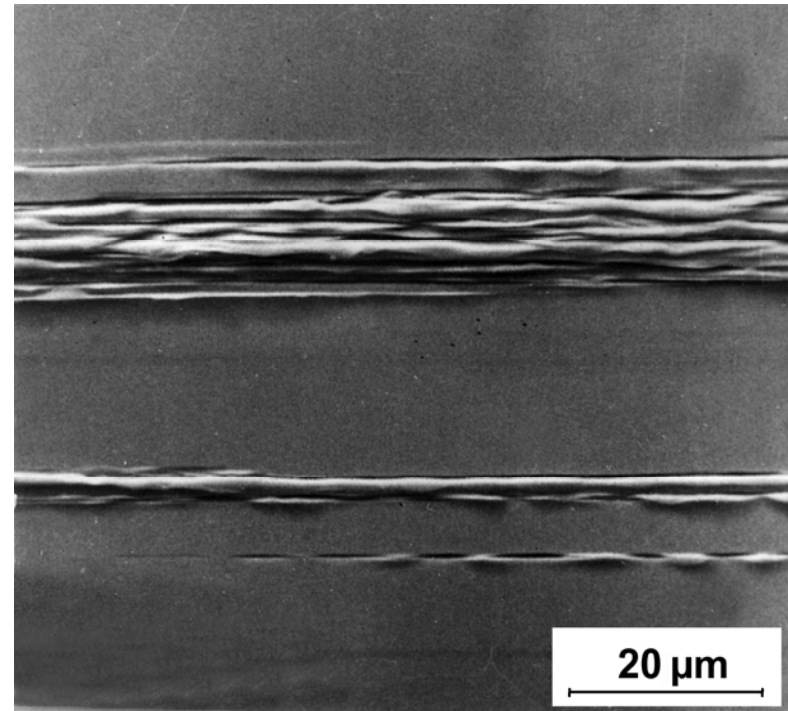
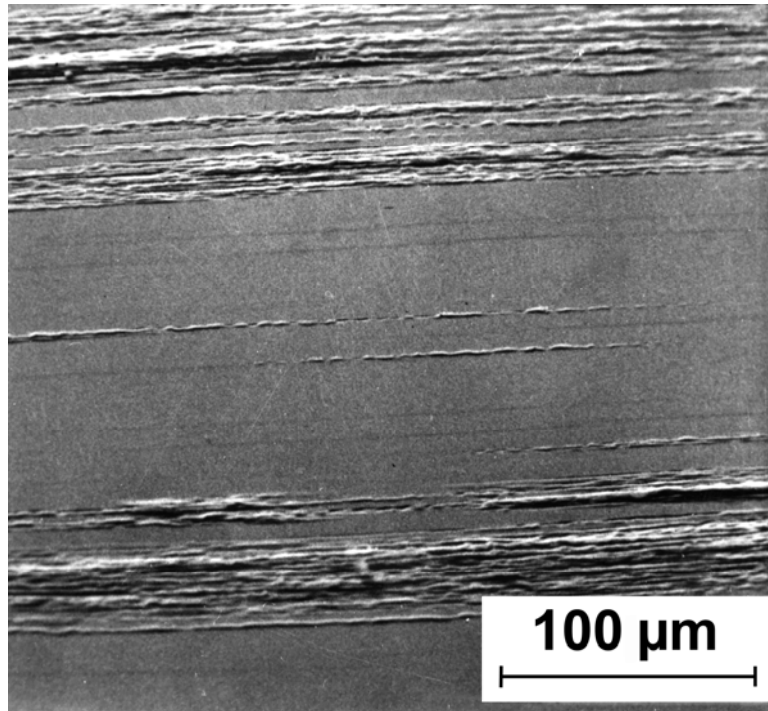


Crack Nucleation

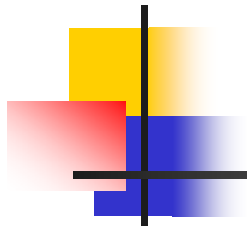




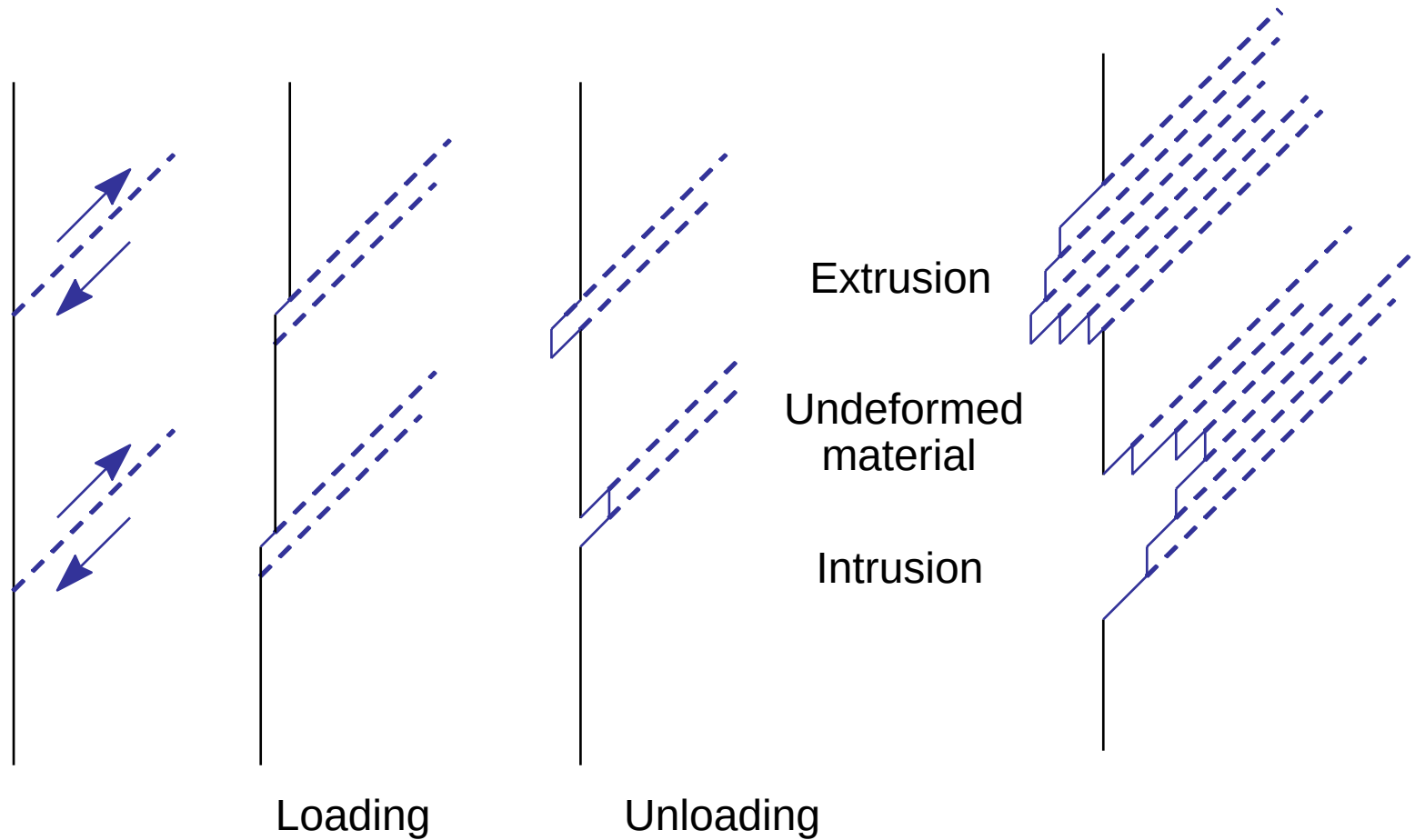
Slip Band in Copper



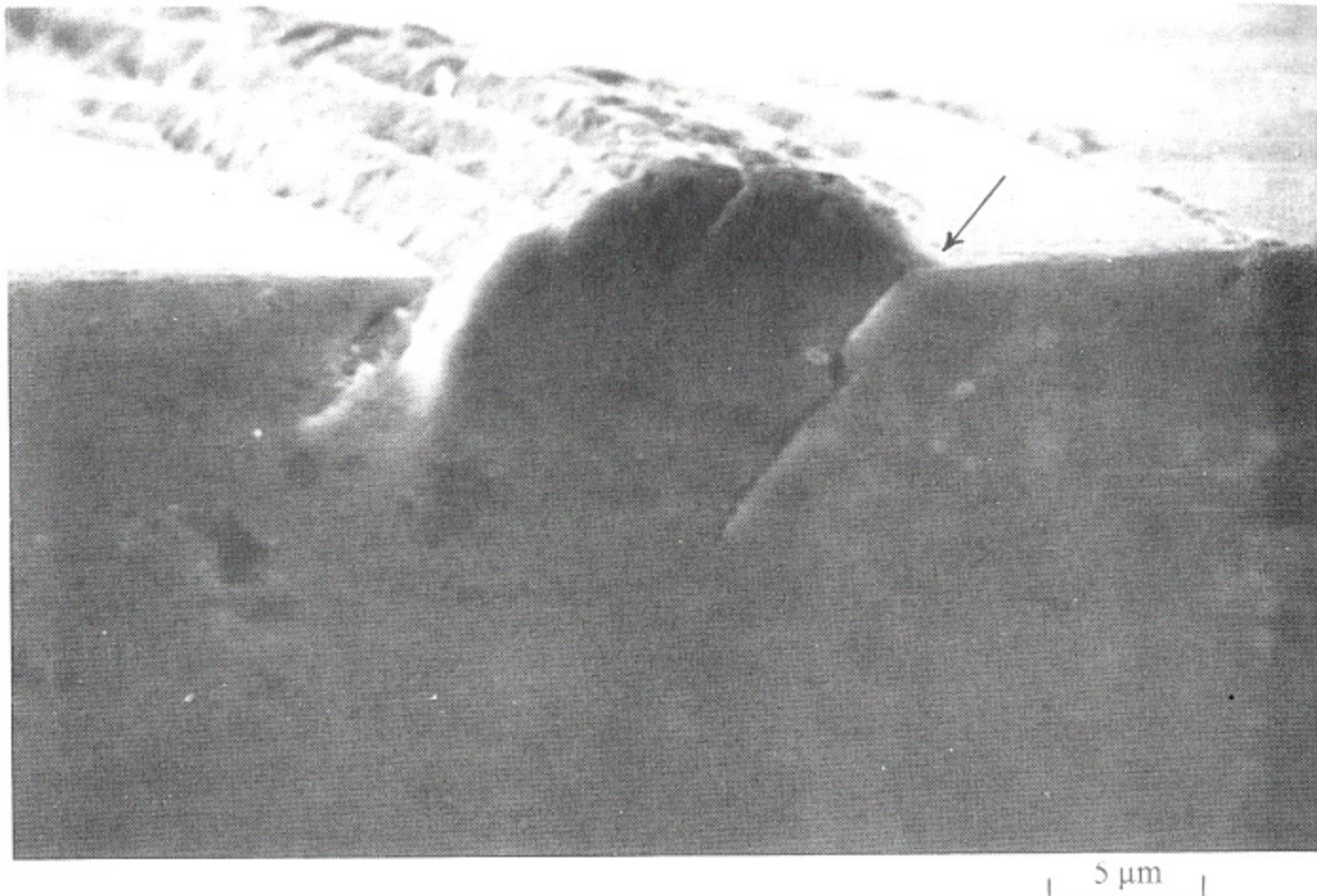
Polak, J. Cyclic Plasticity and Low Cycle Fatigue Life of Metals, Elsevier, 1991



Slip Band Formation

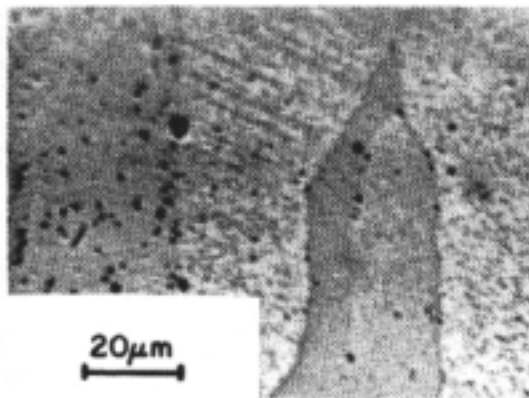


Slip Bands



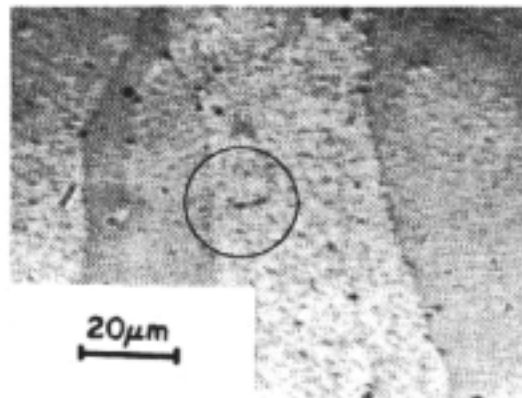
Ma, B-T and Laird C. "Overview of fatigue behavior in copper single crystals –II Population, size, distribution and growth Kinetics of stage I cracks for tests at constant strain amplitude", Acta Metallurgica, Vol 37, 1989, 337-348

2124-T4 Cracking in Slip Bands



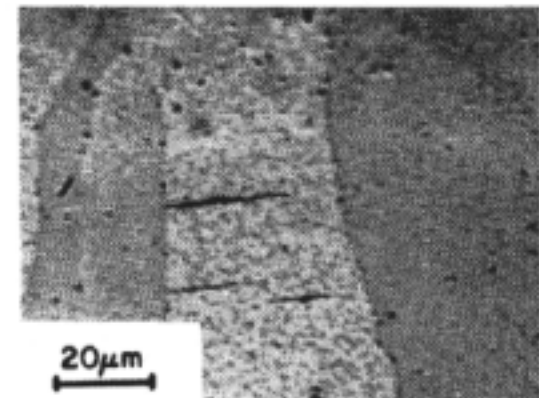
N = 60

(a)



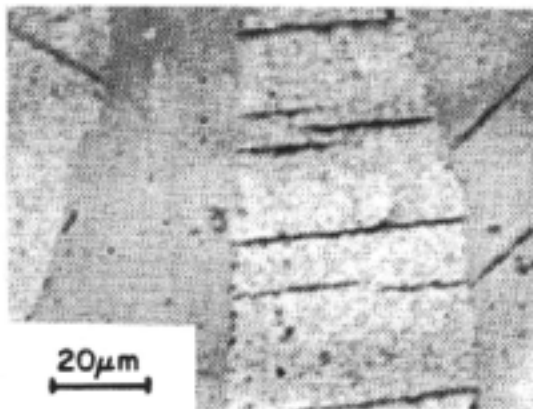
N = 240

(b)



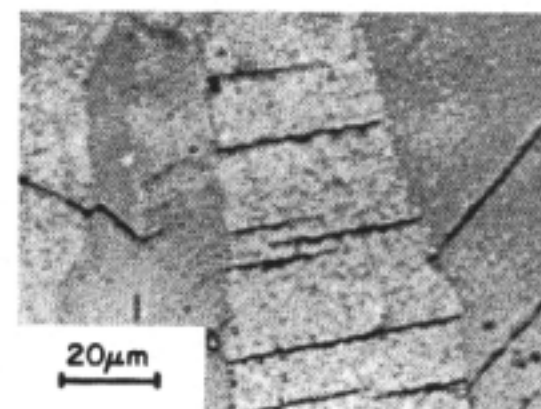
N = 300

(c)



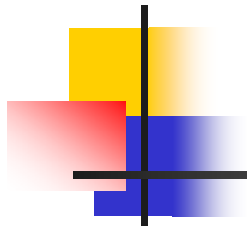
N = 1200

(d)



N = 2000

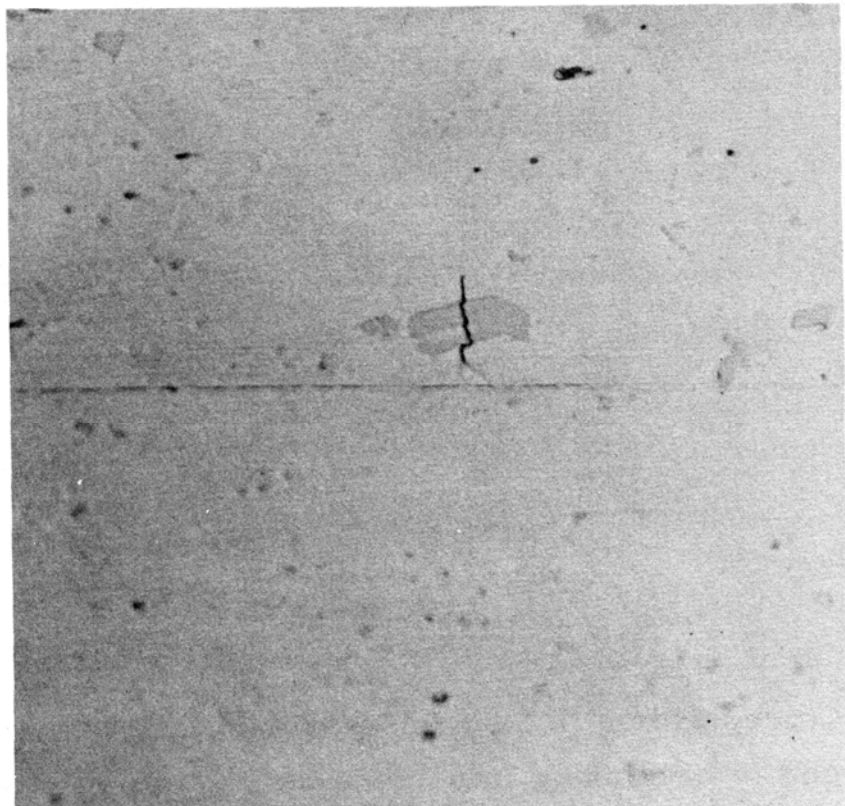
(e)



Crack at Particle

Material: BS L65 Aluminum

Loading: 63 ksi, $R=0$ for 500,000+ cycles, followed by 68 ksi, $R=0$ to failure. Cracks found during 68 ksi loading.



X 1000

S. Pearson, "Initiation of Fatigue Cracks in Commercial Aluminum Alloys and the Subsequent Propagation of Very Short Cracks," RAE TR 72236, Dec 1972.

2219-T851 Cracked Particle

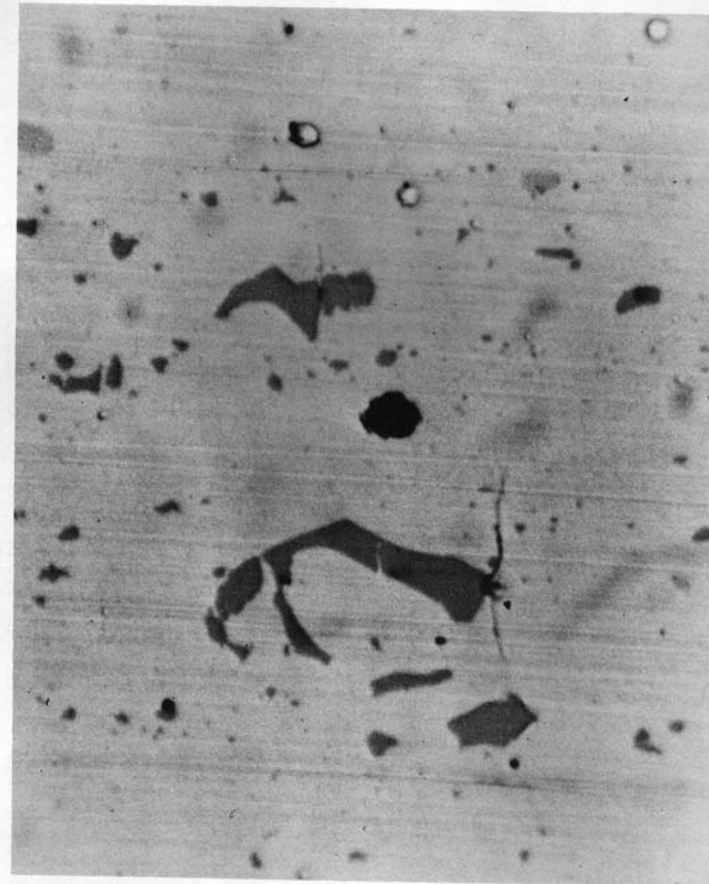


James & Morris, *ASTM STP 811 Fatigue Mechanisms: Advances in Quantitative Measurement of Physical Damage*, pp. 46-70, 1983.

Crack at Bonded Particle

Material: BS L65 Aluminum

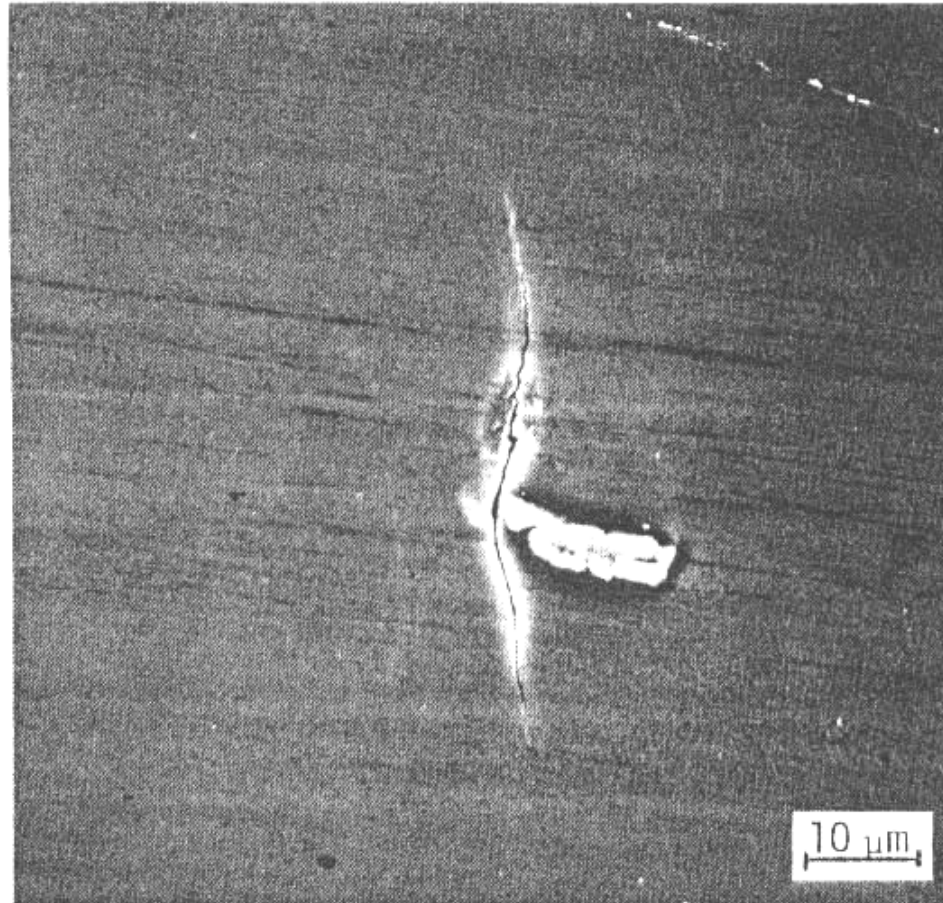
Loading: 63 ksi, $R=0$ for 500,000+ cycles, followed by 68 ksi, $R=0$ to failure. Cracks found during 68 ksi loading.



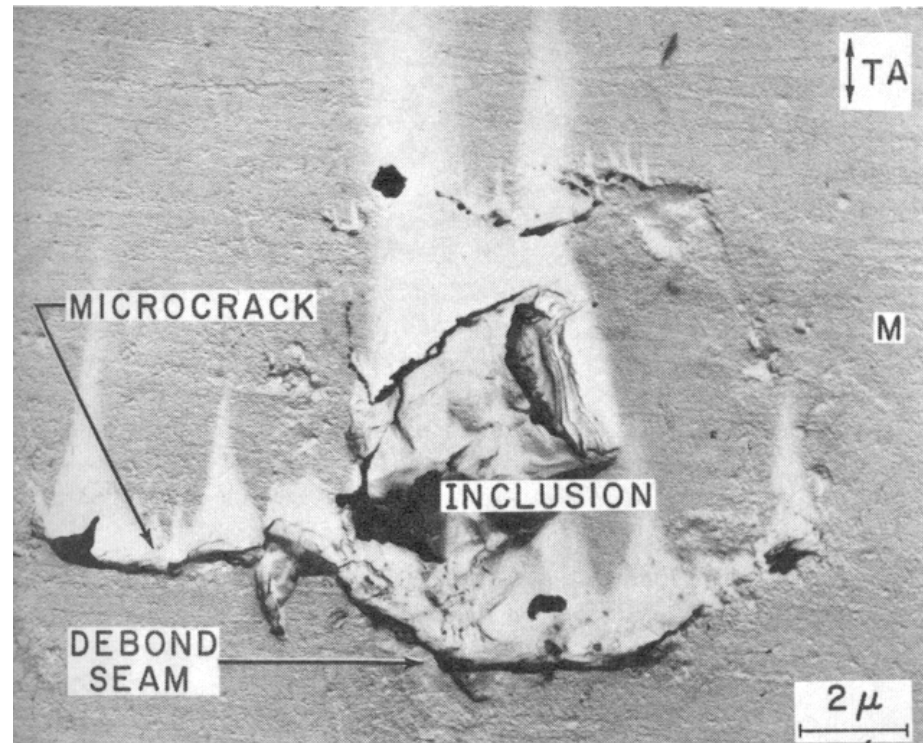
X 1600

S. Pearson, "Initiation of Fatigue Cracks in Commercial Aluminum Alloys and the Subsequent Propagation of Very Short Cracks," RAE TR 72236, Dec 1972.

7075-T6 Cracking at Inclusion

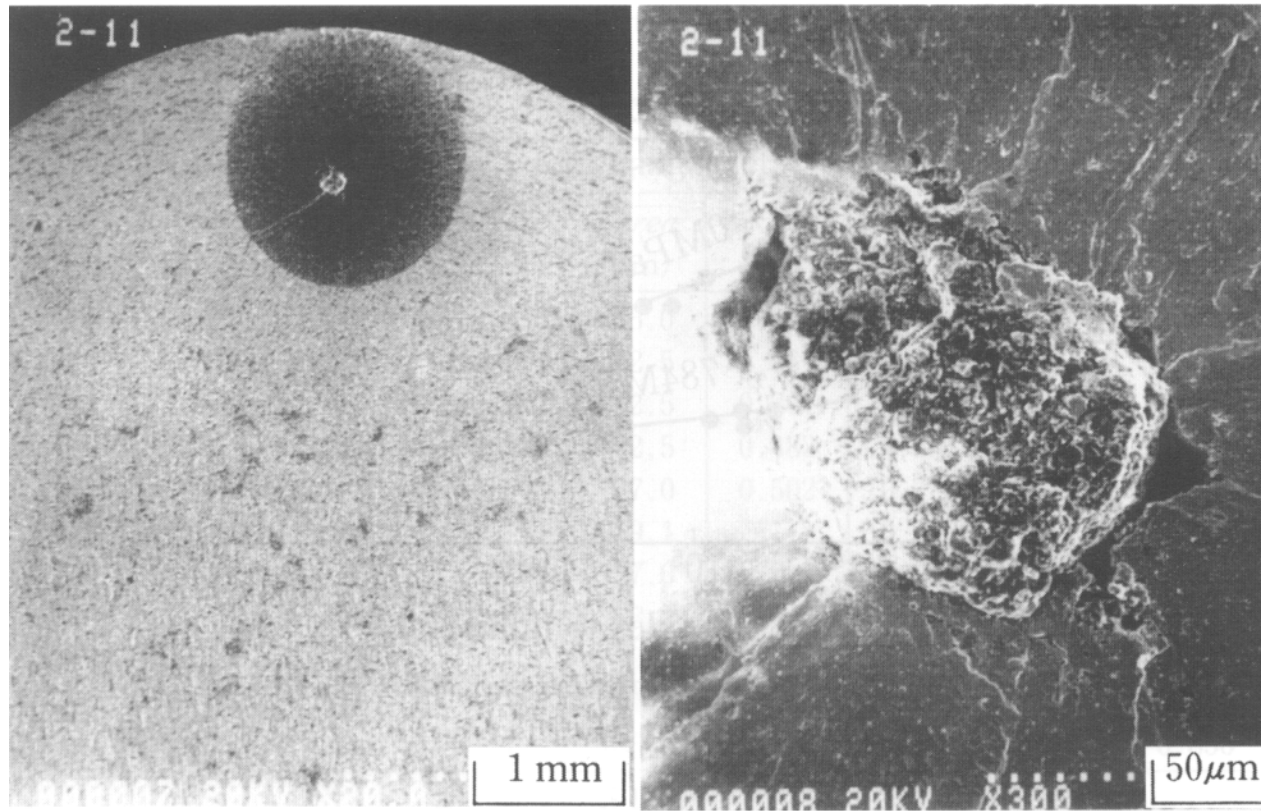


Crack Initiation at Inclusions



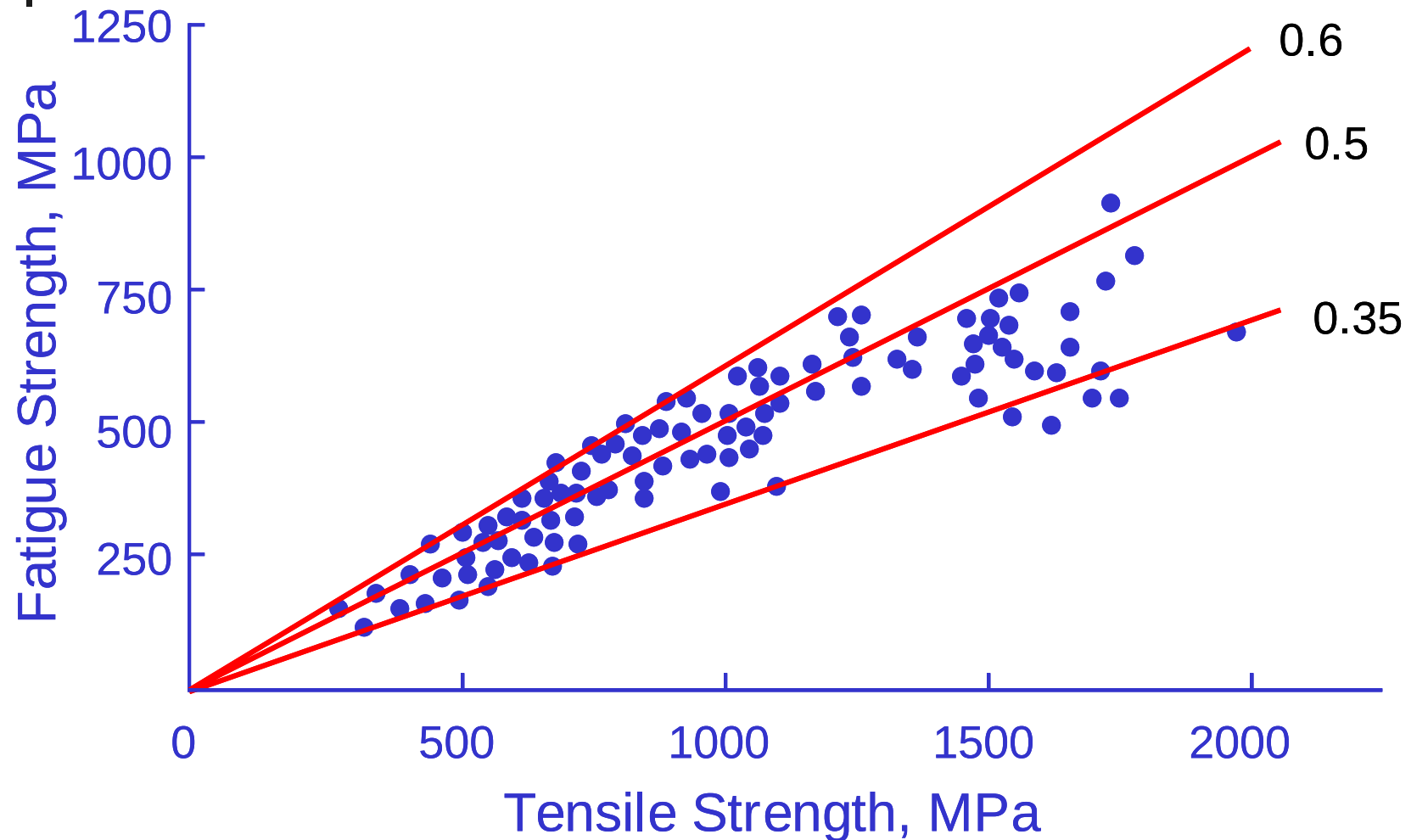
Langford and Kusenberger, "Initiation of Fatigue Cracks in 4340 Steel", *Metallurgical Transactions*, Vol 4, 1977, 553-559

Subsurface Crack Initiation

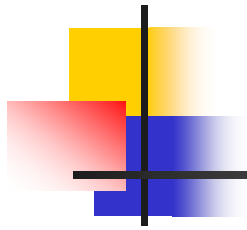


Y. Murakami, Metal Fatigue: *Effects of Small Defects and Nonmetallic Inclusions*, 2002

Fatigue Limit and Strength Correlation



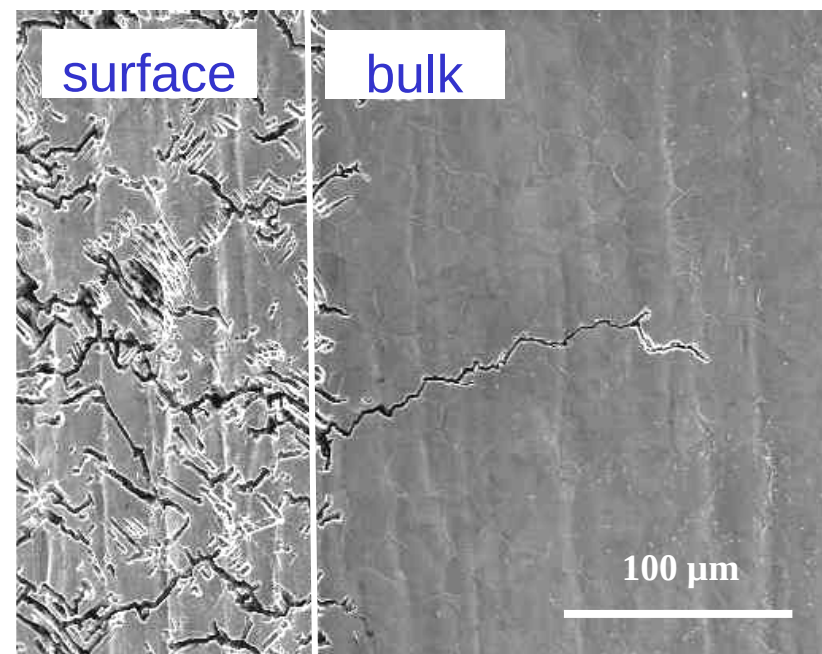
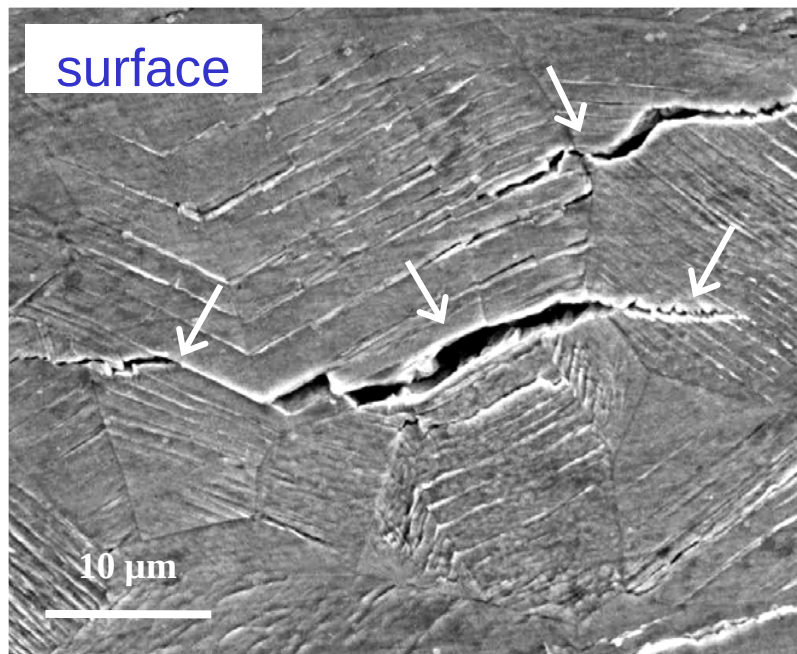
From Forrest, *Fatigue of Metals*, Pergamon Press, London, 1962



Crack Nucleation Summary

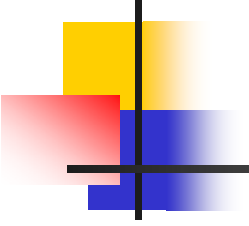
- Highly localized plastic deformation
- Surface phenomena
- Stochastic process

Surface Damage

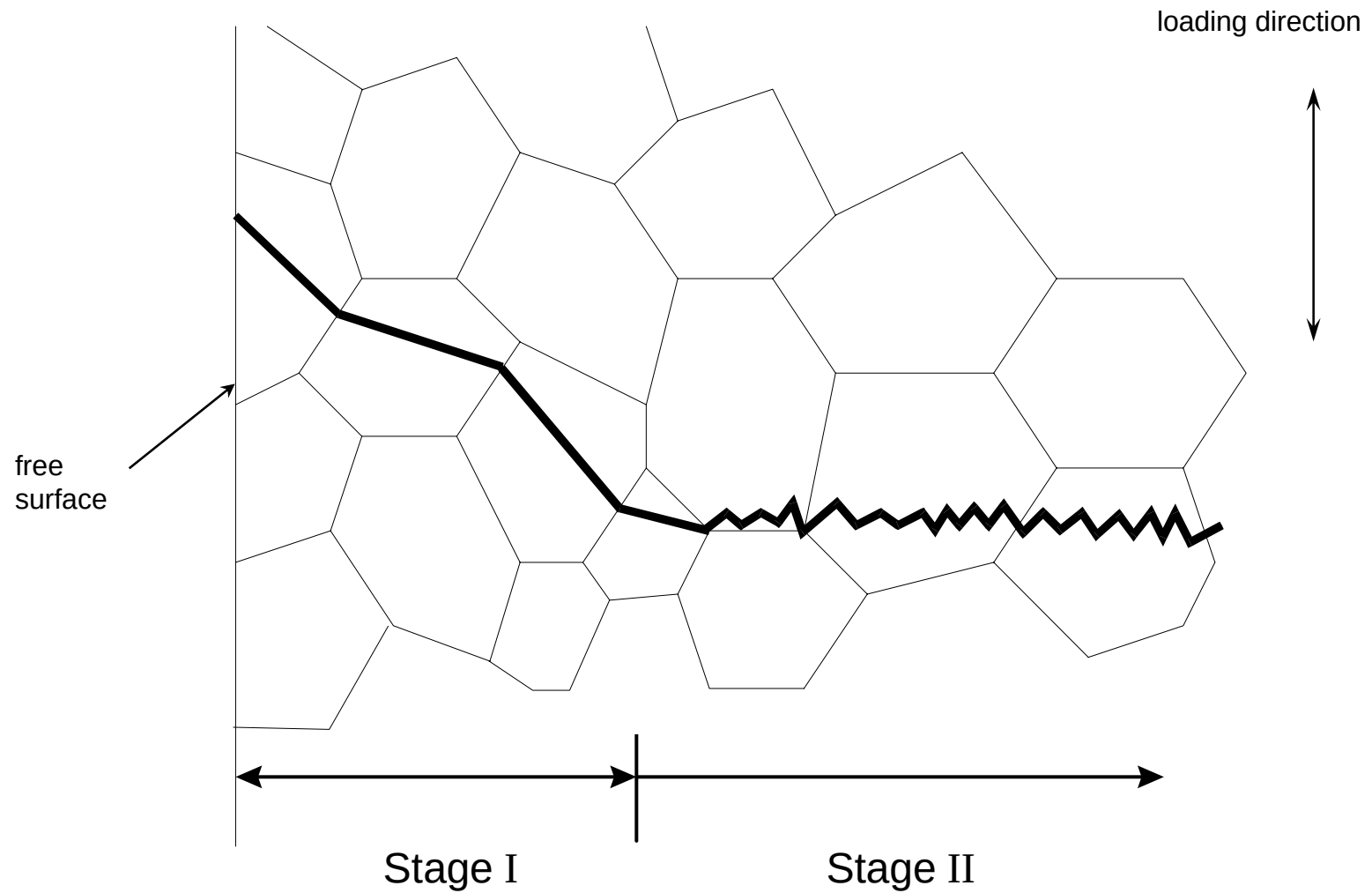


**20-25 austenitic steel in symmetrical push-pull fatigue
(20°C, $\Delta\epsilon_p/2 = \pm 0.4\%$) : short cracks on the surface and in the bulk**

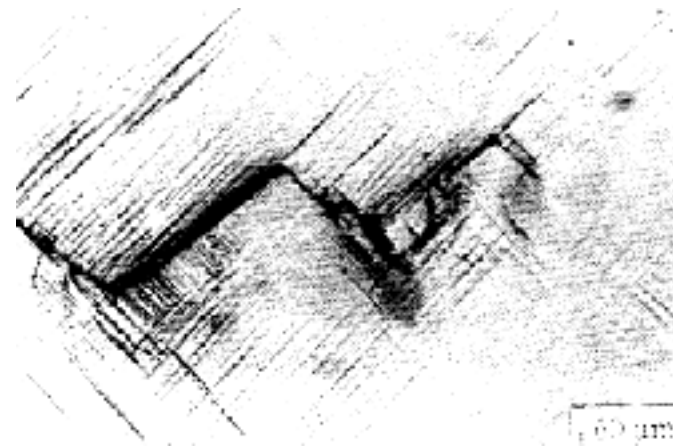
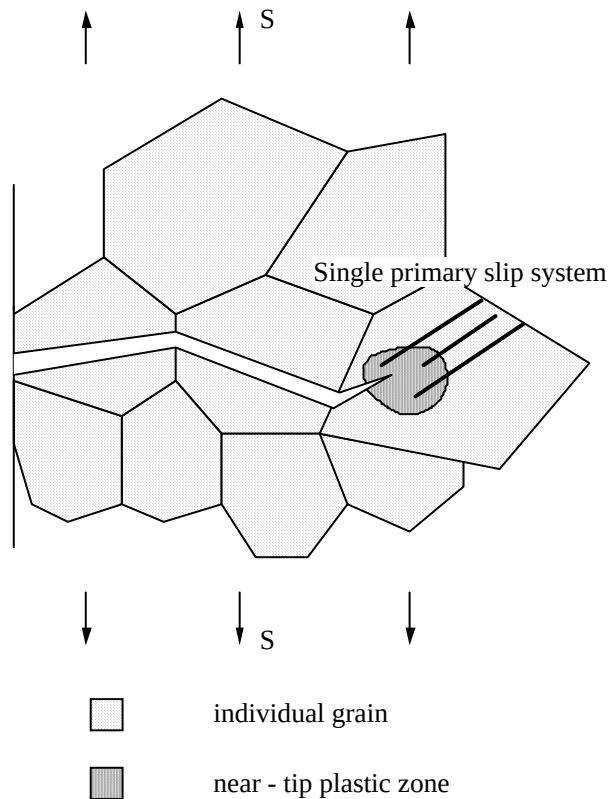
From Jacques Stolarz, Ecole Nationale Supérieure des Mines
Presented at LCF 5 in Berlin, 2003



Stage I and Stage II

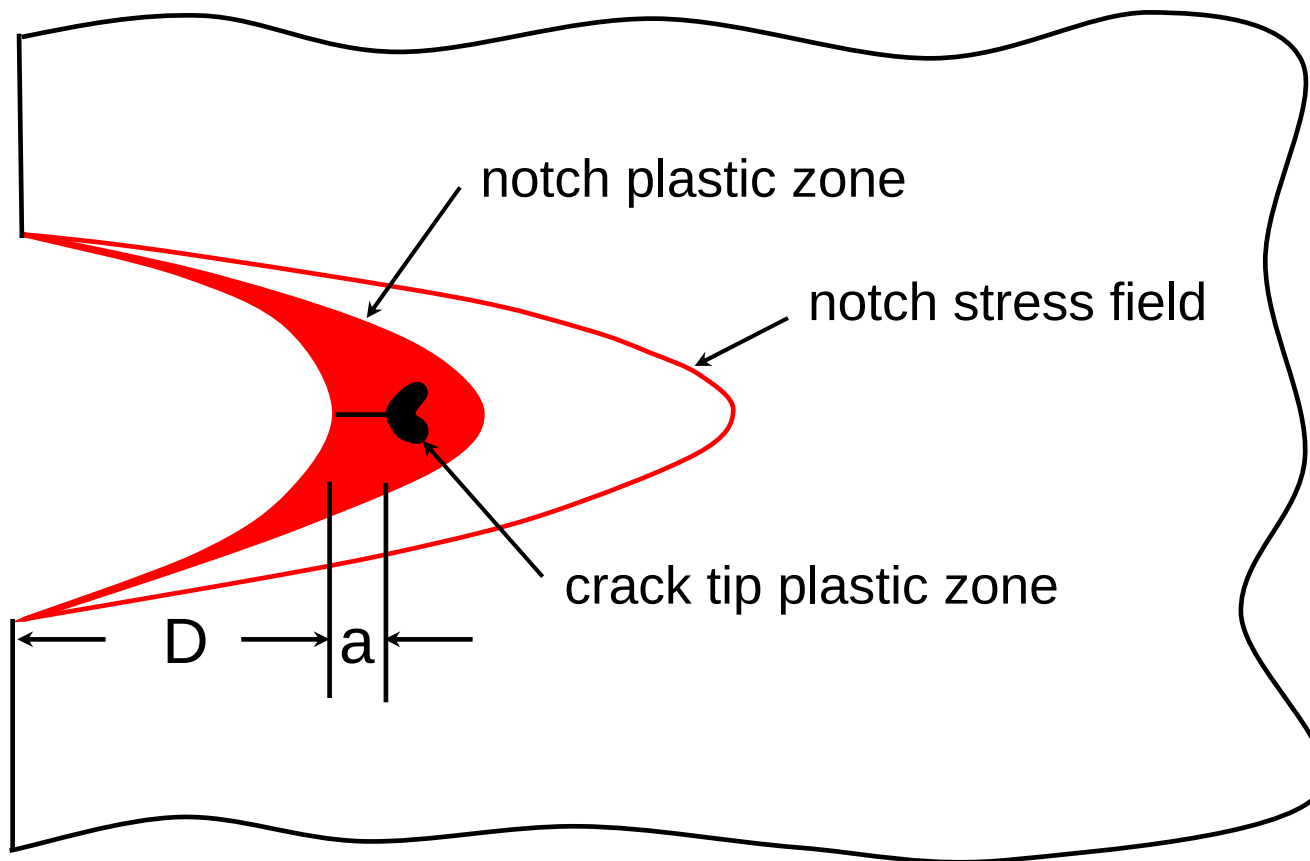


Stage I Crack Growth



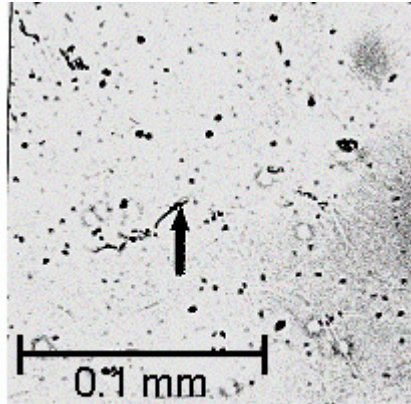
Stage I crack is strongly affected by slip characteristics, microstructure dimensions, stress level, extent of near tip plasticity

Small Cracks at Notches

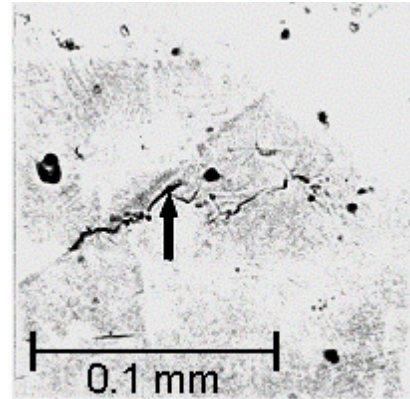


Crack growth controlled by the notch plastic strains

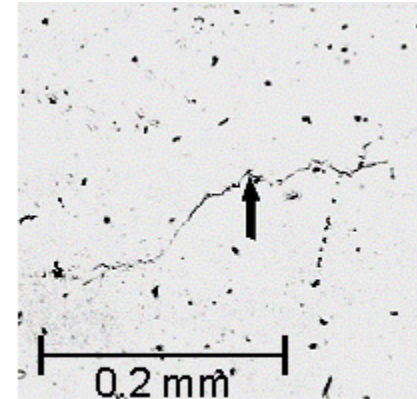
Small Crack Growth



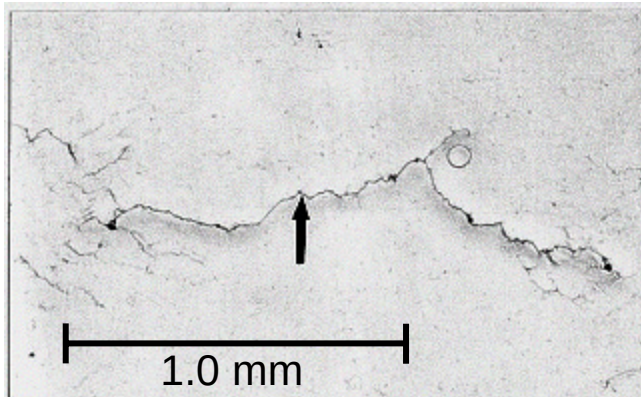
N = 160



N = 240



N = 520



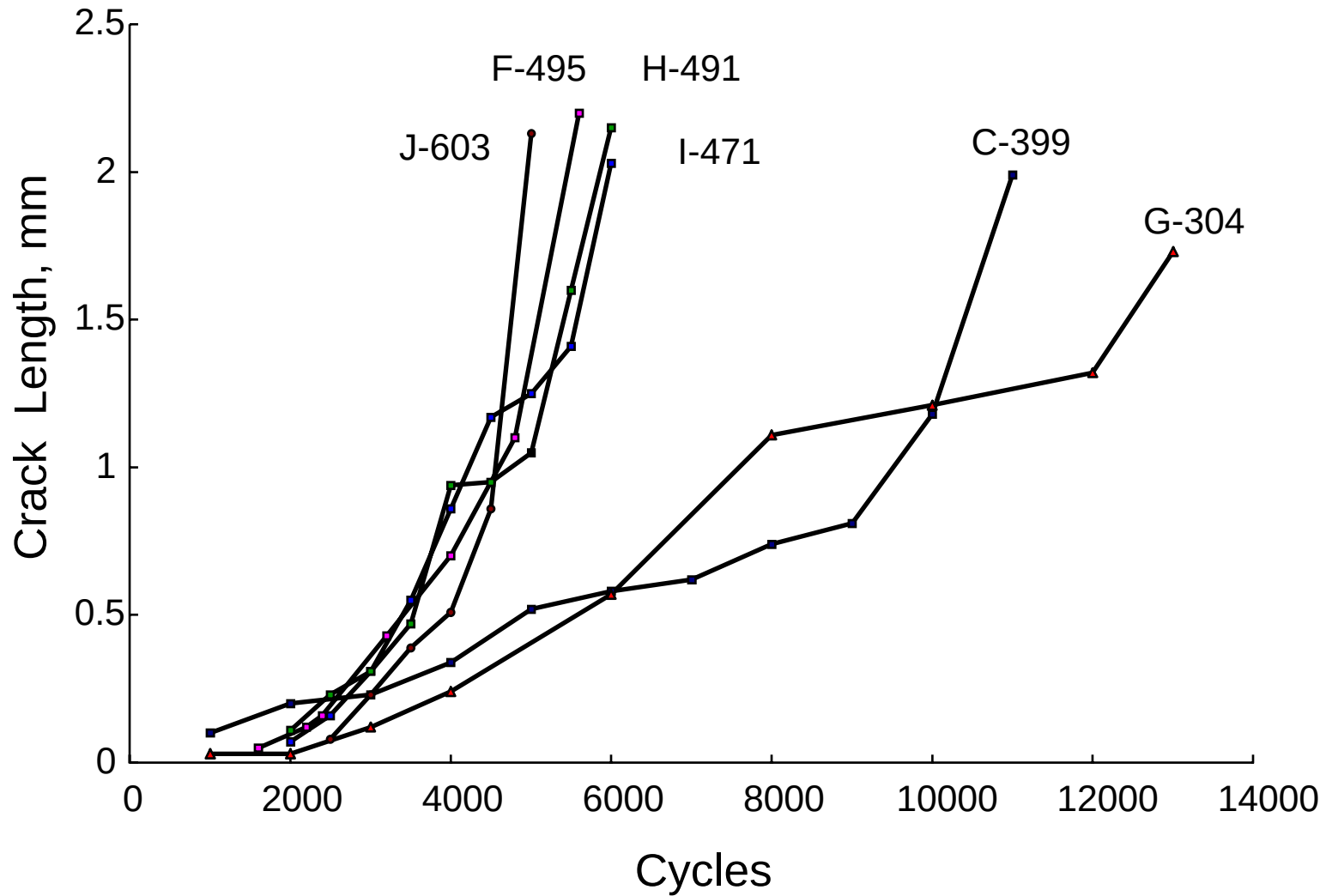
N = 900

Inconel 718

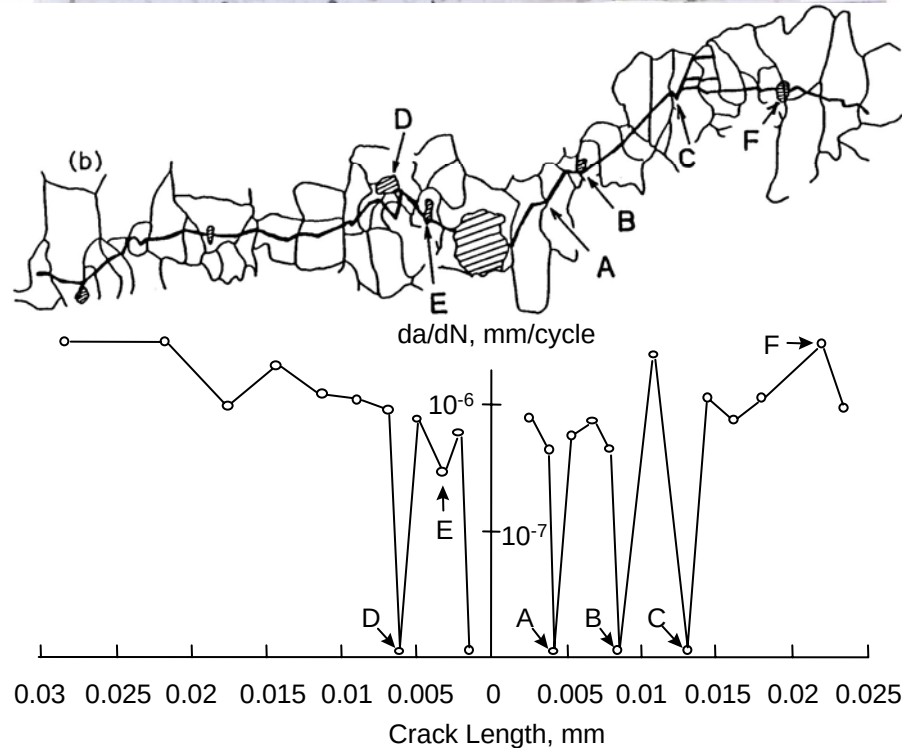
$\Delta\varepsilon = 0.02$

$N_f = 936$

Crack Length Observations

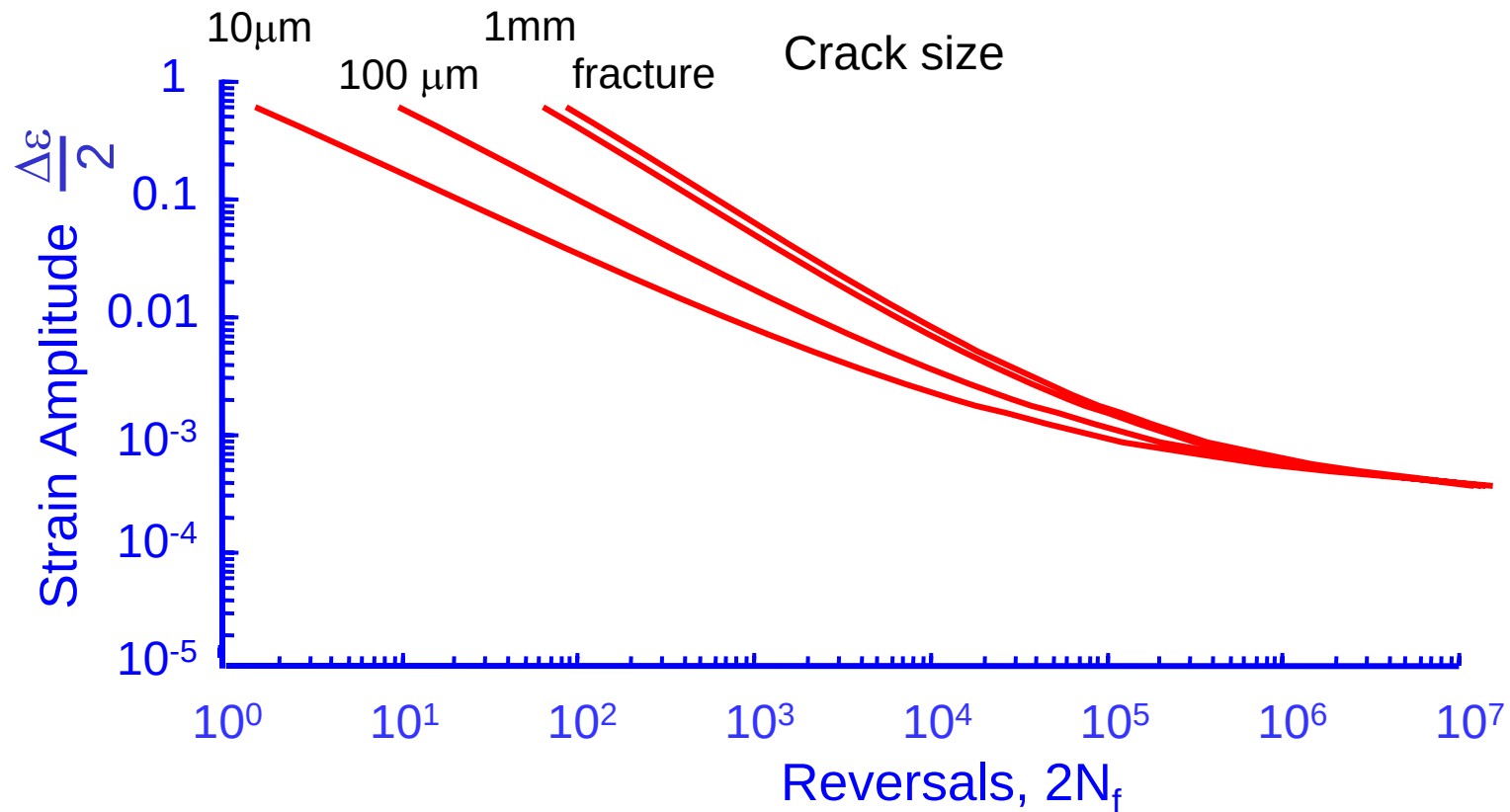


Crack - Microstructure Interactions

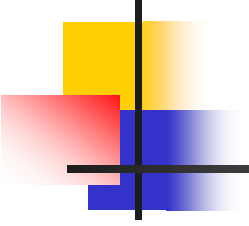


Akiniwa, Y., Tanaka, K., and Matsui, E., "Statistical Characteristics of Propagation of Small Fatigue Cracks in Smooth Specimens of Aluminum Alloy 2024-T3, *Materials Science and Engineering*, Vol. A104, 1988, 105-115

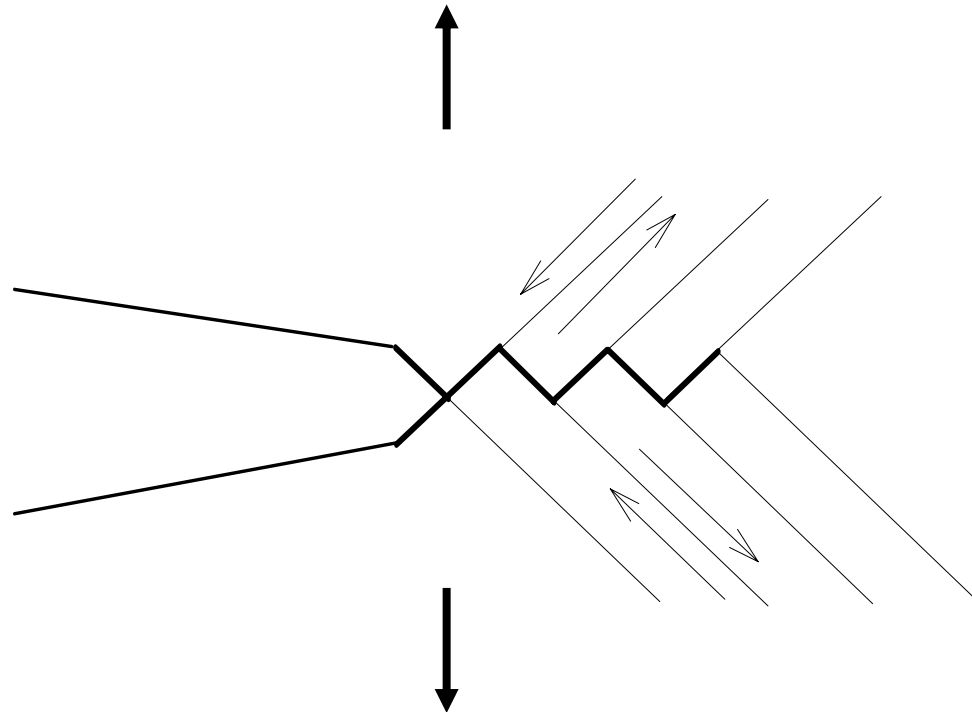
Strain-Life Data



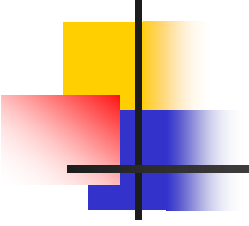
Most of the life is spent in microcrack growth in the plastic strain dominated region



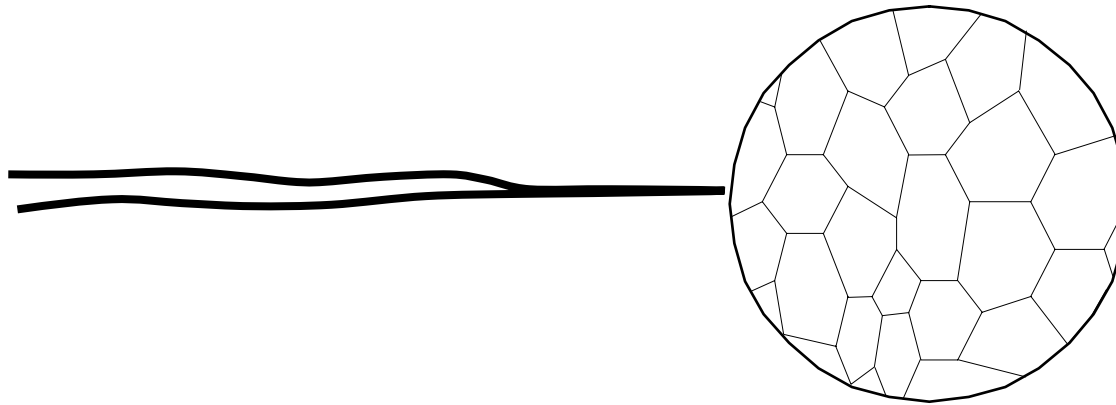
Stage II Crack Growth



Locally, the crack grows in shear
Macroscopically it grows in tension

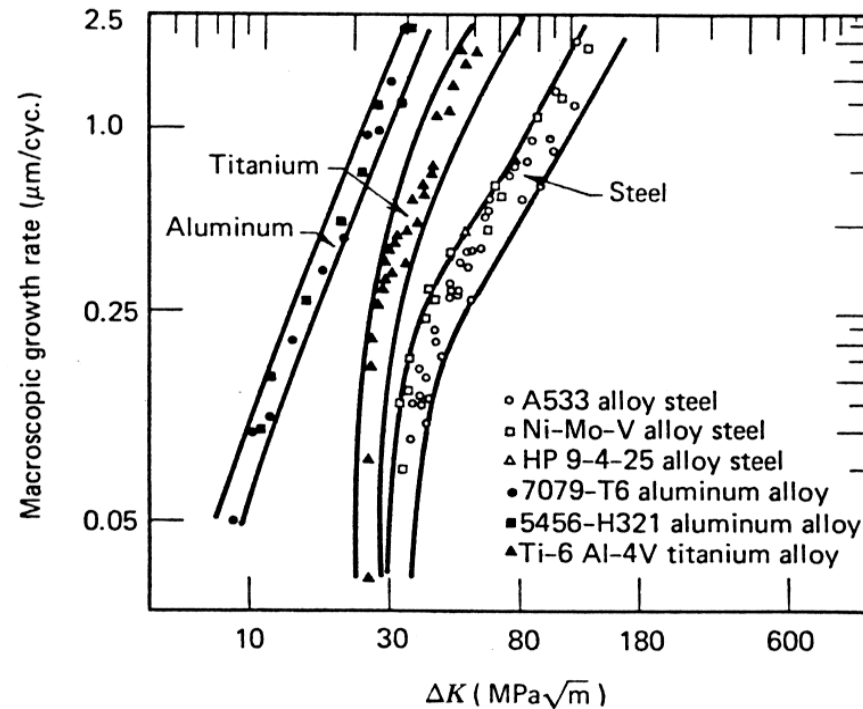


Long Crack Growth

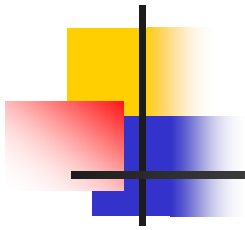


Plastic zone size is much larger than the material microstructure so that the microstructure does not play such an important role.

Crack Growth Rates of Metals

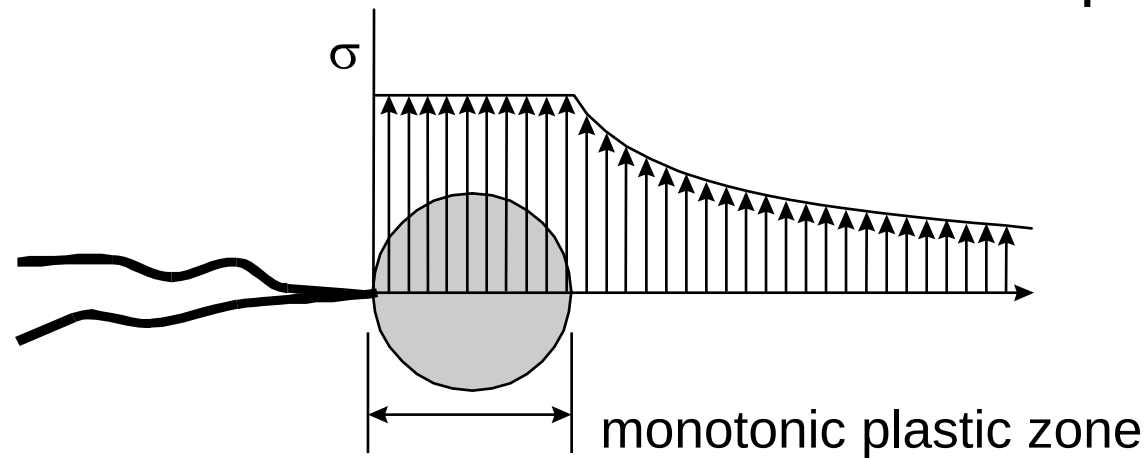
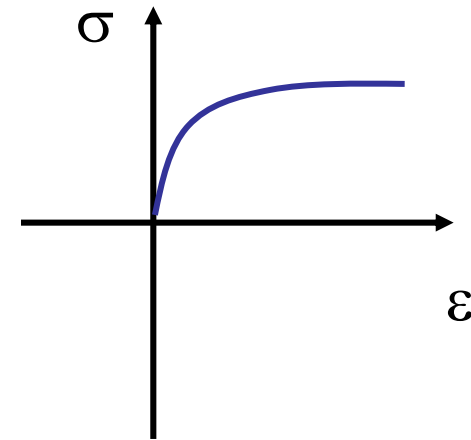
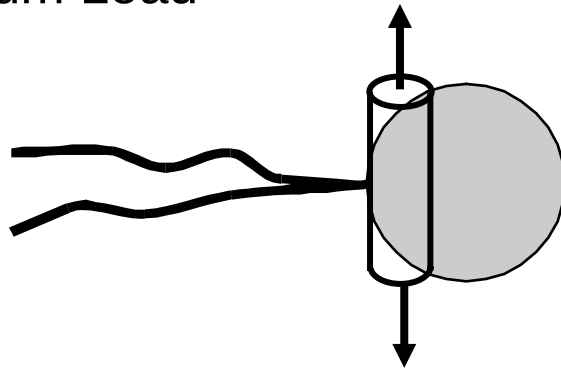


Material strength does not play a major role in fatigue crack growth



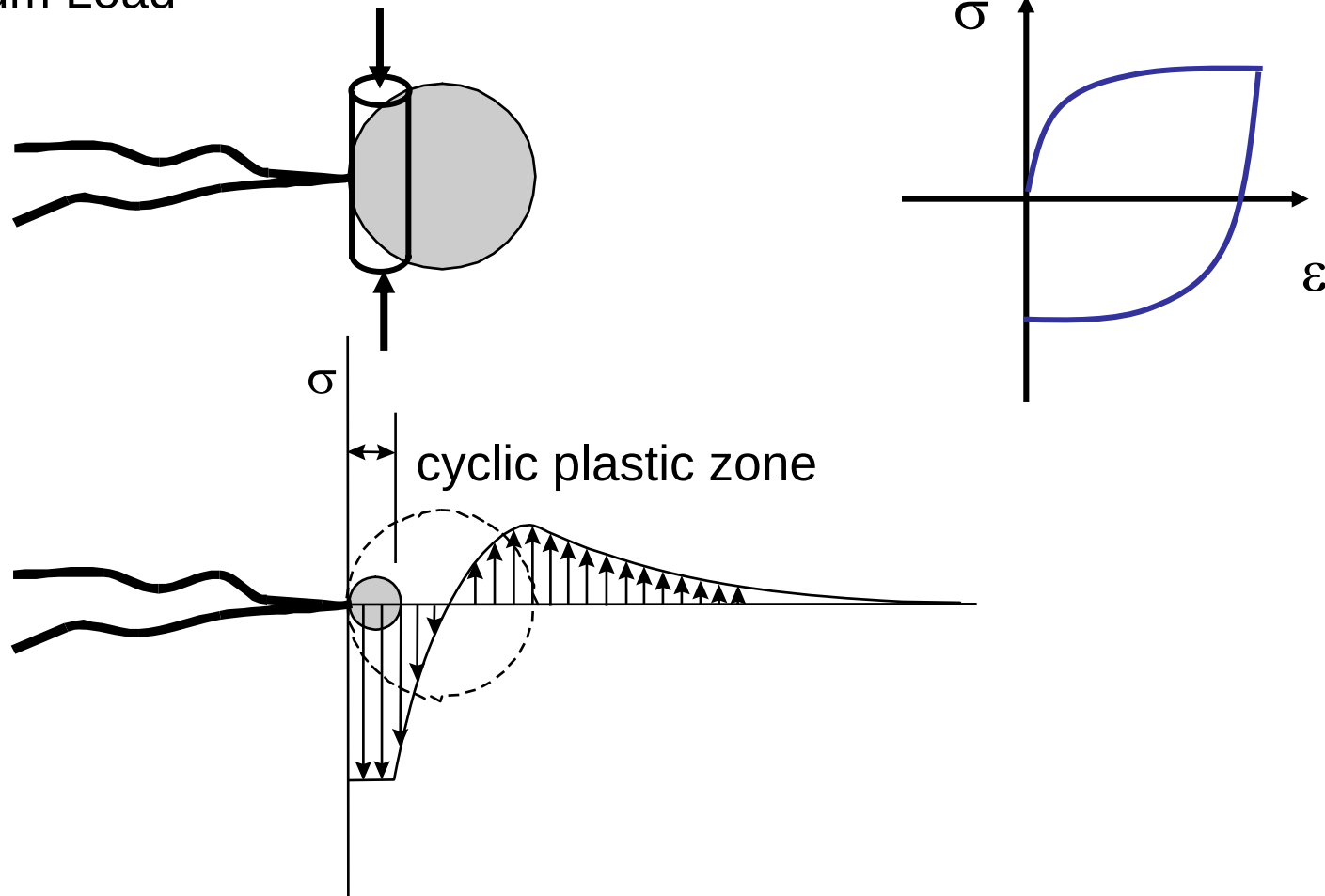
Stresses Around a Crack

Maximum Load

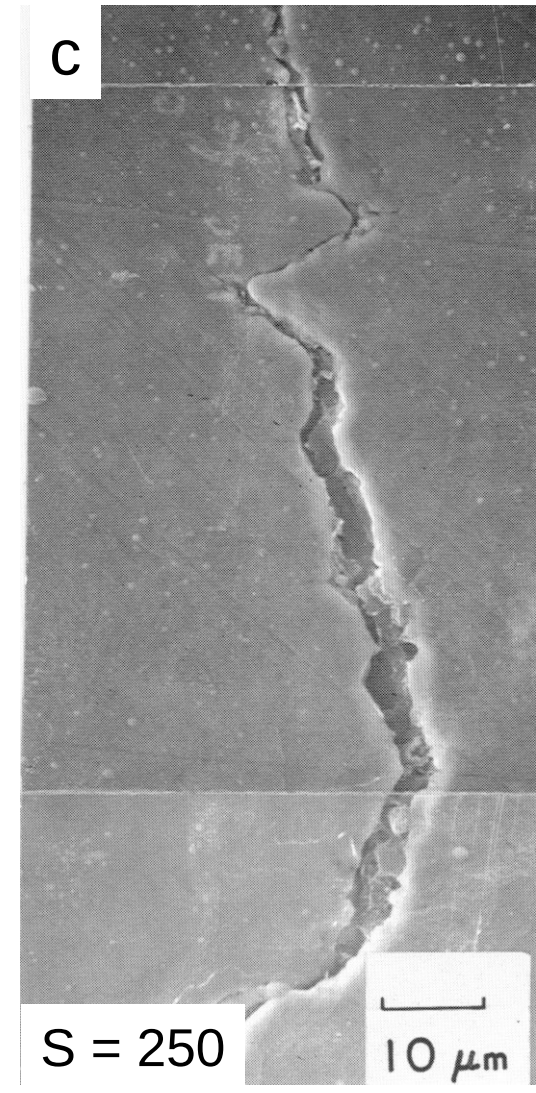


Stresses Around a Crack (continued)

Minimum Load

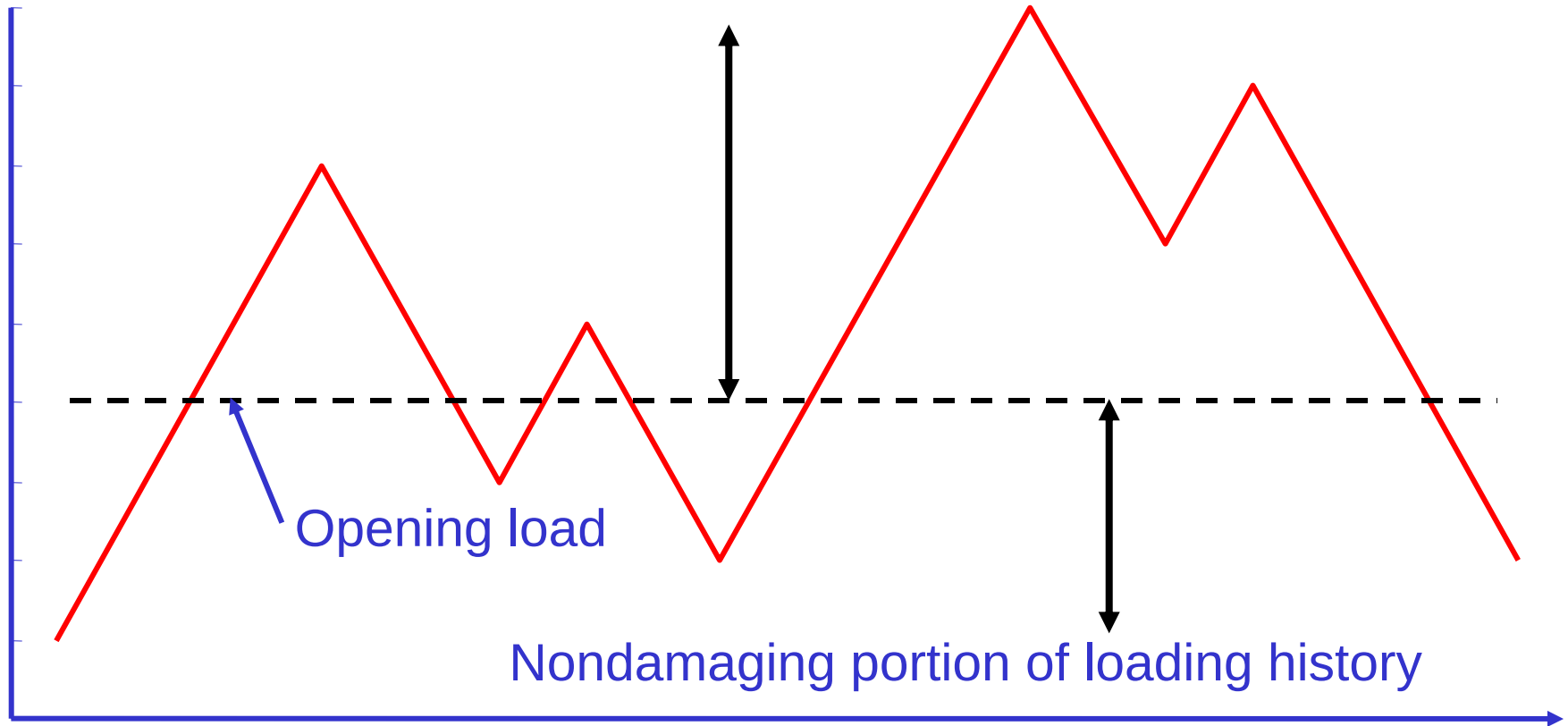


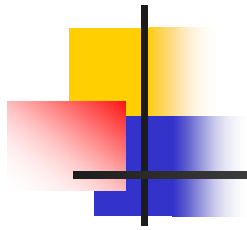
Crack Closure



Crack Opening Load

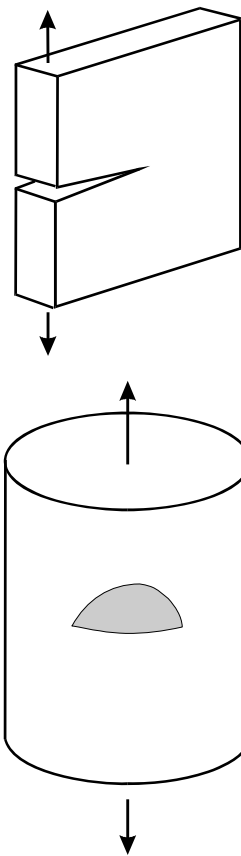
Damaging portion of loading history



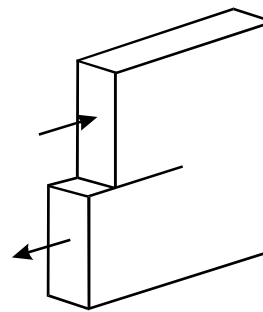


Mode I, Mode II, and Mode III

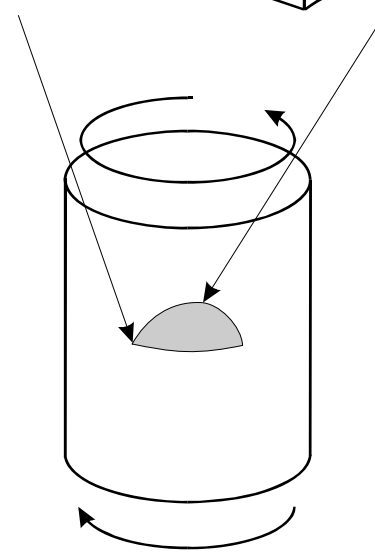
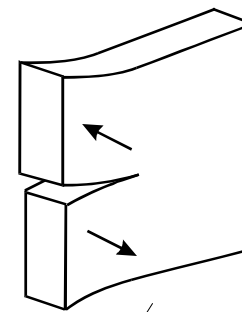
Mode I
opening

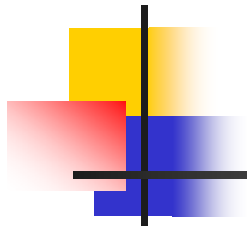


Mode II
in-plane shear

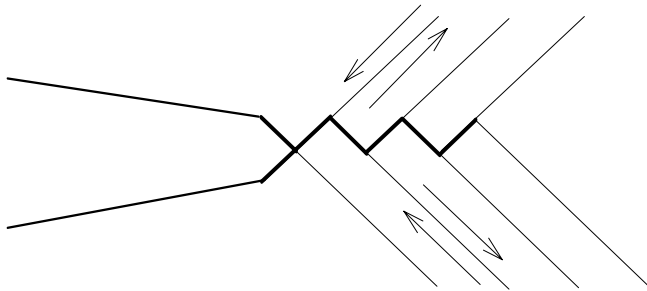


Mode III
out-of-plane shear

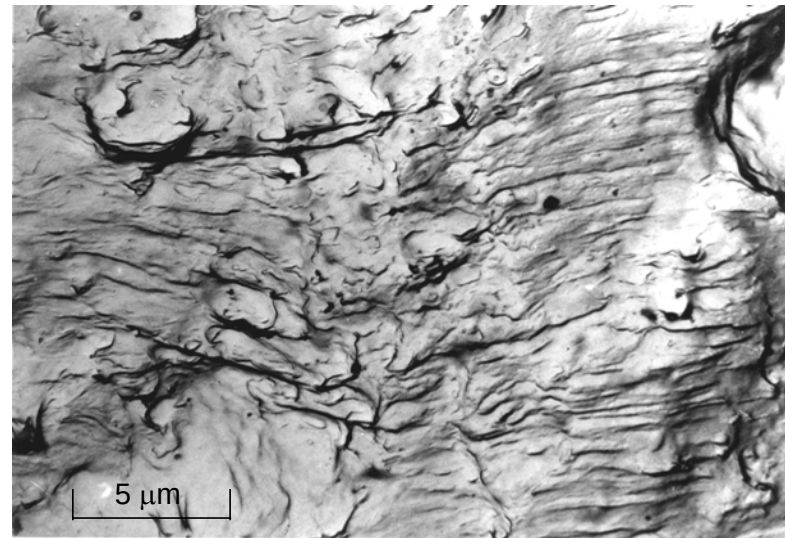




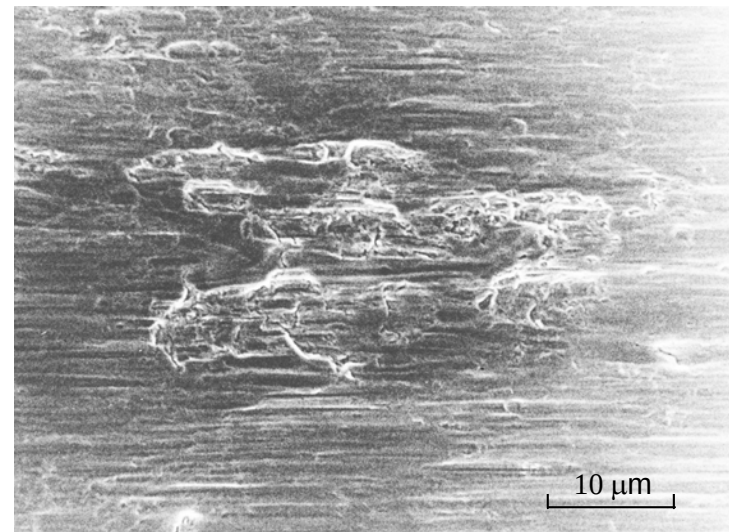
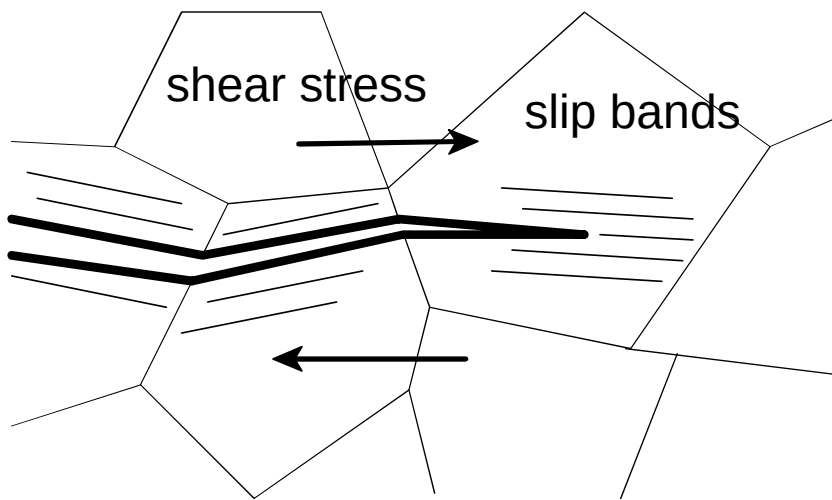
Mode I Growth



crack growth direction

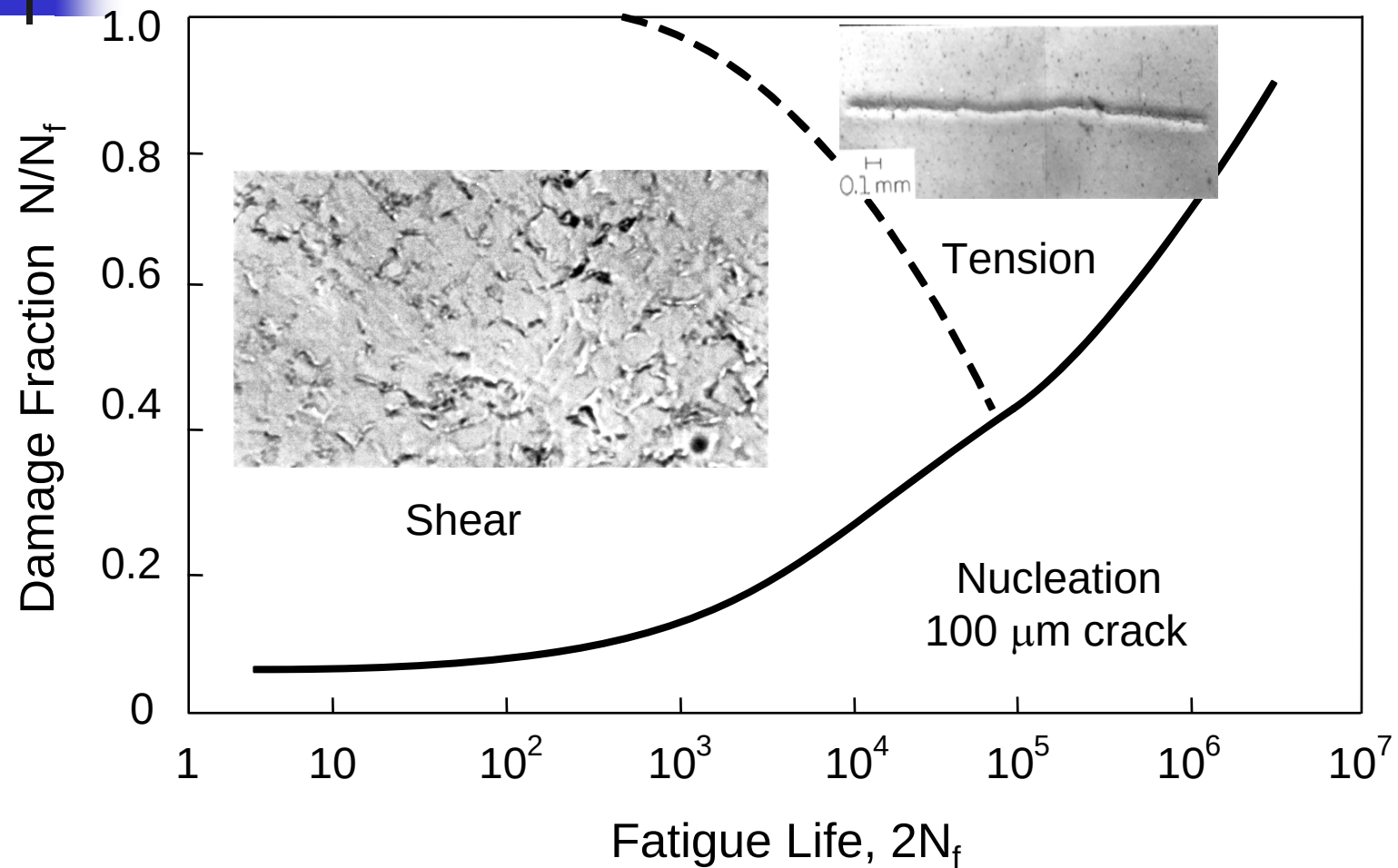


Mode II Growth

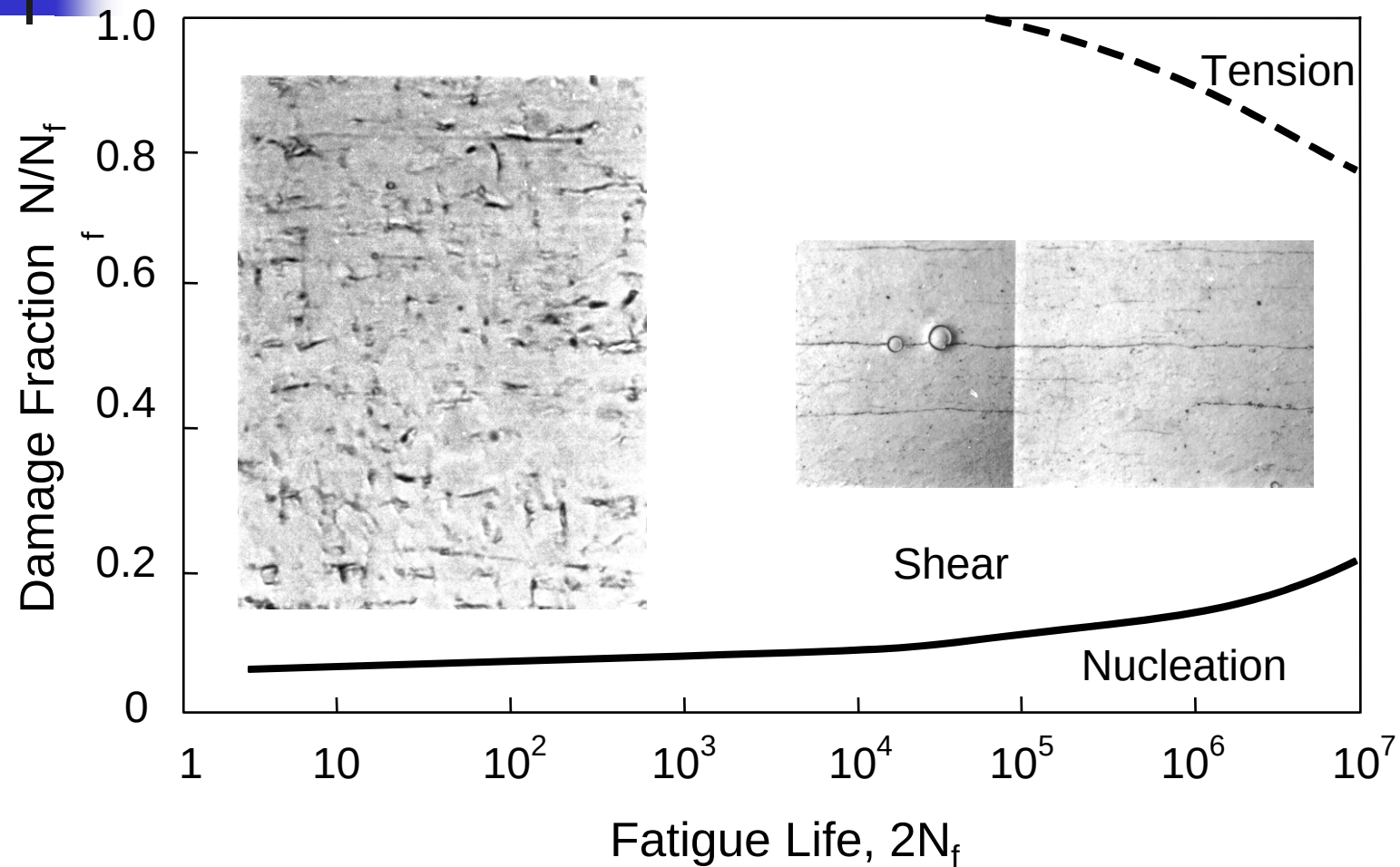


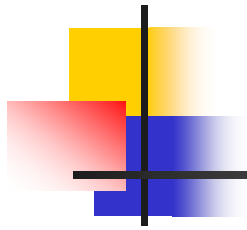
← crack growth direction

1045 Steel - Tension



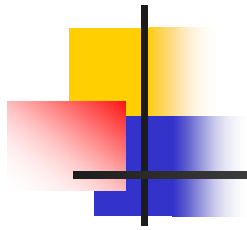
1045 Steel - Torsion





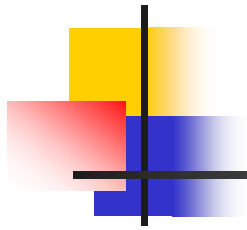
Things Worth Remembering

- Fatigue is a localized process involving the nucleation and growth of cracks to failure.
- Fatigue is caused by localized plastic deformation.
- Most of the fatigue life is consumed growing microcracks in the finite life region
- Crack nucleation is dominate at long lives.



Fatigue, How and Why

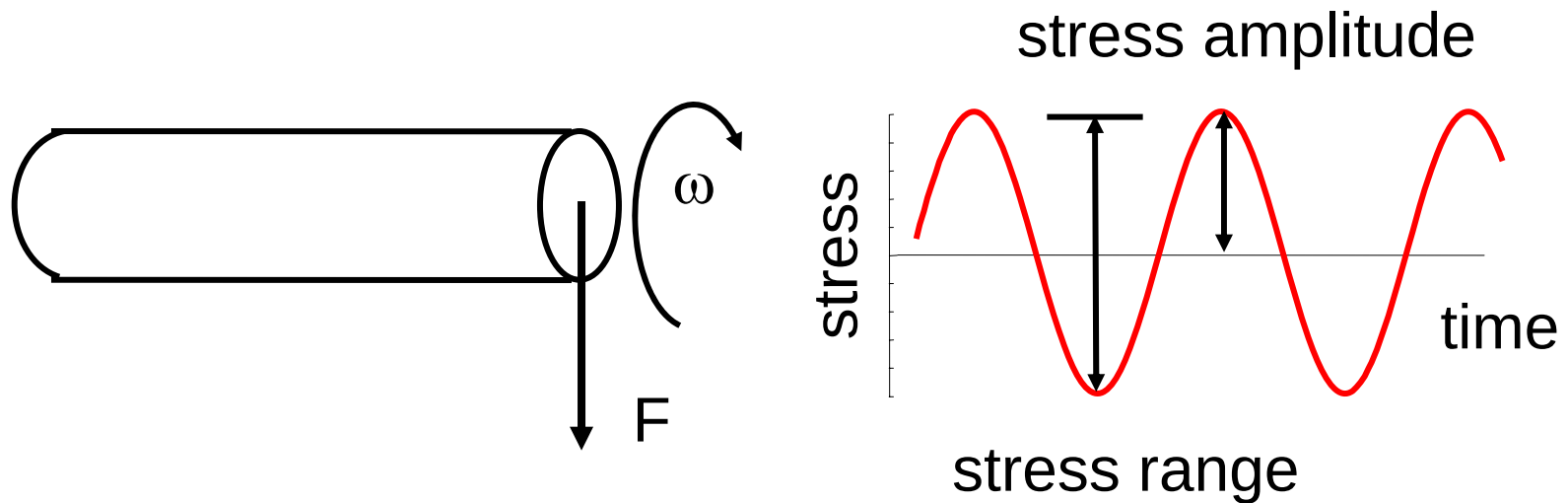
- Physics of Fatigue
- **Material Properties**
- Similitude
- Fatigue Calculator



Characterization

- Stress Life Curve
 - Fatigue Limit
- Strain Life Curve
 - Cyclic Stress Strain Curve
- Crack Growth Curve
 - Threshold Stress Intensity

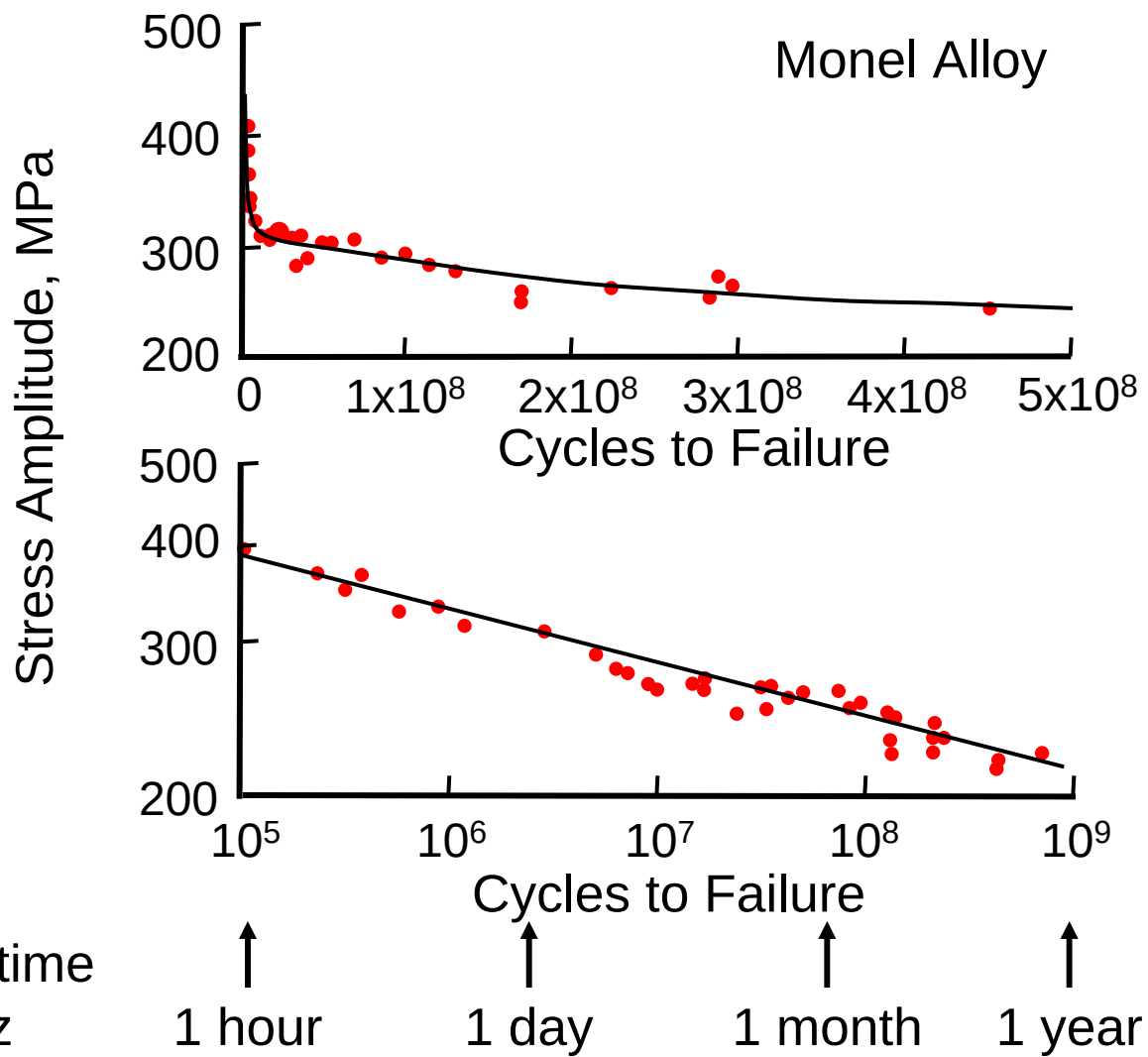
Bending Fatigue

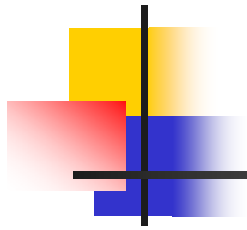


Bending stress:

$$\sigma = \frac{Mc}{I}$$

SN Curve

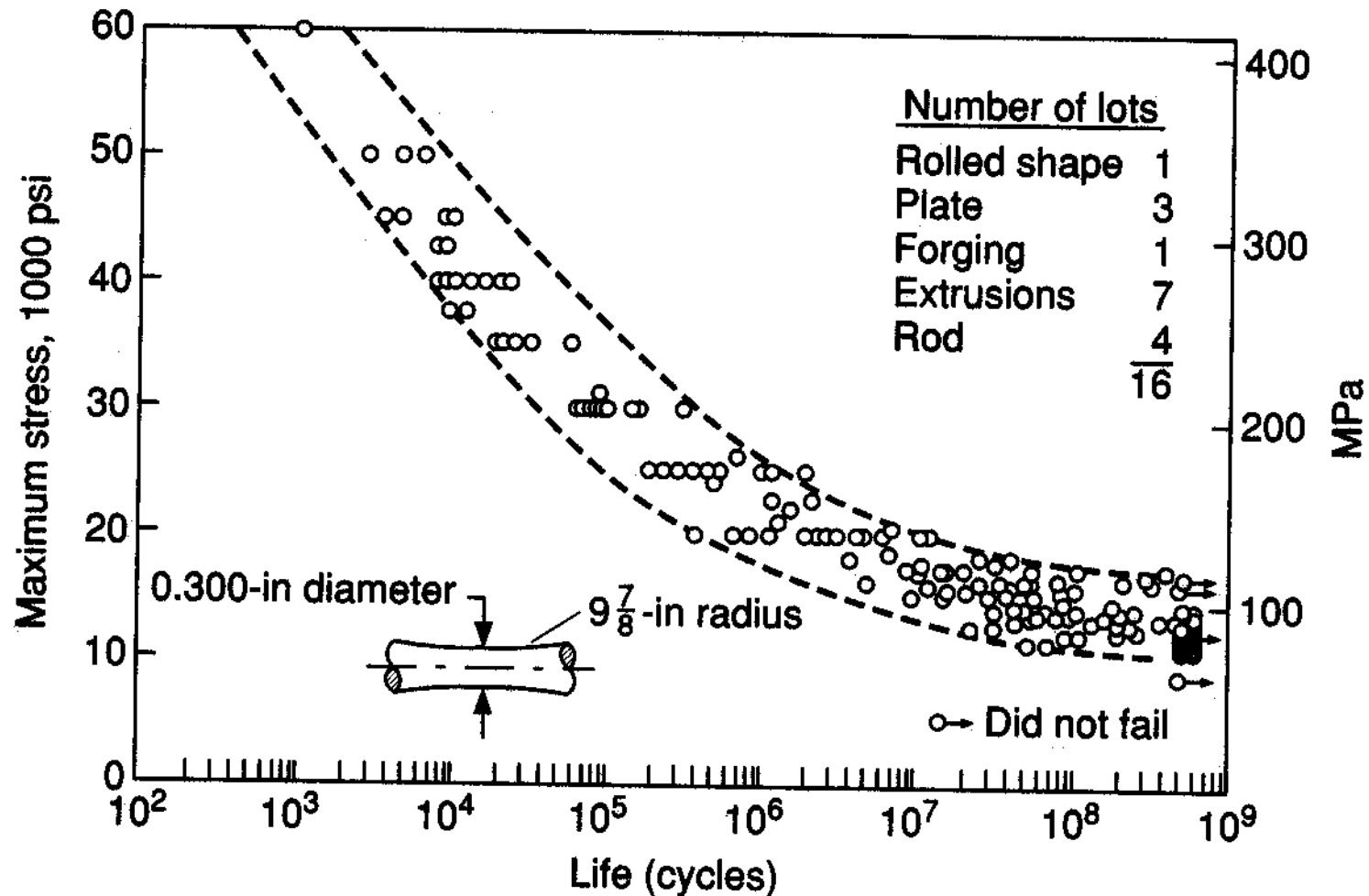




Fatigue Strength

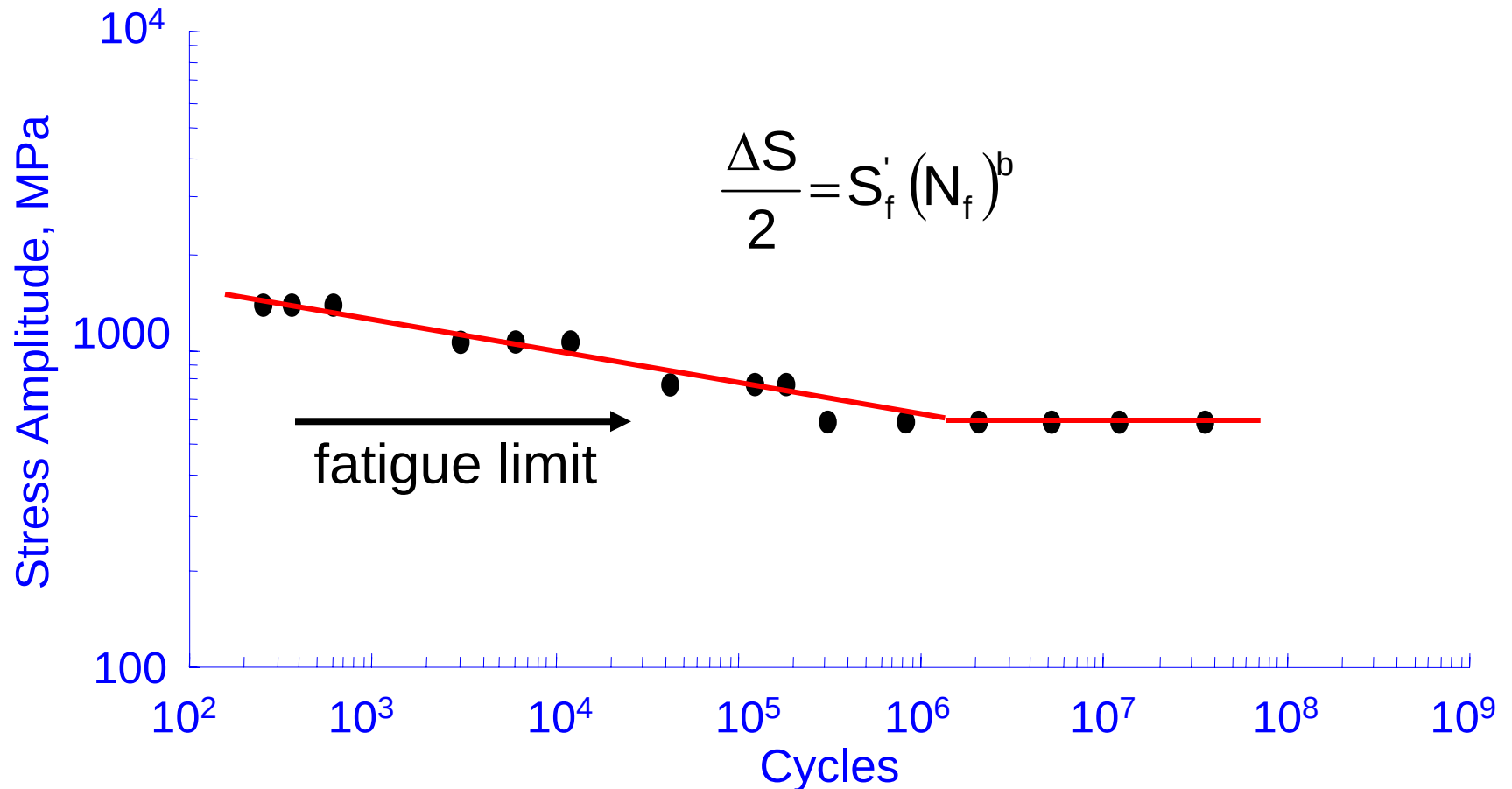
Alloy	Fatigue Life				
	10^5	10^6	10^7	10^8	10^9
2014-T4	290	235	186	152	138
2024-T4	297	214	166	145	138
6061-T6	186	152	117	104	90
7075-T6	276	200	166	152	145

6061-T6 Aluminum Test Data



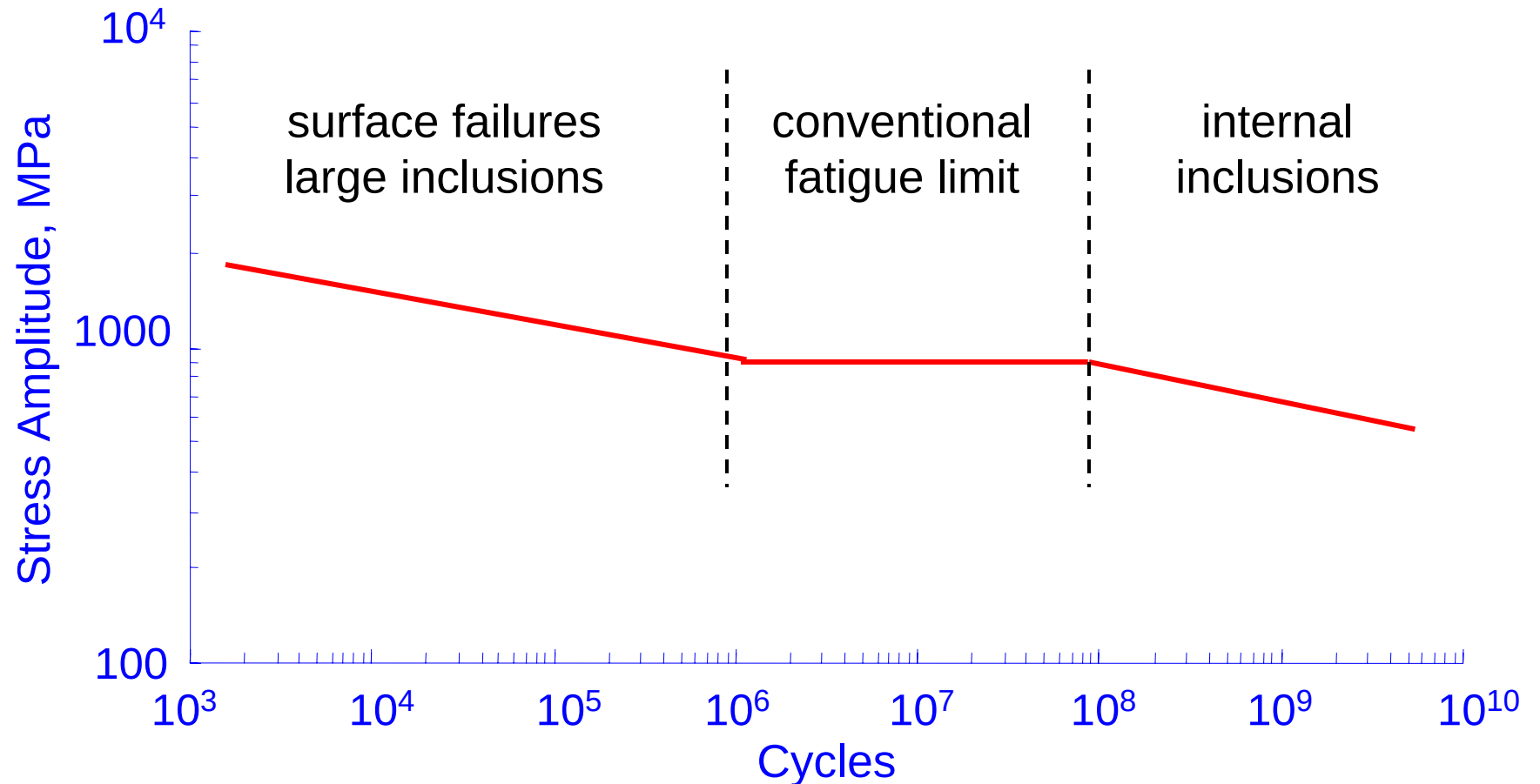
Sharpe et. al. Fatigue Design of Aluminum Components and Structures , 1996

SN Curve for Steel

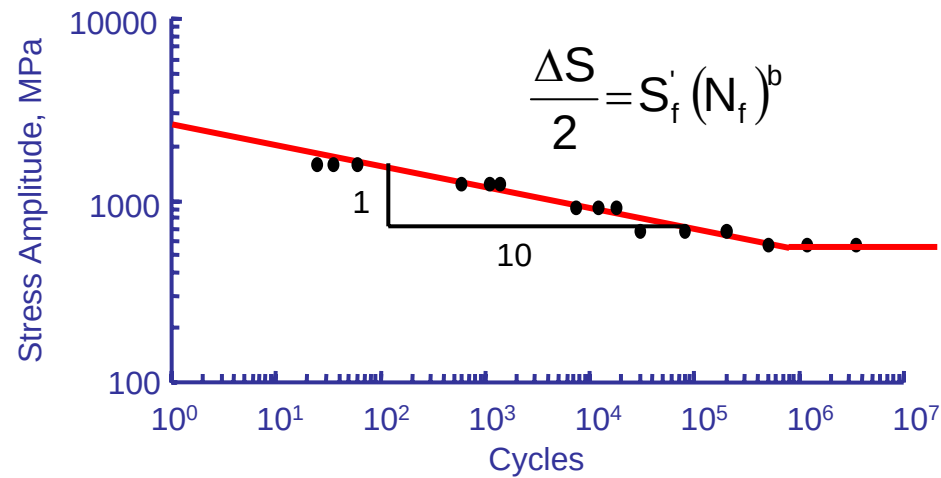


The fatigue limit is usually only found in steel laboratory specimens

Very High Cycle Fatigue of Steel



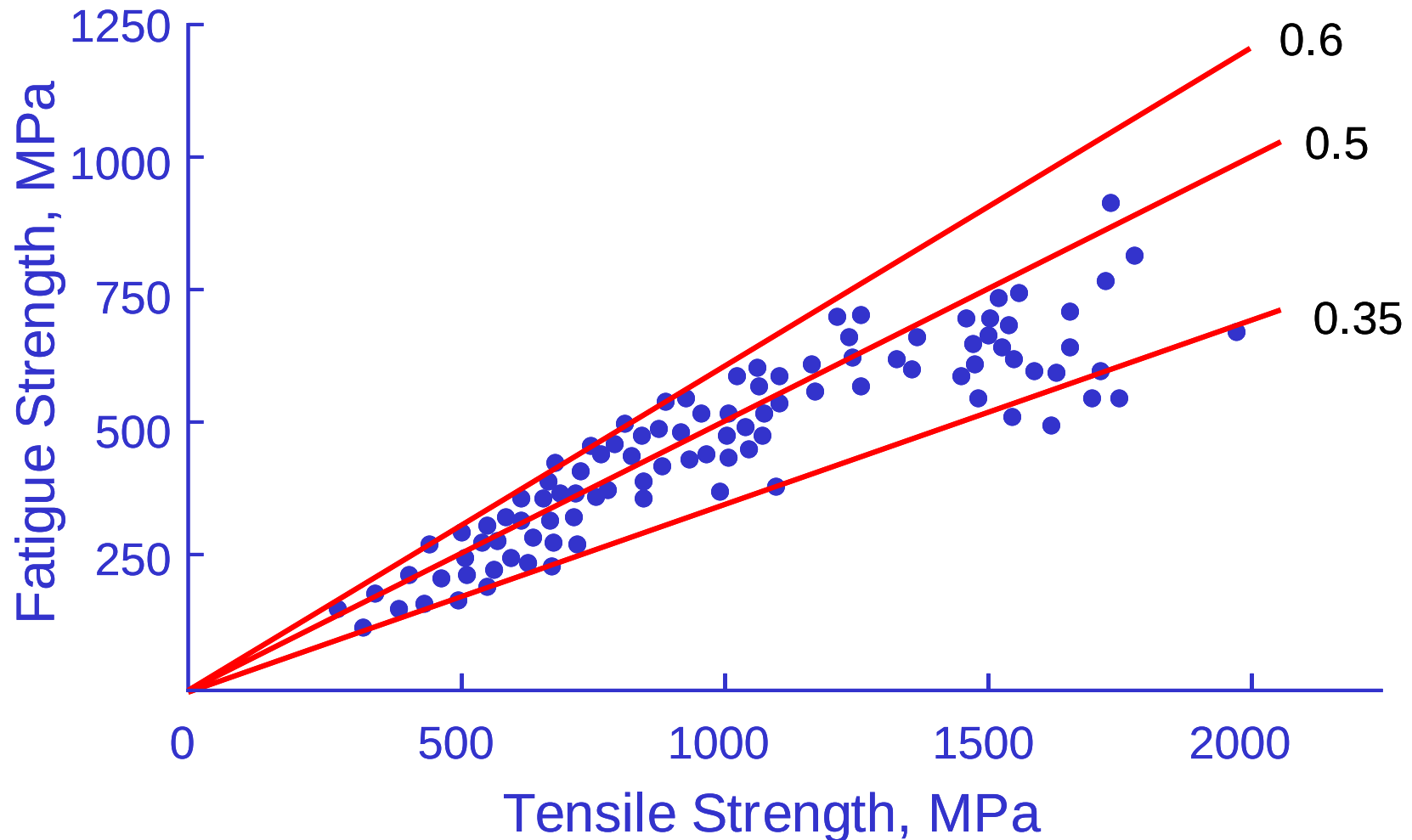
Fatigue Damage



$$N_f = \left(\frac{\Delta S}{2 S'_f} \right)^{\frac{1}{b}}$$

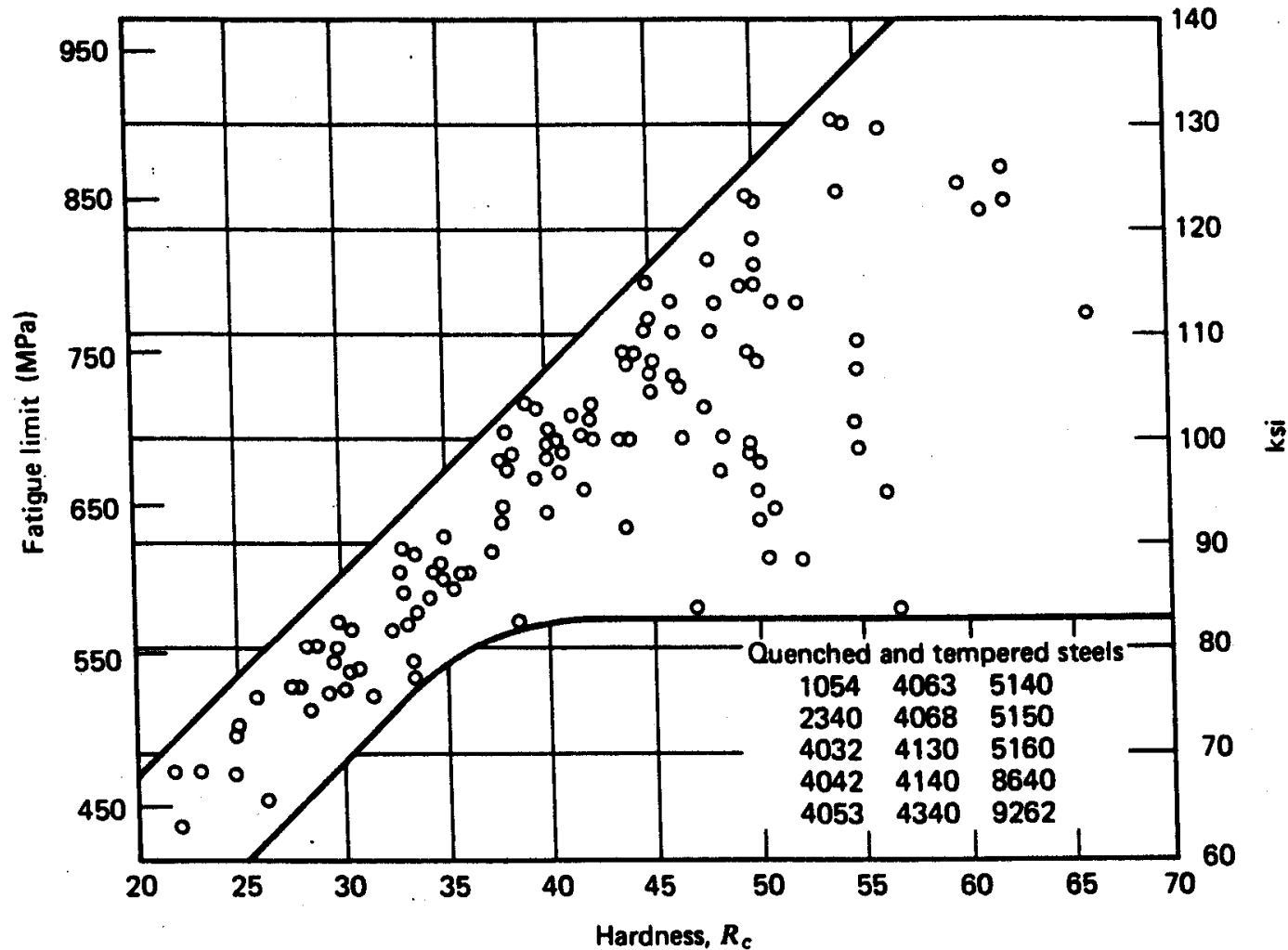
$$\text{Damage} \propto \Delta S^{10}$$

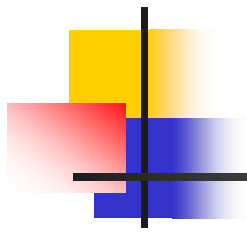
Fatigue Limit Strength Correlation



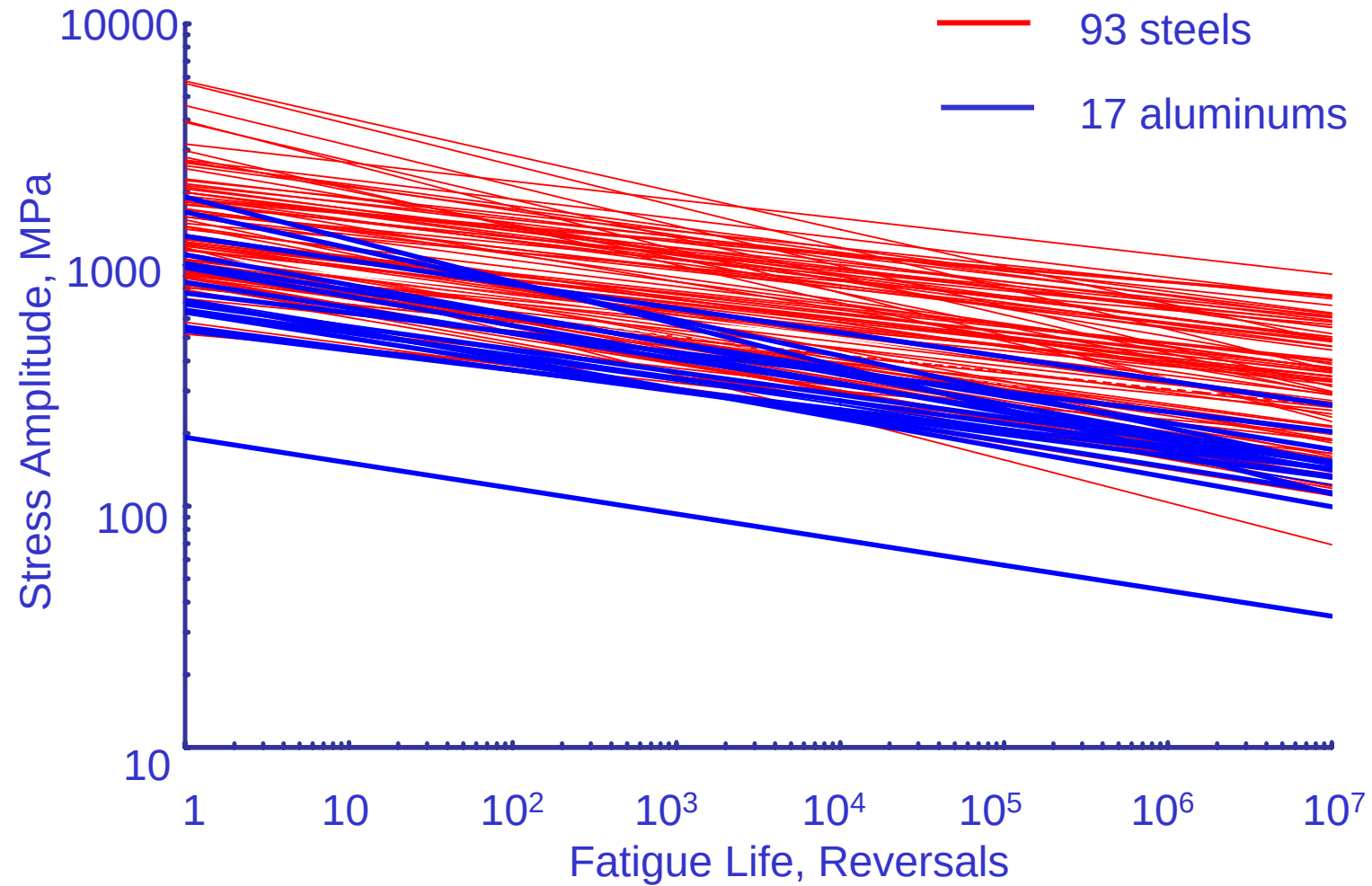
From Forrest, *Fatigue of Metals*, Pergamon Press, London, 1962

Fatigue Limit Strength Correlation

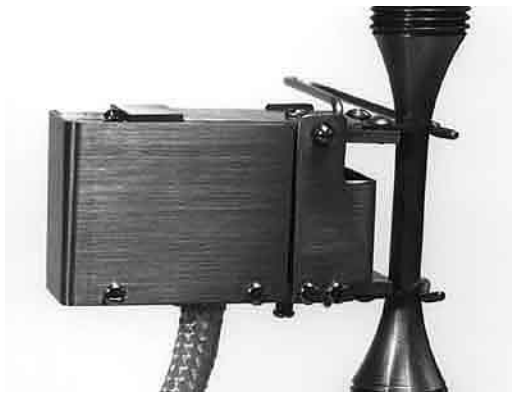




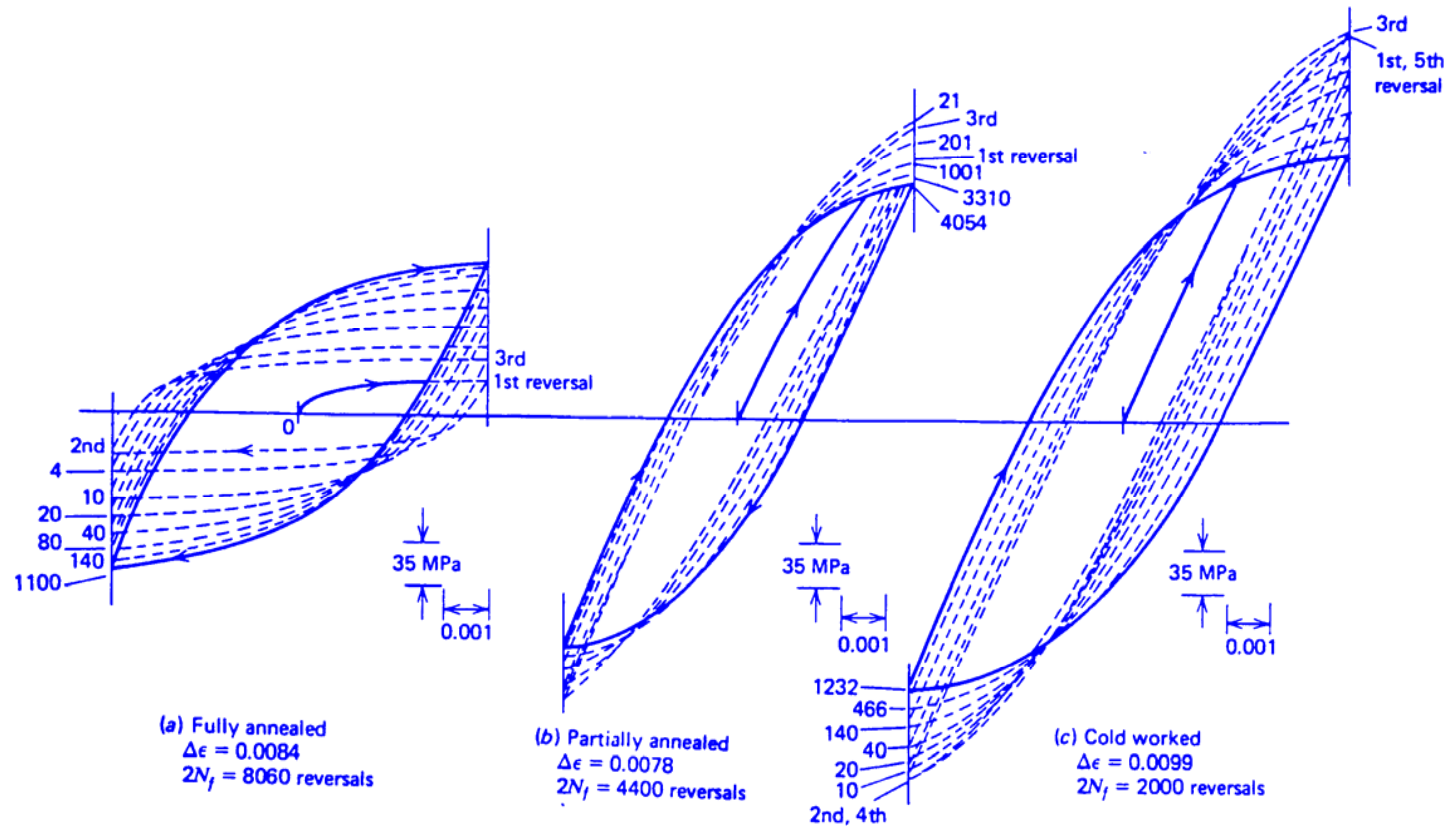
SN Materials Data



Strain Controlled Testing

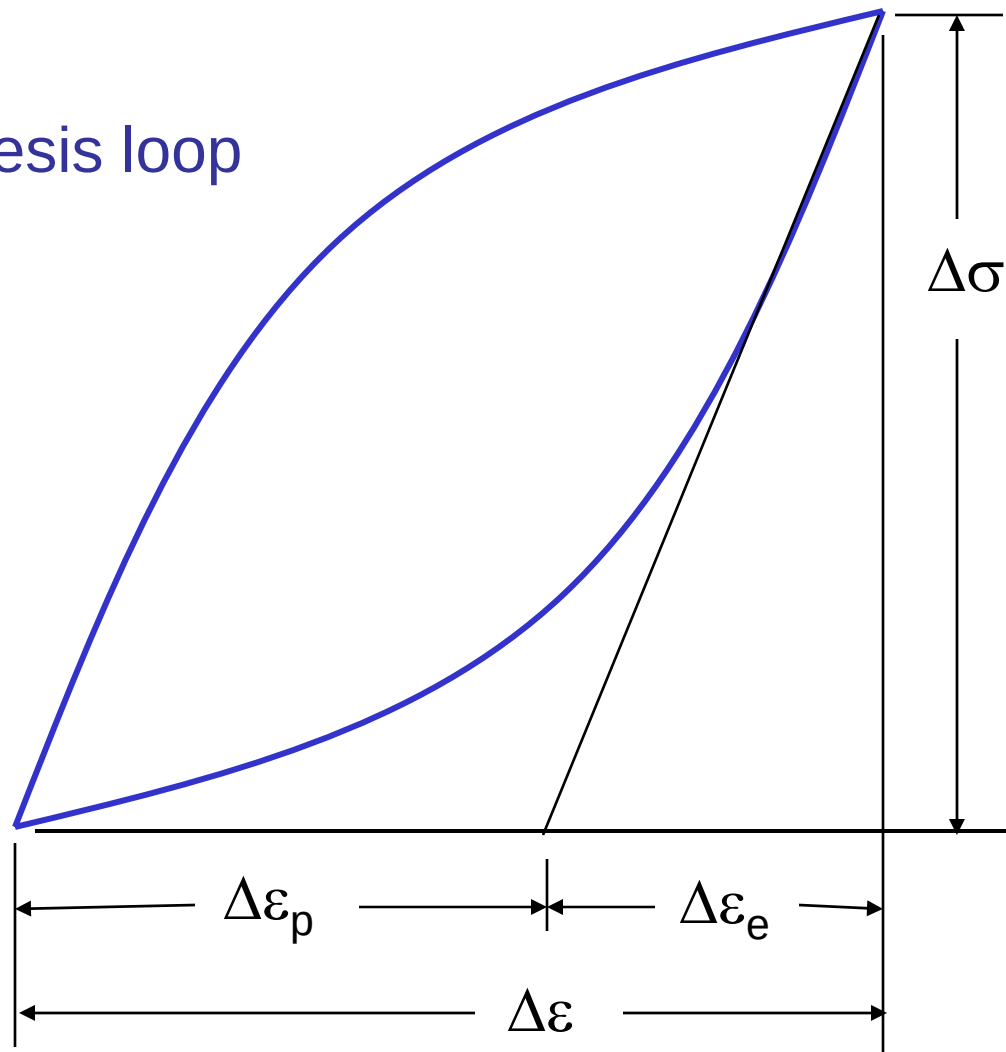


Cyclic Hardening / Softening

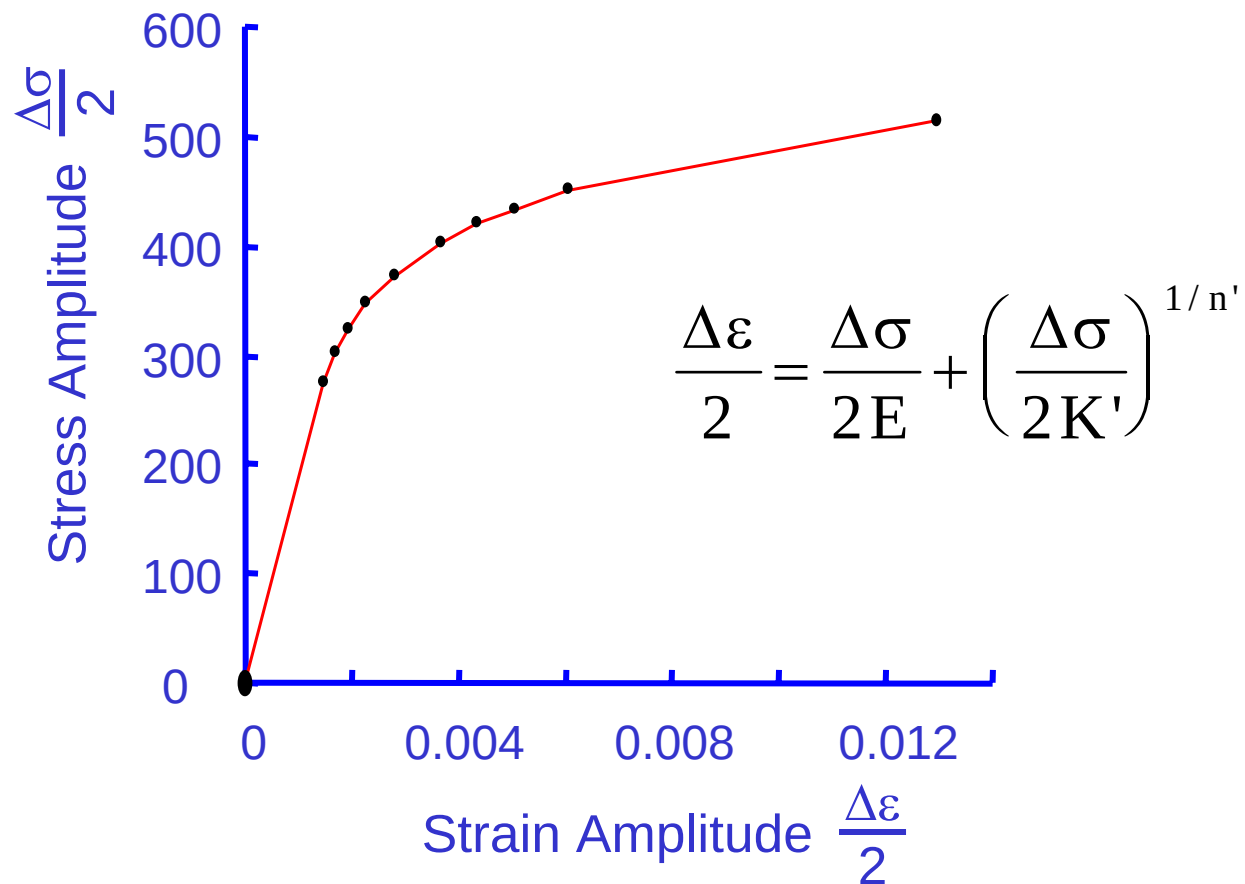


Stable Hysteresis Loop

Hysteresis loop

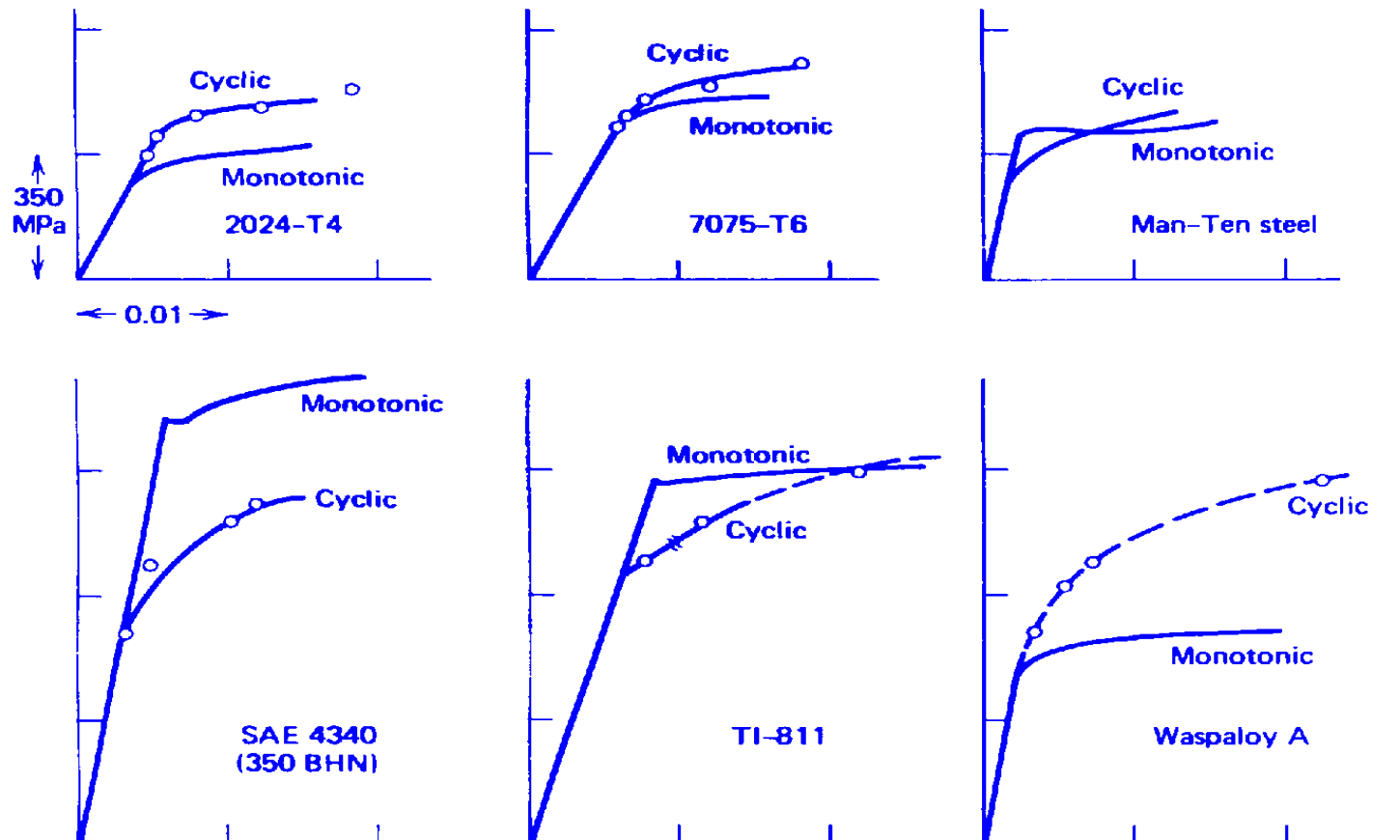


Strain-Life Data $\sigma - \varepsilon$

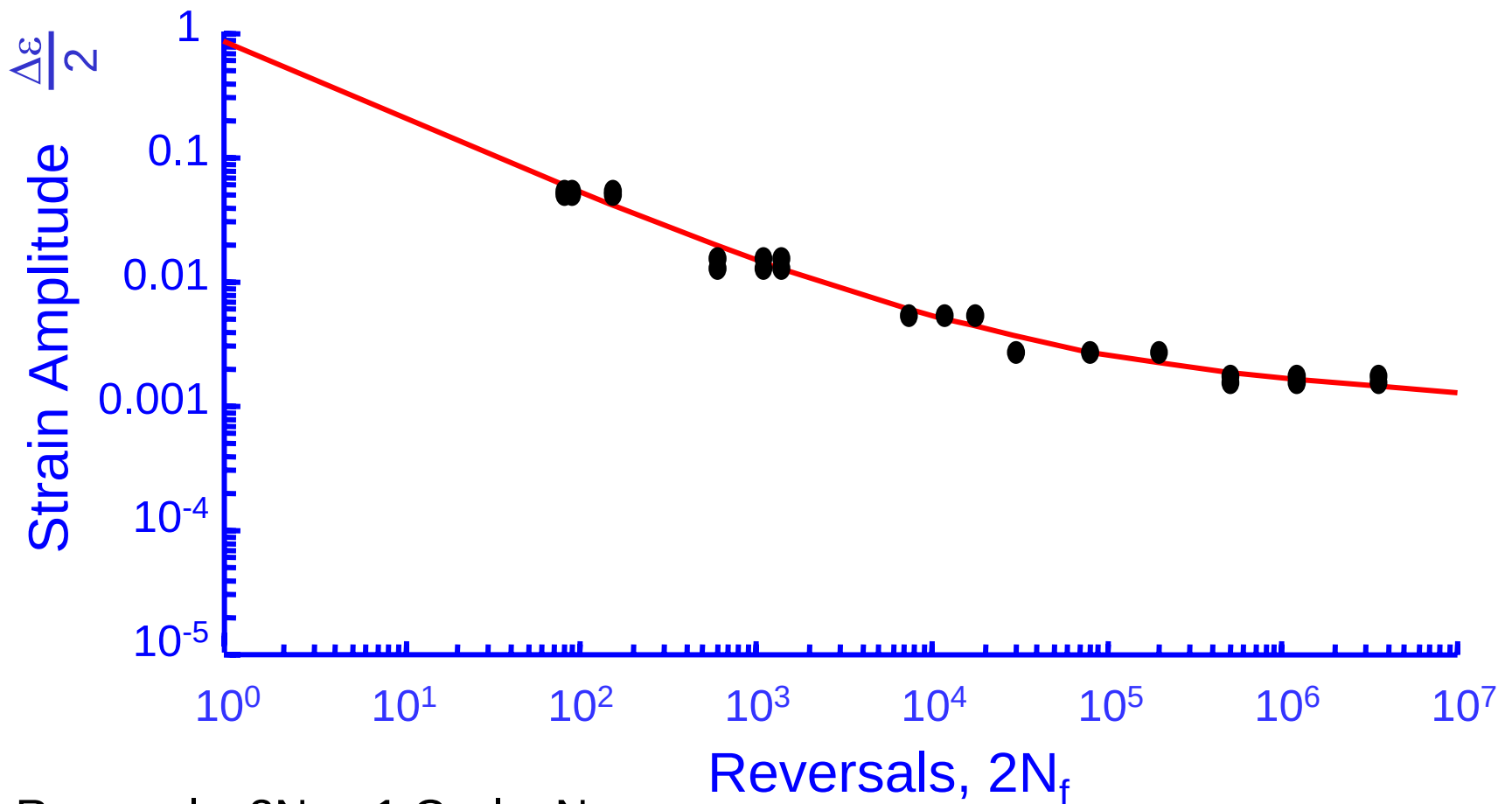


During cyclic deformation, the material deforms on a path described by the cyclic stress strain curve

Cyclic Stress Strain Curve

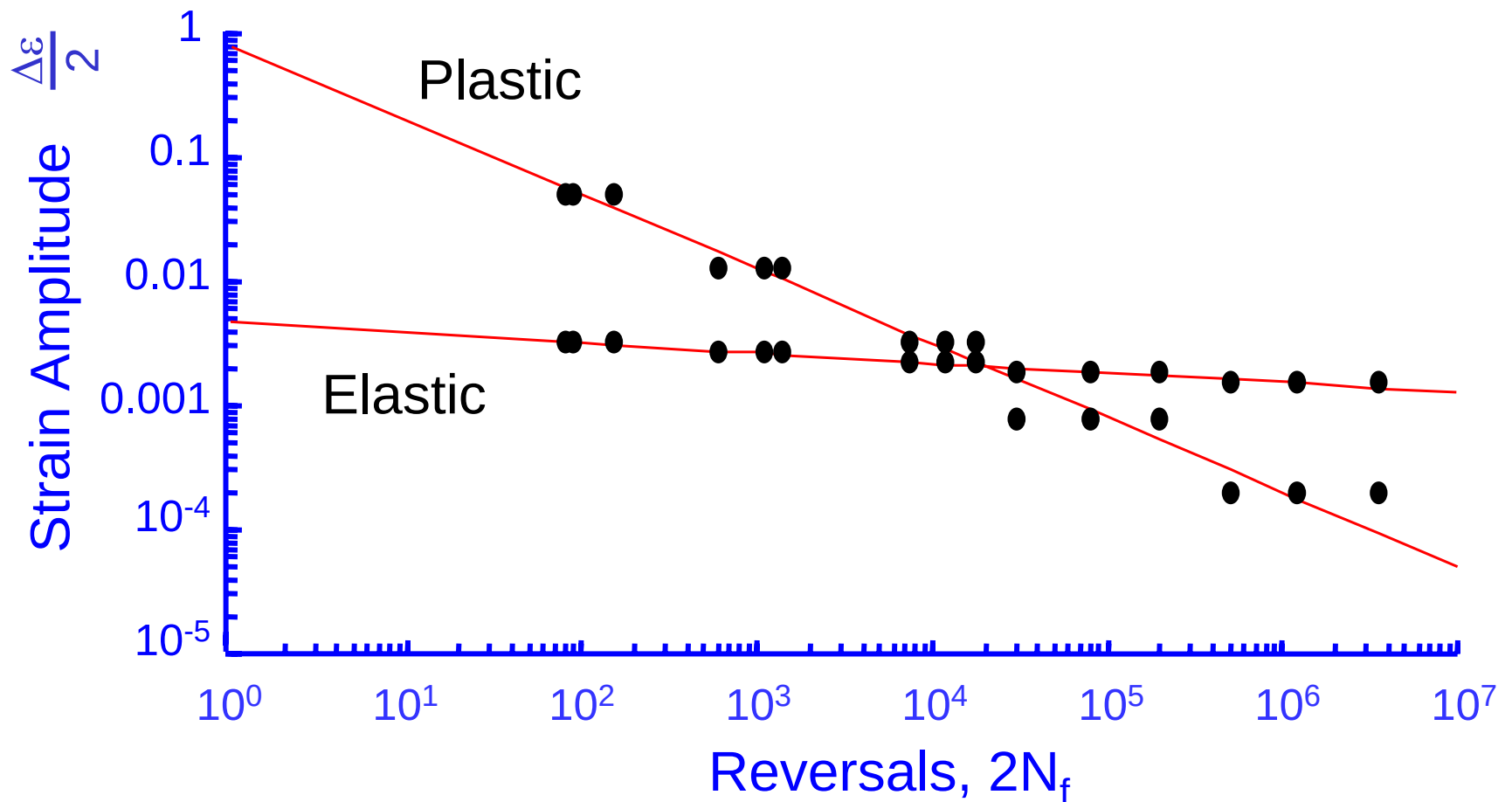


Strain-Life Data $\Delta\varepsilon - 2N_f$

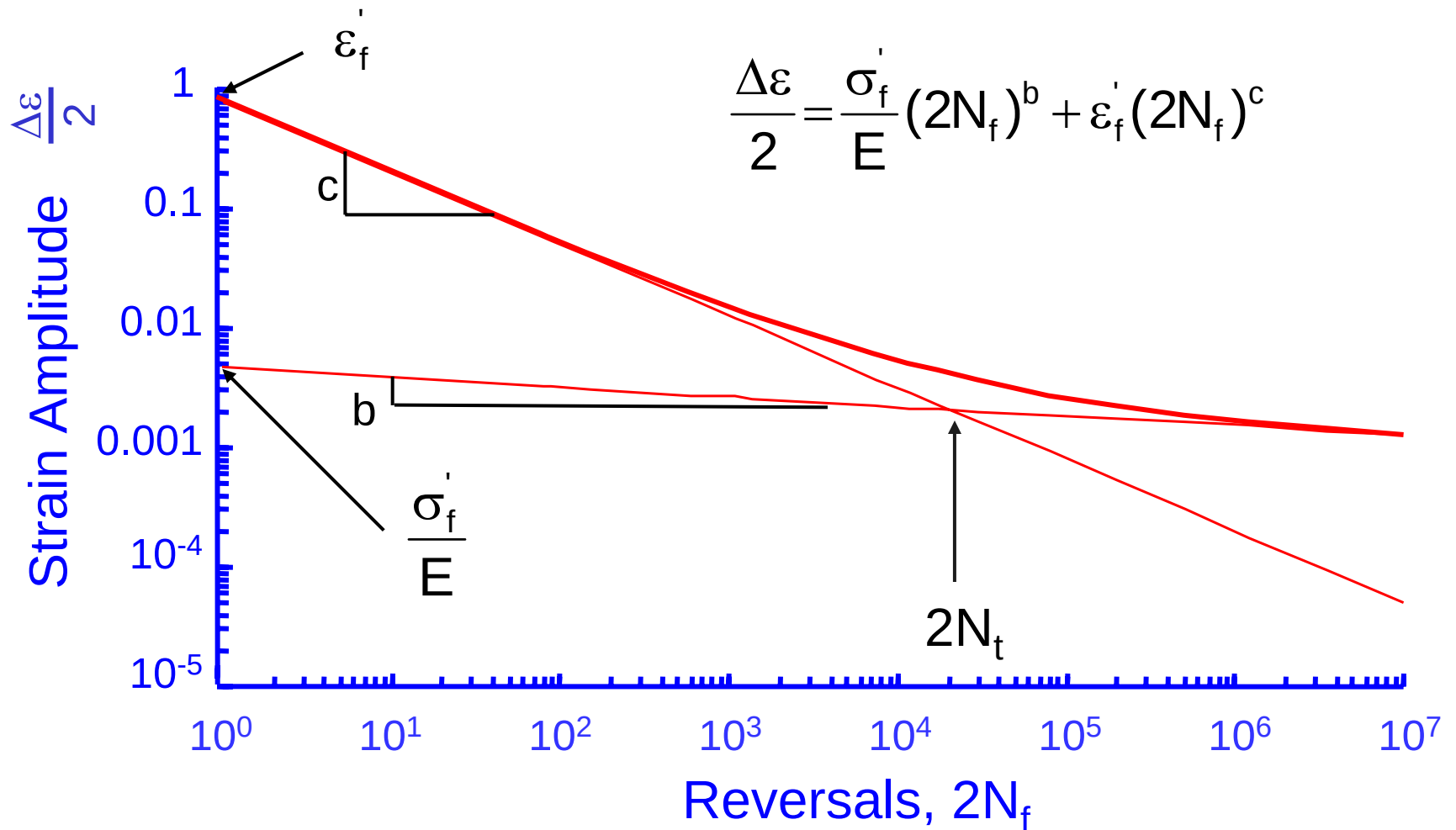


2 Reversals, $2N_f = 1$ Cycle, N_f

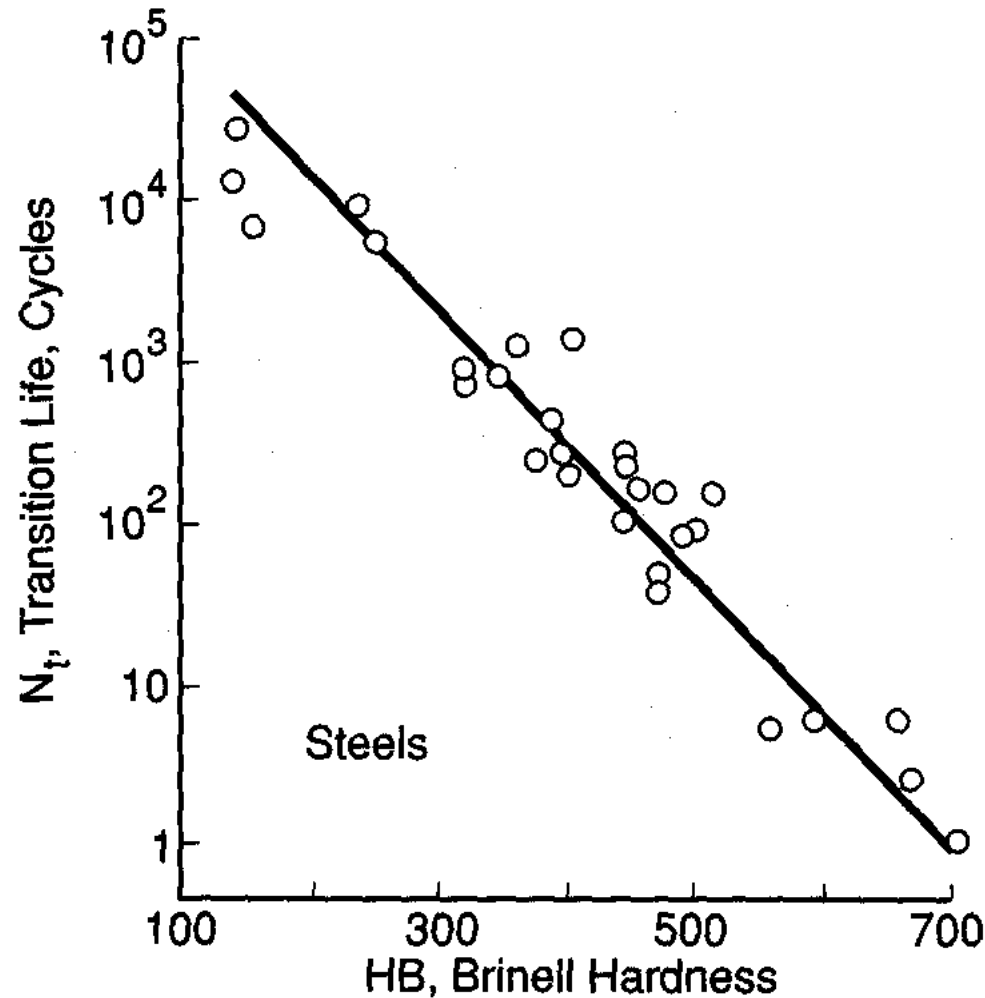
Elastic and Plastic Strain-Life Data



Strain-Life Curve

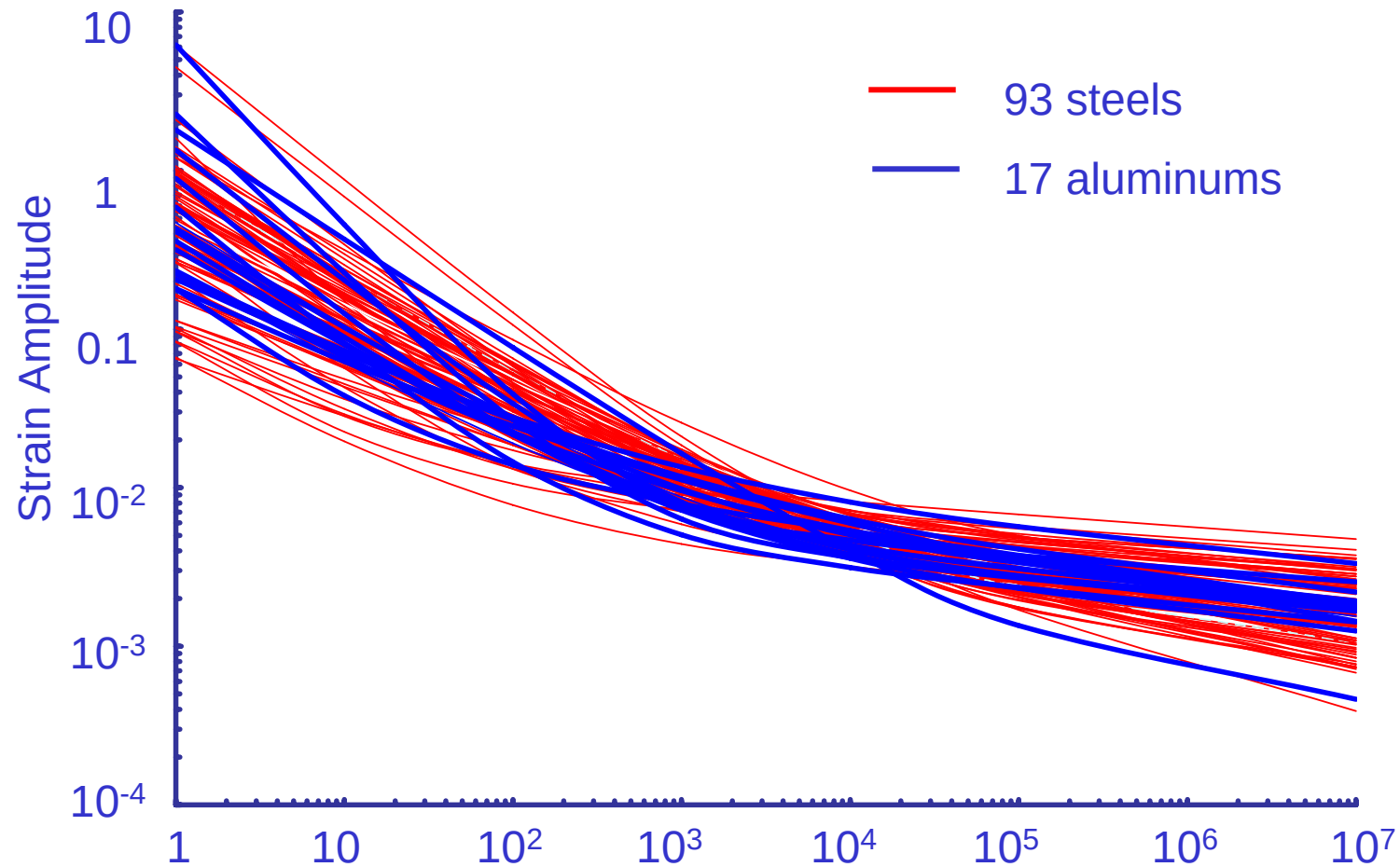


Transition Fatigue Life



From Dowling, Mechanical Behavior of Materials, 1999

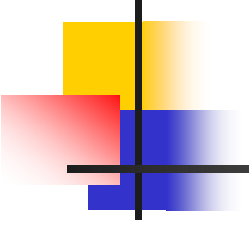
ϵN Materials Data



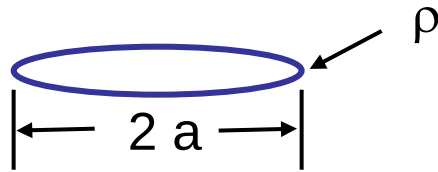
Fatigue Life, Reversals

Crack Growth Testing





Stress Concentration of a Crack



$$K_T = 1 + 2\sqrt{\frac{a}{\rho}}$$

$$K_T \sim 2000$$

for a crack

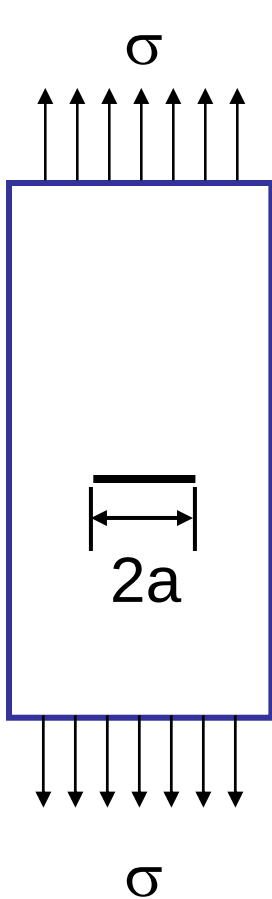
$$a \sim 10^{-3}$$

$$\sigma_{\text{local}} = 2000 \sigma_{\text{applied}}$$

$$\rho \sim 10^{-9}$$

Traditional material properties like tensile strength are not very useful for cracked structures

Stress Intensity Factor

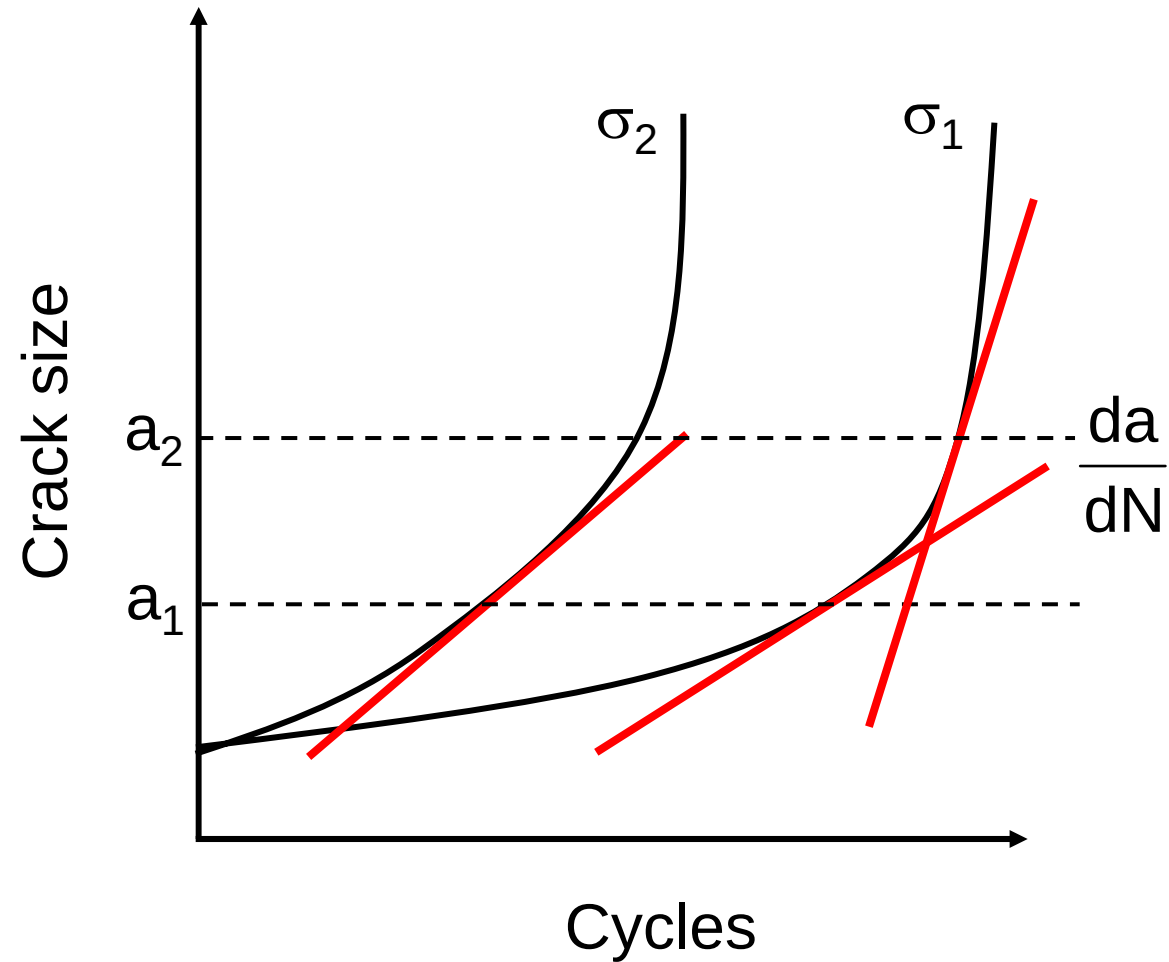
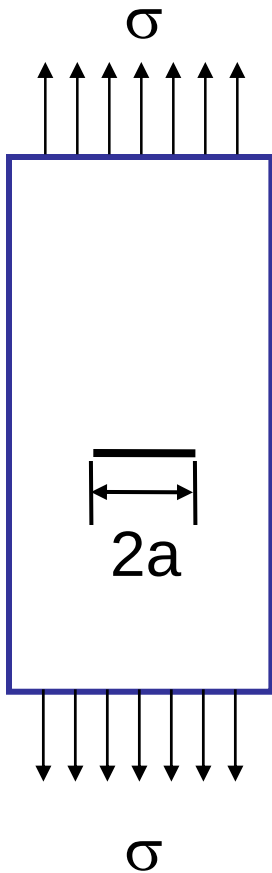


$$K = \sigma \sqrt{\pi a}$$

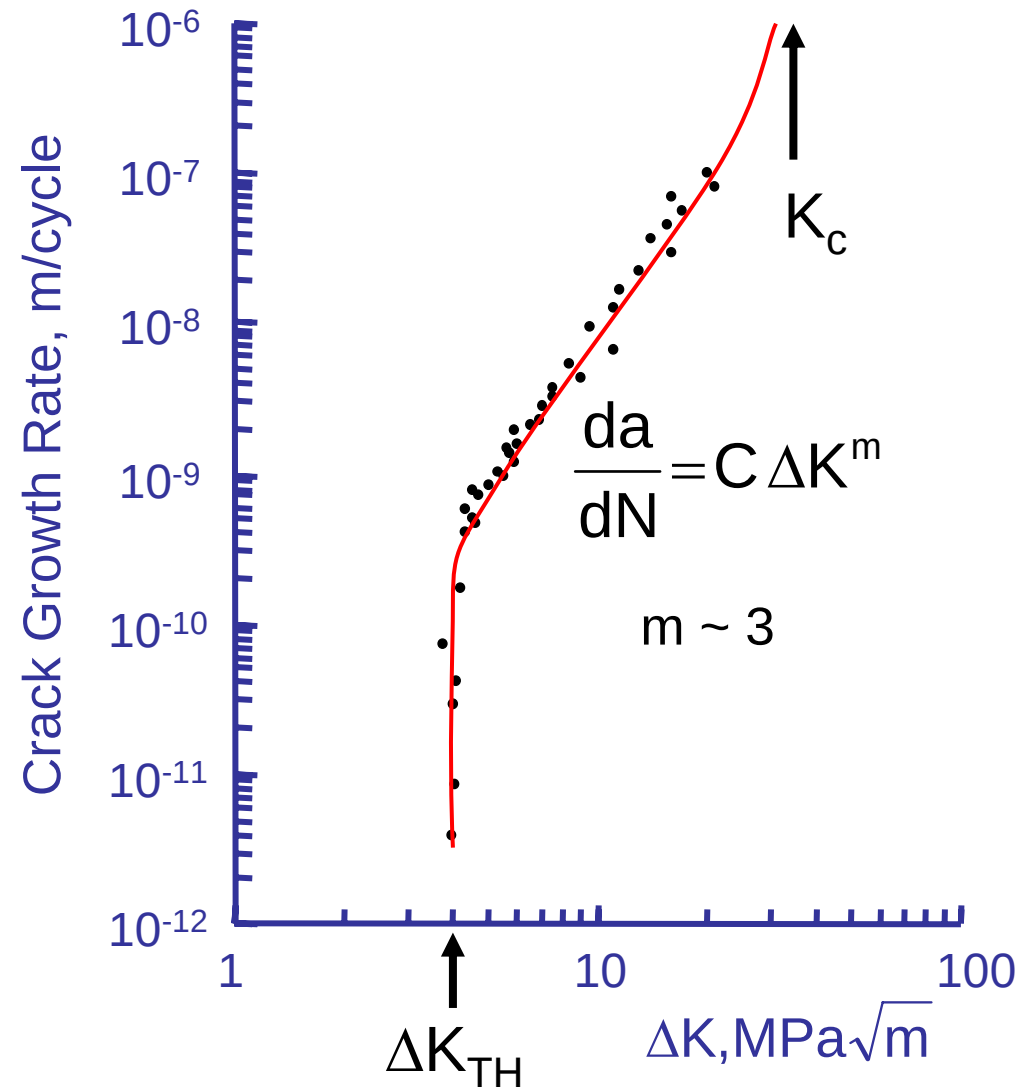
K characterizes the magnitude of the stresses, strains, and displacements in the neighborhood of a crack tip

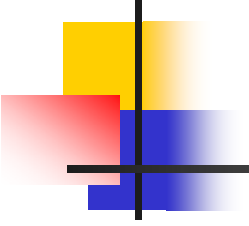
Two cracks with the same K will have the same behavior

Crack Growth Measurements



Crack Growth Data





Threshold Region

$$\Delta K_{TH} > \Delta \sigma \sqrt{\pi a} f\left(\frac{a}{w}\right)$$

threshold stress intensity

↑

operating stresses

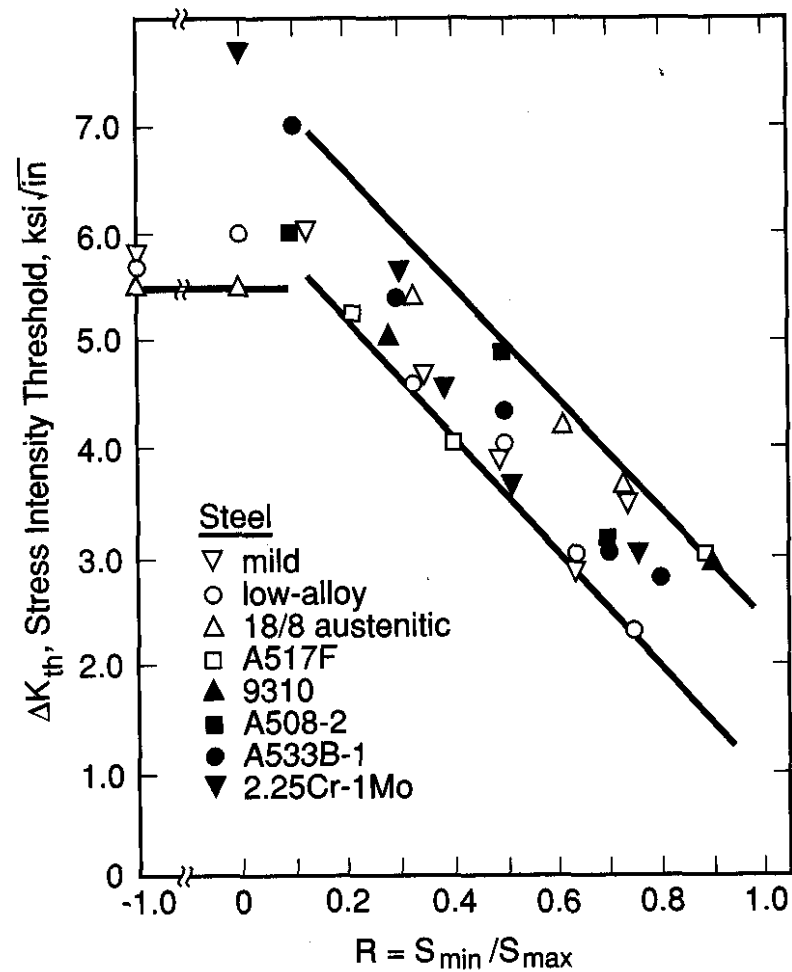
↑

flaw size

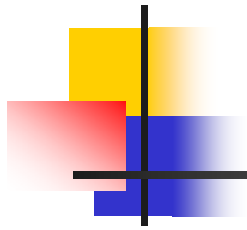
↑

flaw shape

Threshold Stress Intensity



From Dowling, Mechanical Behavior of Materials, 1999



Non-propagating Crack Sizes

Small cracks are frequently semielliptical surface cracks

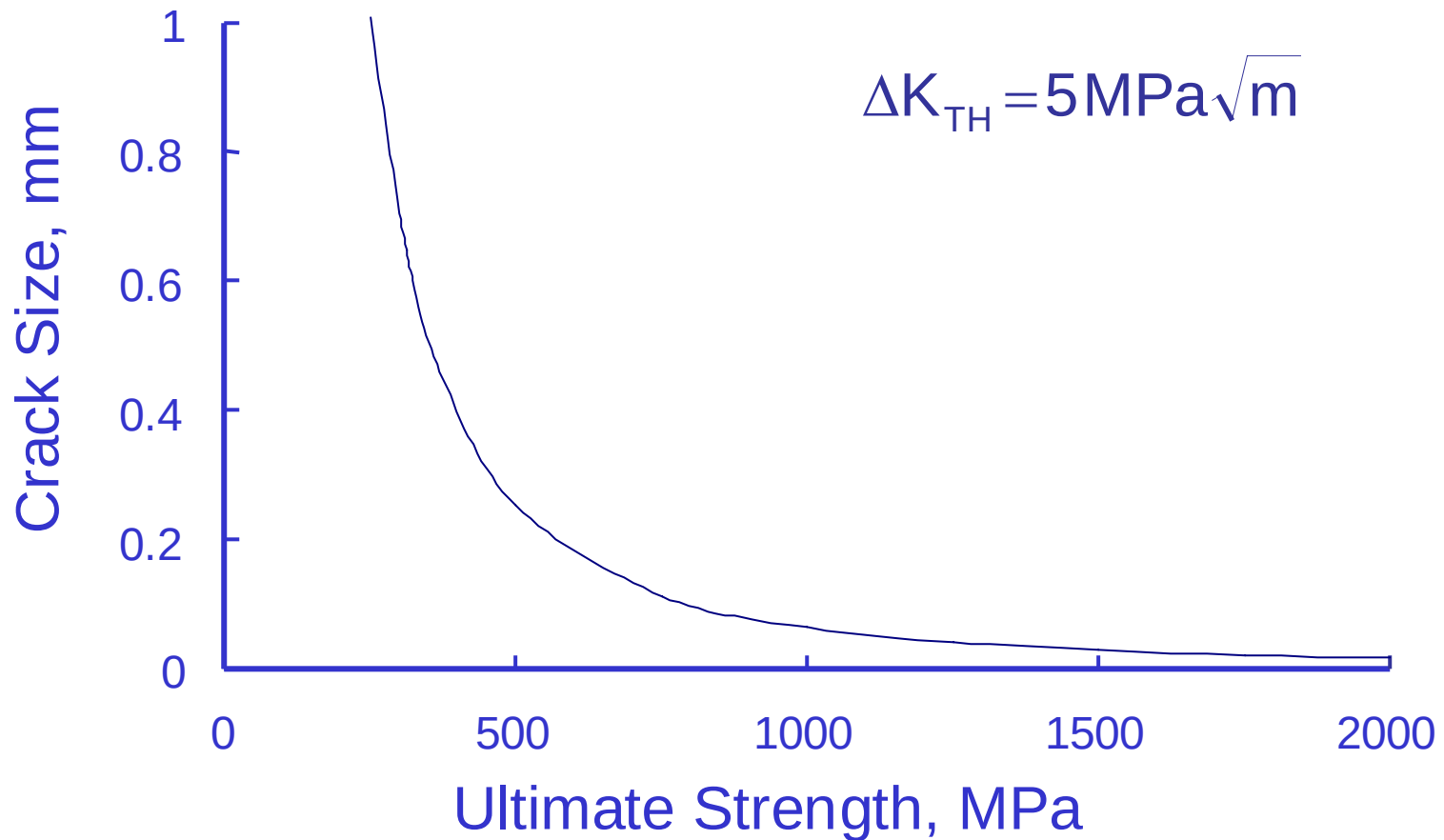
$$\Delta K_{TH} > \Delta \sigma 1.12 \frac{2}{\pi} \sqrt{\pi a}$$

$$a_c = 0.63 \left(\frac{\Delta K_{TH}}{\Delta \sigma} \right)^2$$

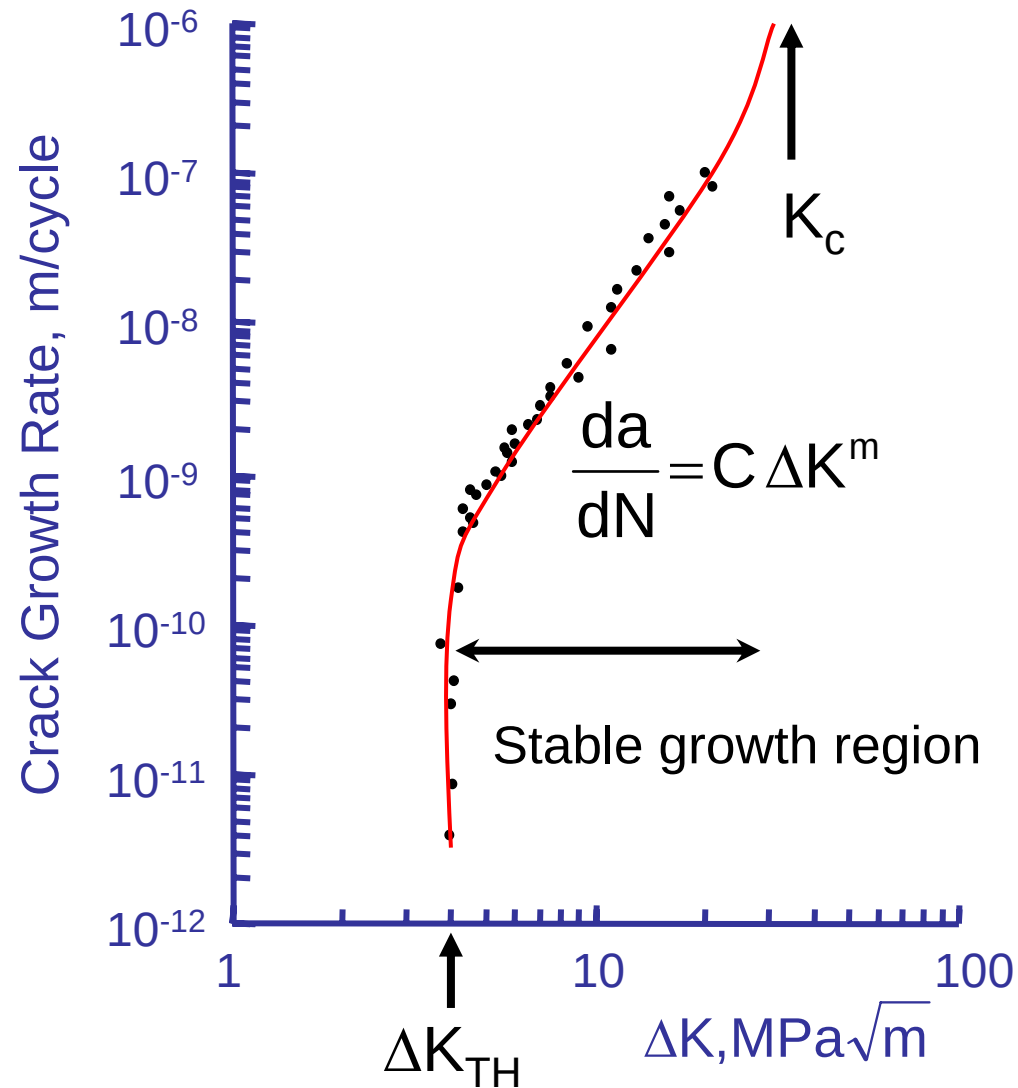
Smooth specimen fatigue limit $\approx \frac{\sigma_u}{2}$

$$a_c = 2.52 \left(\frac{\Delta K_{TH}}{\sigma_u} \right)^2$$

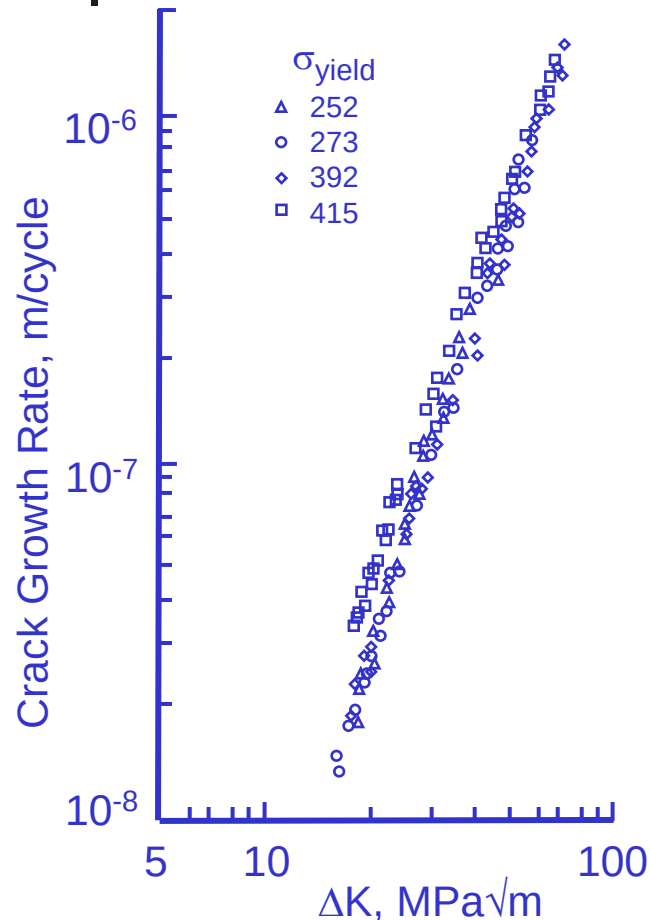
Non-propagating Crack Sizes



Stable Crack Growth



Crack Growth Data



Ferritic-Pearlitic Steel:

$$\frac{da}{dN} = 6.9 \times 10^{-12} (\Delta K \text{ MPa}\sqrt{\text{m}})^{3.0}$$

Martensitic Steel:

$$\frac{da}{dN} = 1.4 \times 10^{-10} (\Delta K \text{ MPa}\sqrt{\text{m}})^{2.25}$$

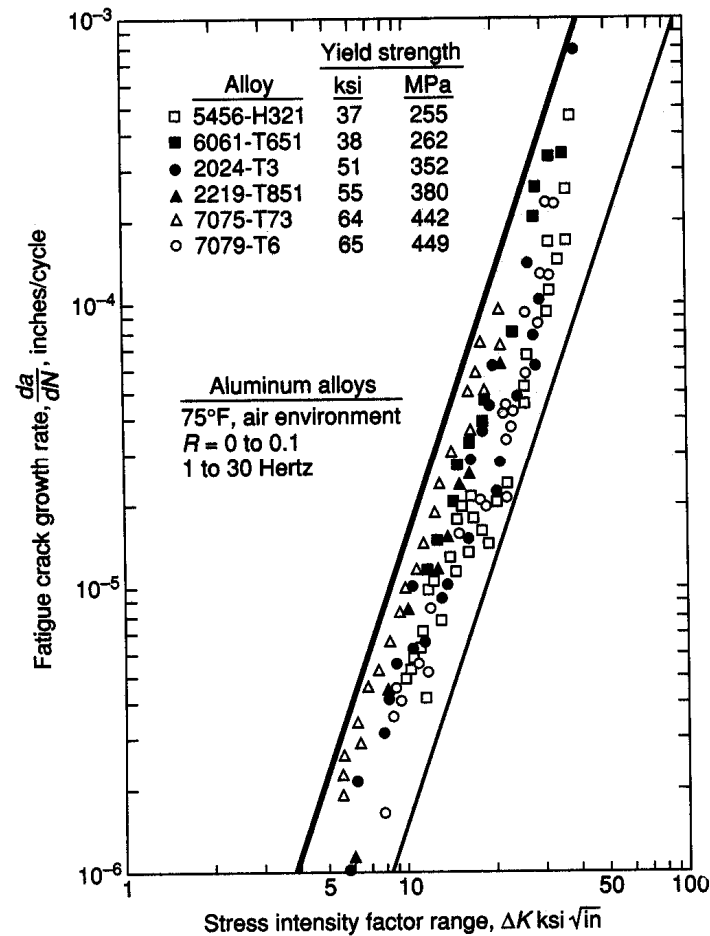
Austenitic Stainless Steel:

$$\frac{da}{dN} = 5.6 \times 10^{-12} (\Delta K \text{ MPa}\sqrt{\text{m}})^{3.25}$$

Barsom, "Fatigue Crack Propagation in Steels of Various Yield Strengths"

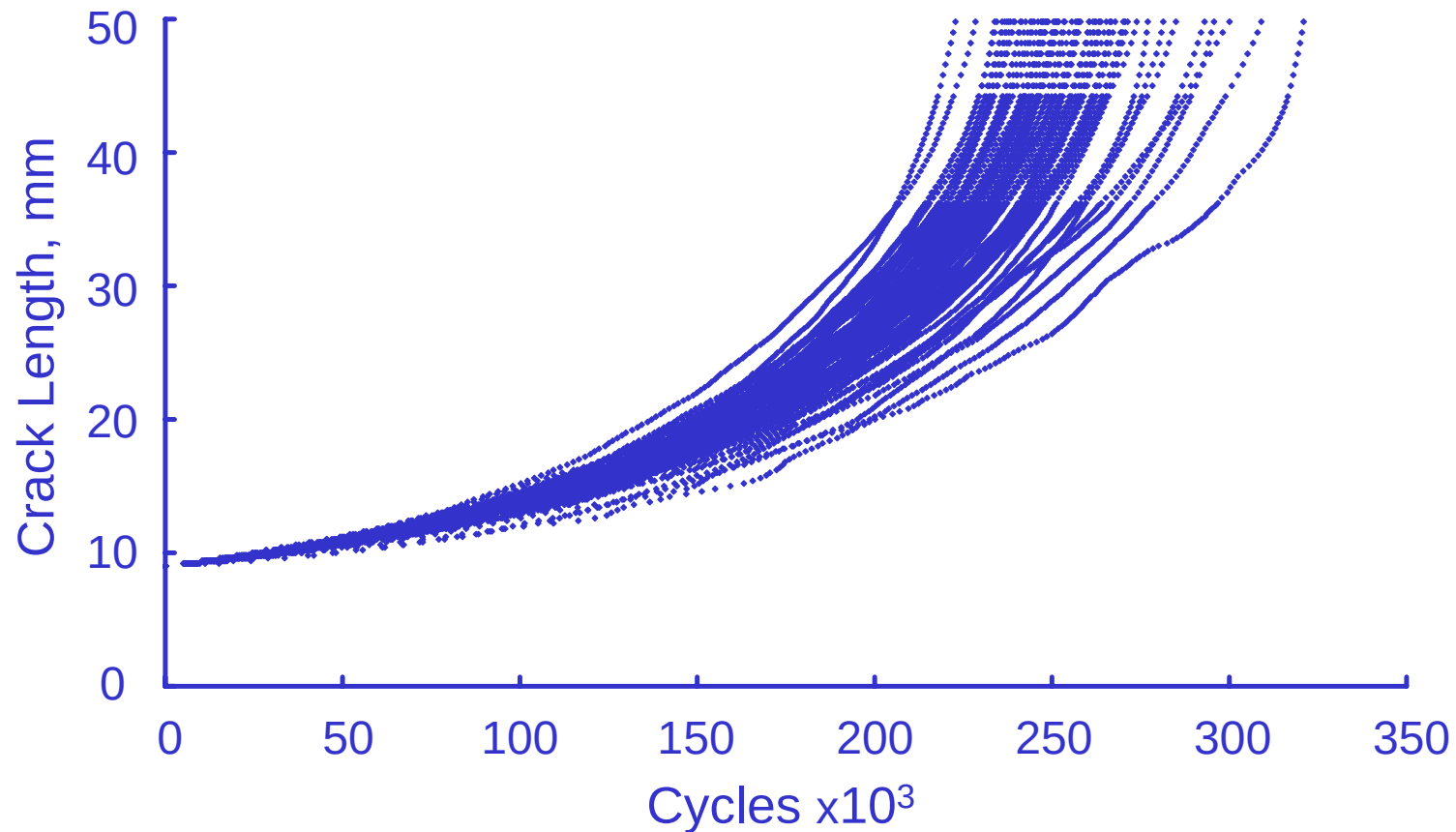
Journal of Engineering for Industry, Trans. ASME, Series B, Vol. 93, No. 4, 1971, 1190-1196

Aluminum Crack Growth Rate Data

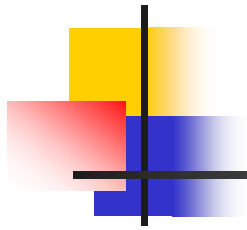


Sharp, Nordmark and Menzemer, *Fatigue Design of Aluminum Components and Structures*, 1996

Crack Growth Data

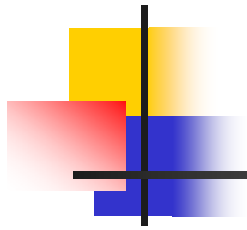


Virkler, Hillberry and Goel, "The Statistical Nature of Fatigue Crack Propagation", Journal of Engineering Materials and Technology, Vol. 101, 1979, 148-153



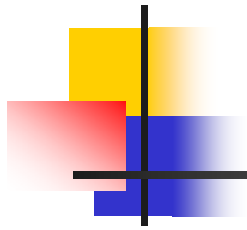
Things Worth Remembering

<u>Method</u>	<u>Physics</u>	<u>Size</u>
Stress-Life	Crack Nucleation	0.01 mm
Strain-Life	Microcrack Growth	0.1 - 1 mm
Crack Growth	Macrocrack Growth	> 1mm

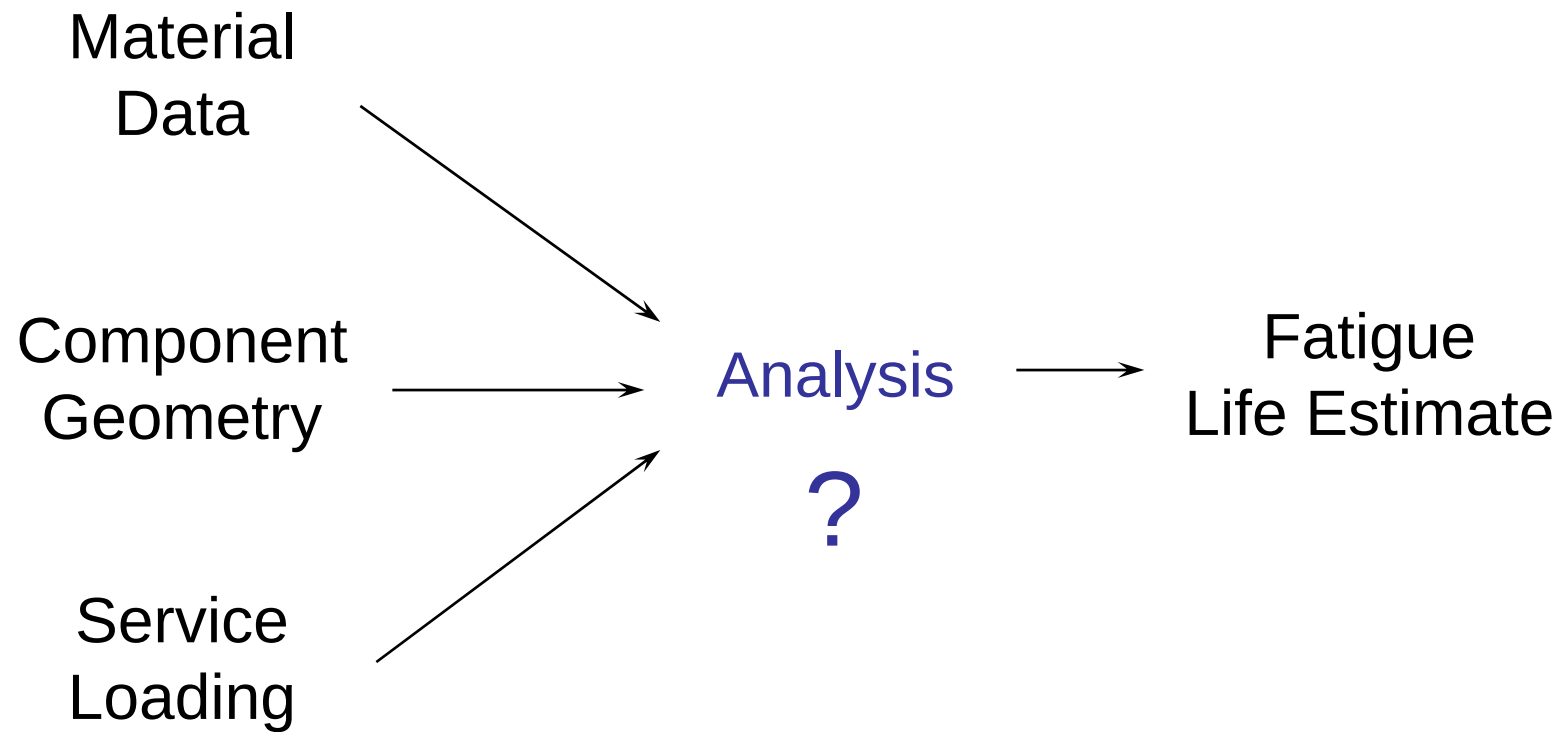


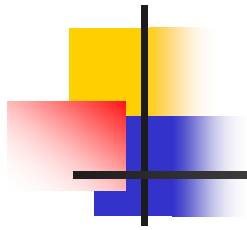
Fatigue, How and Why

- Physics of Fatigue
- Material Properties
- **Similitude**
- Fatigue Calculator



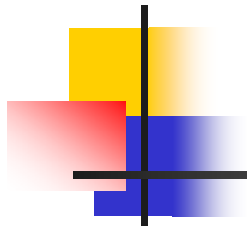
Fatigue Analysis





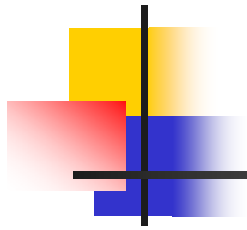
The Similitude Concept

Why Fatigue Modeling Works !



What is the Similitude Concept

The “Similitude Concept” allows engineers to relate the behavior of small-scale cyclic material test specimens, defined under carefully controlled conditions, to the likely performance of real structures subjected to variable amplitude fatigue loads under either simulated or actual service conditions.



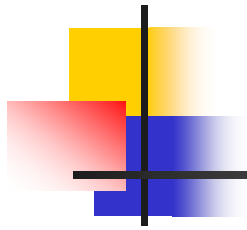
Fatigue Analysis Techniques

Stress - Life

BS 7608, Eurocode 3

Strain - Life

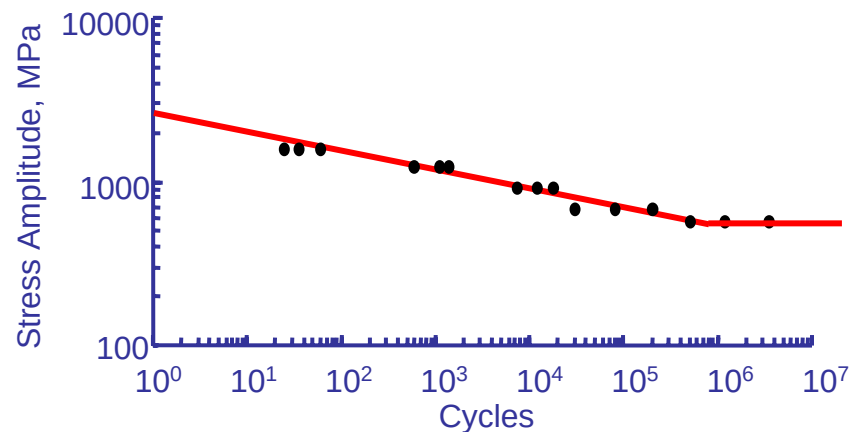
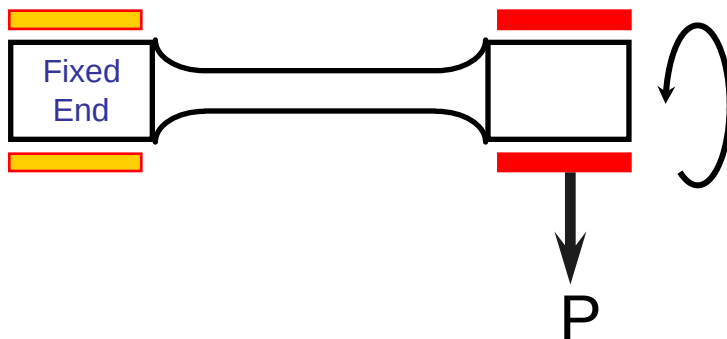
Crack Growth



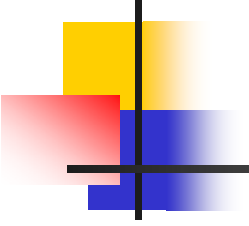
Life Estimation

<u>Method</u>	<u>Physics</u>	<u>Size</u>
Stress-Life	Crack Nucleation	0.01 mm
BS 7608	Crack Growth	1 - 10 mm
Strain-Life	Microcrack Growth	0.1 - 1 mm
Crack Growth	Macrocrack Growth	> 1mm

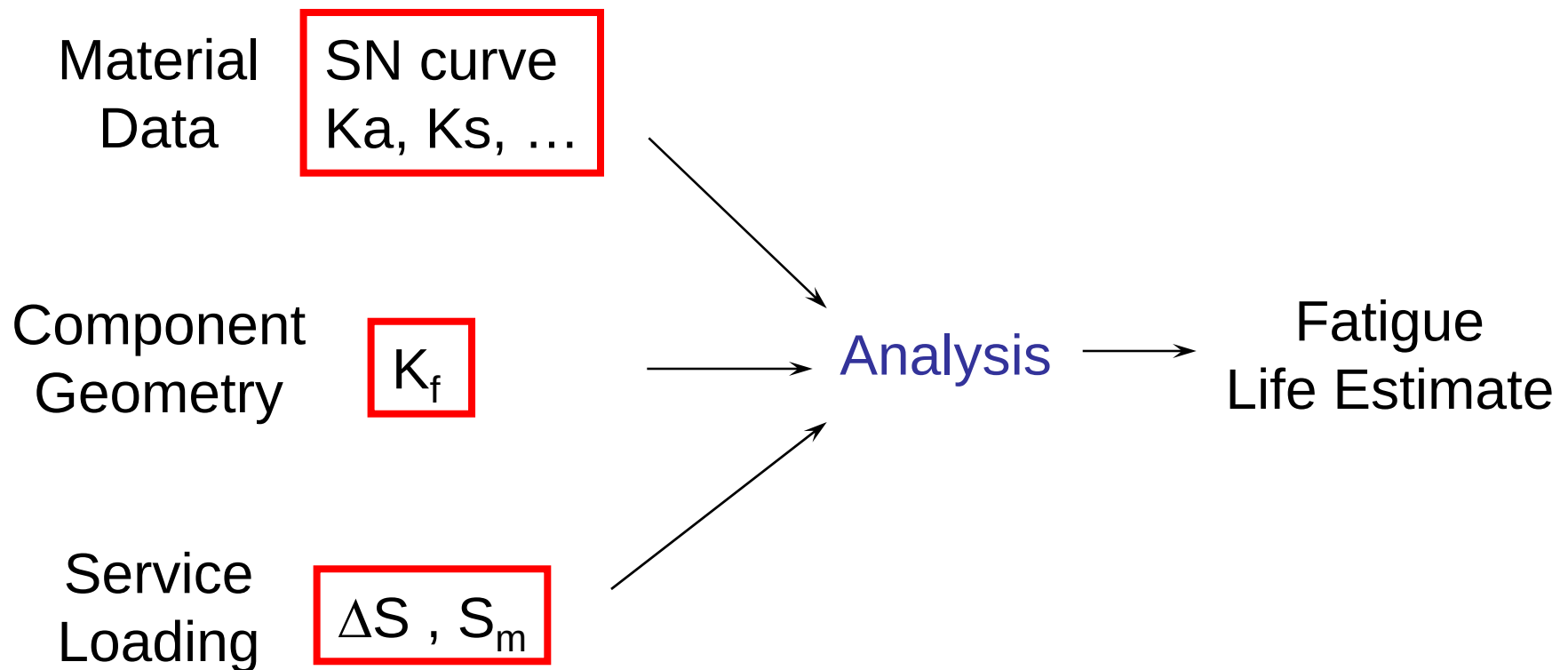
Stress-Life Fatigue Modeling

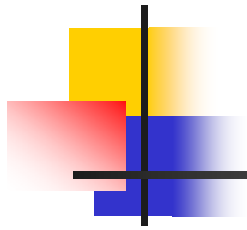


The Similitude Concept states that if the instantaneous loads applied to the 'test' structure (wing spar, say) and the test specimen are the same, then the response in each case will also be the same and can be described by the material's S-N curve.



Fatigue Analysis: Stress-Life

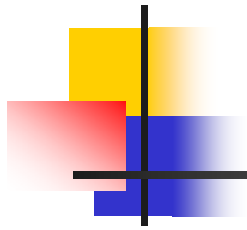




Stress-Life

■ Major Assumptions:

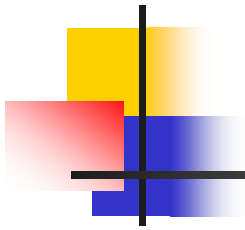
- Most of the life is consumed nucleating cracks
- Elastic deformation
- Nominal stresses and material strength control fatigue life
- Accurate determination of K_f for each geometry and material



Stress-Life

■ Advantages:

- Changes in material and geometry can easily be evaluated
- Large empirical database for steel with standard notch shapes

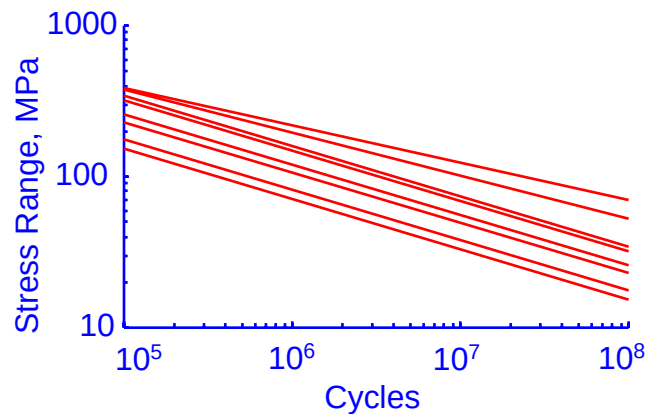
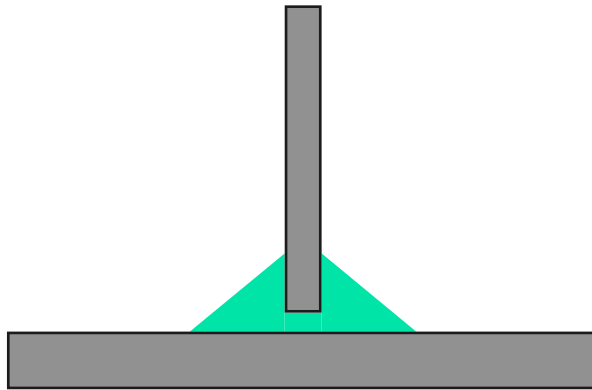


Stress-Life

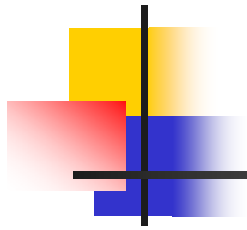
■ Limitations:

- Does not account for notch root plasticity
- Mean stress effects are often in error
- Requires empirical K_f for good results

BS 7608 Fatigue Modeling

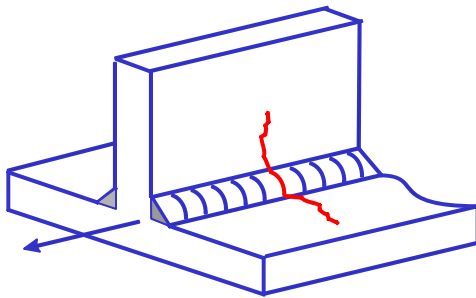


The Similitude Concept states that if the instantaneous loads applied to the 'test' structure (welded beam on a bulldozer, say) and the test specimen (standard fillet weld) are the same, then the response in each case will also be the same and can be described by one of the standard BS 7608 Weld Classification S-N curves.

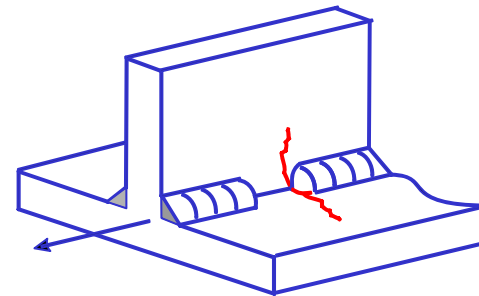


Weld Classifications

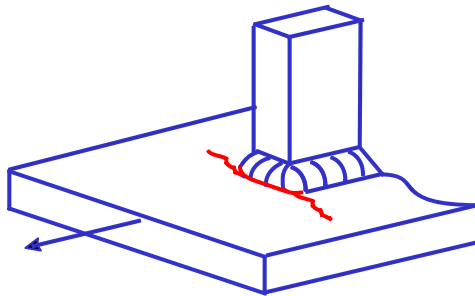
D



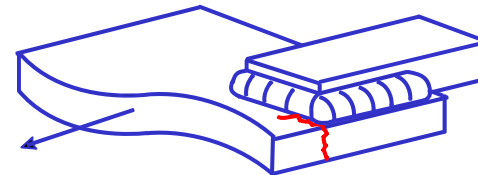
E

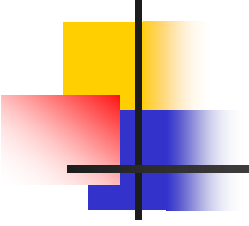


F2

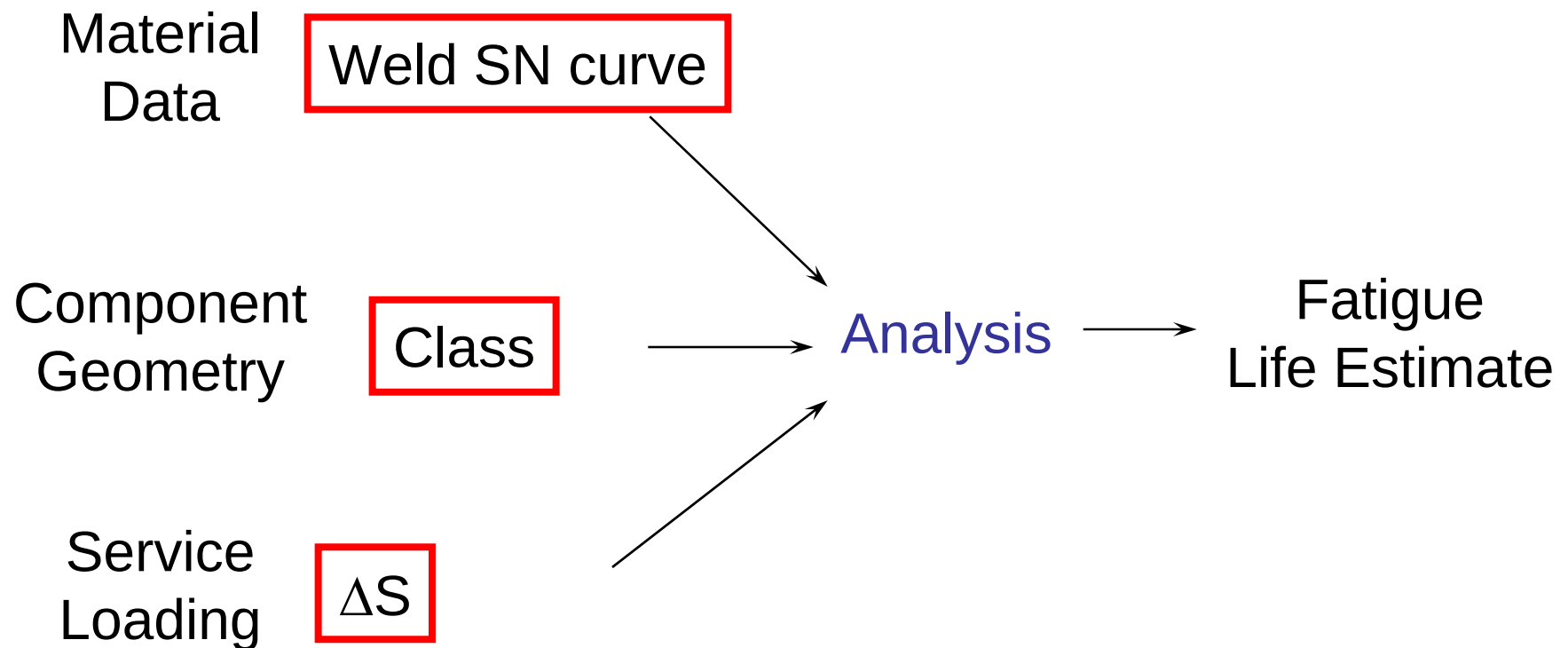


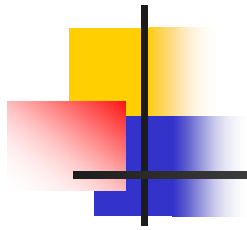
G





Fatigue Analysis: BS 7608

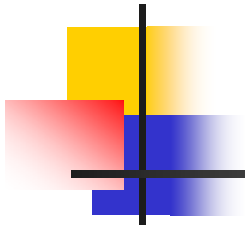




BS 7608

■ Major Assumptions:

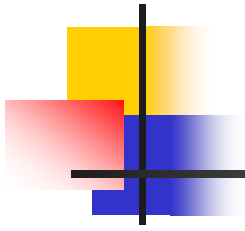
- Crack growth dominates fatigue life
- Complex weld geometries can be described by a standard classification
- Results independent of material and mean stress for structural steels



BS 7608

- Advantages:

- Manufacturing effects are directly included
- Large empirical database exists

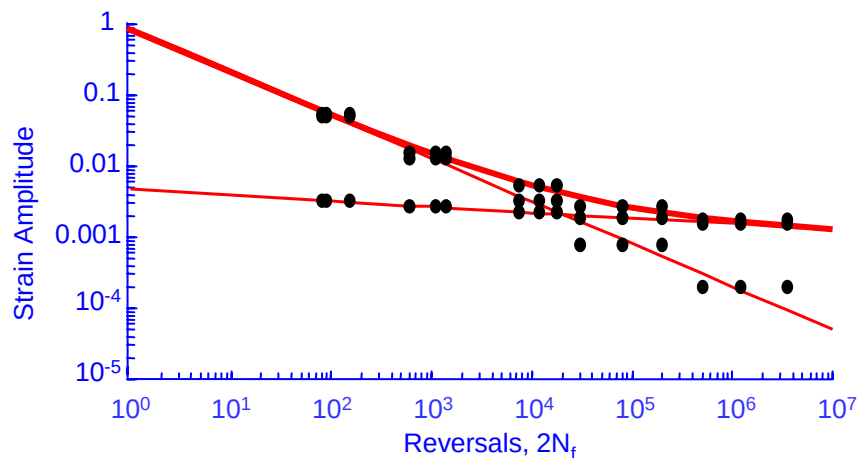
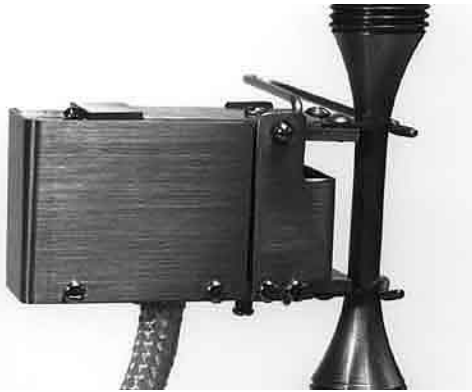


BS 7608

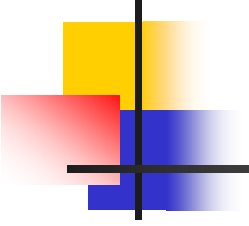
■ Limitations:

- Difficult to determine weld class for complex shapes
- No benefit for improving manufacturing process

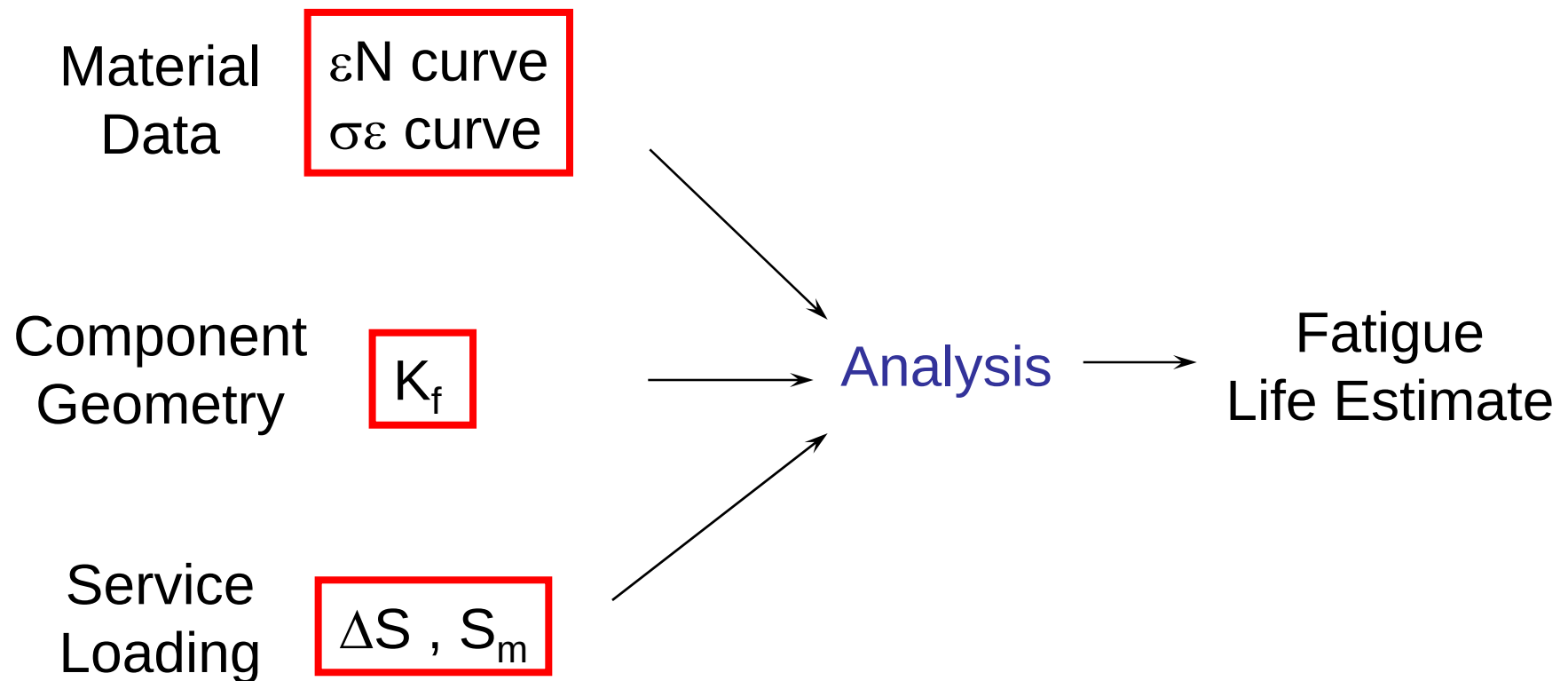
Strain-Life Fatigue Modeling

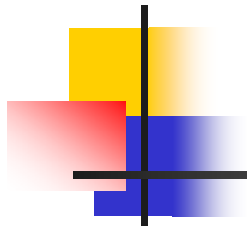


The Similitude Concept states that if the instantaneous strains applied to the 'test' structure (vehicle suspension, say) and the test specimen are the same, then the response in each case will also be the same and can be described by the material's e-N curve. Due account can also be made for stress concentrations, variable amplitude loading etc.



Fatigue Analysis: Strain-Life

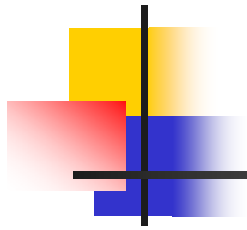




Strain-Life

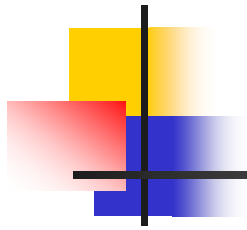
■ Major Assumptions:

- Local stresses and strains control fatigue behavior
- Plasticity around stress concentrations
- Accurate determination of K_f



Strain-Life

- Advantages:
 - Plasticity effects
 - Mean stress effects

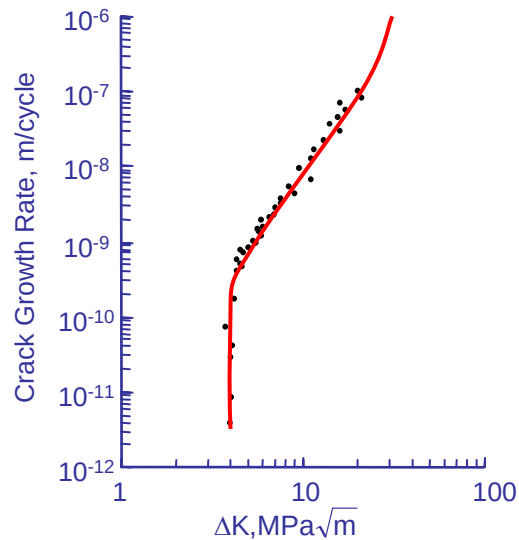
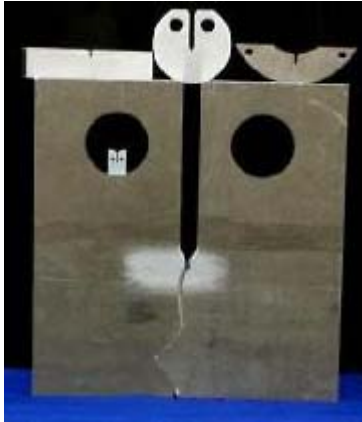


Strain-Life

■ Limitations:

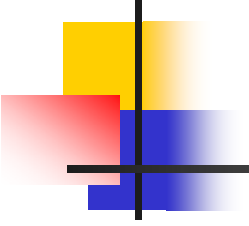
- Requires empirical K_f
- Long life situations where surface finish and processing variables are important

Crack Growth Fatigue Modeling

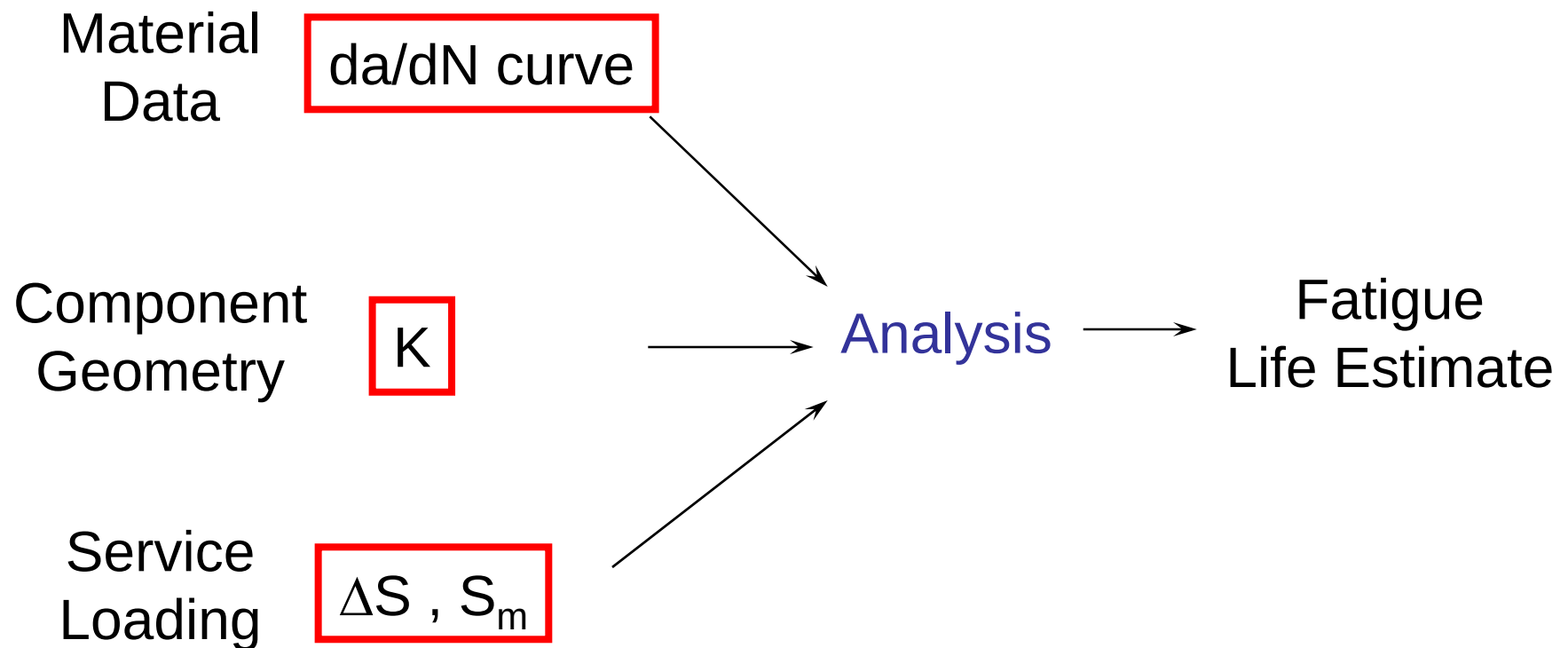


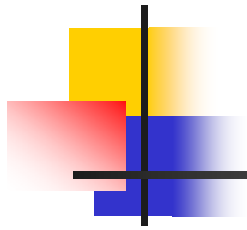
The Similitude Concept states that if the stress intensity (K) at the tip of a crack in the 'test' structure (welded connection on an oil platform leg, say) and the test specimen are the same, then the crack growth response in each case will also be the same and can be described by the Paris relationship. Account can also be made for local chemical environment, if necessary.





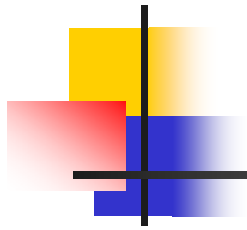
Fatigue Analysis: Crack Growth





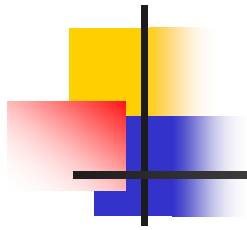
Crack Growth

- Major Assumptions:
 - Nominal stress and crack size control fatigue life
 - Accurate determination of initial crack size



Crack Growth

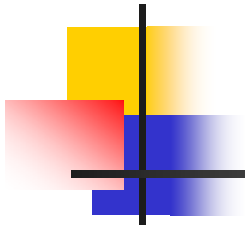
- Advantage:
 - Only method to directly deal with cracks



Crack Growth

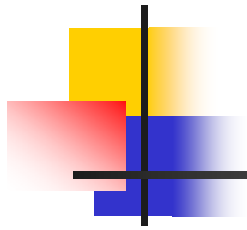
- Limitations:

- Complex sequence effects
- Accurate determination of initial crack size



Choose the Right Model

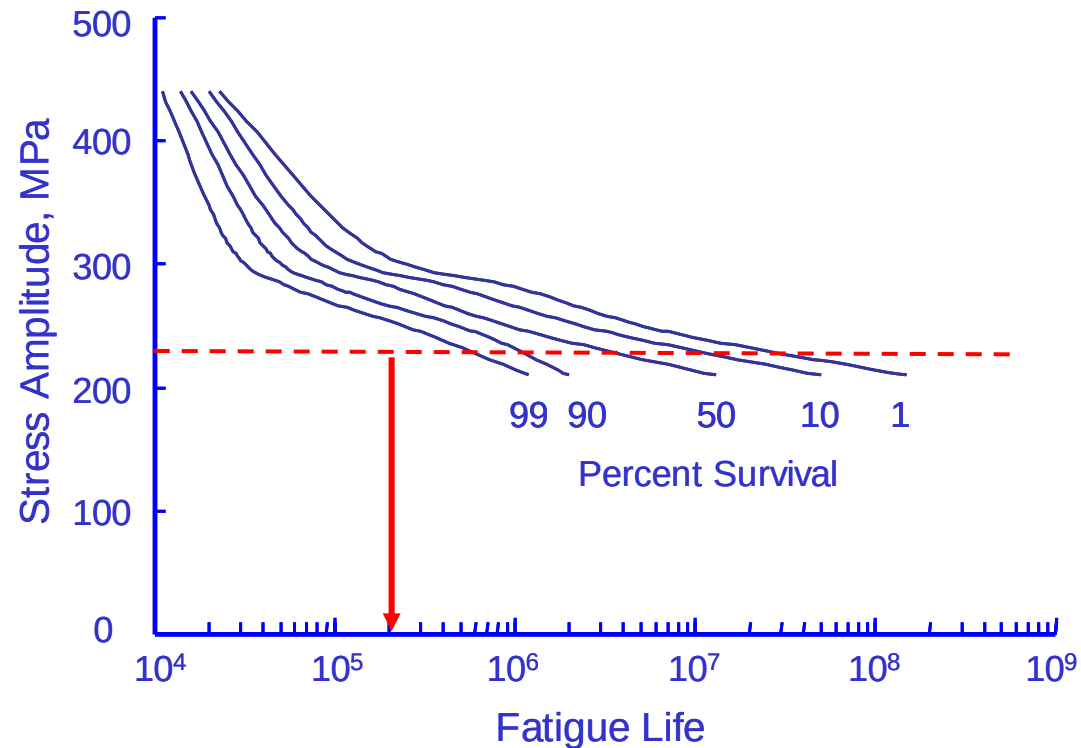
- Similitude
 - Failure mechanism
 - Size scale



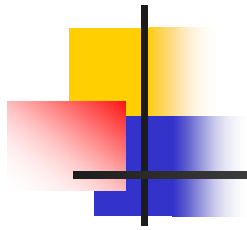
Design Philosophy

- Safe Life
- Damage Tolerant

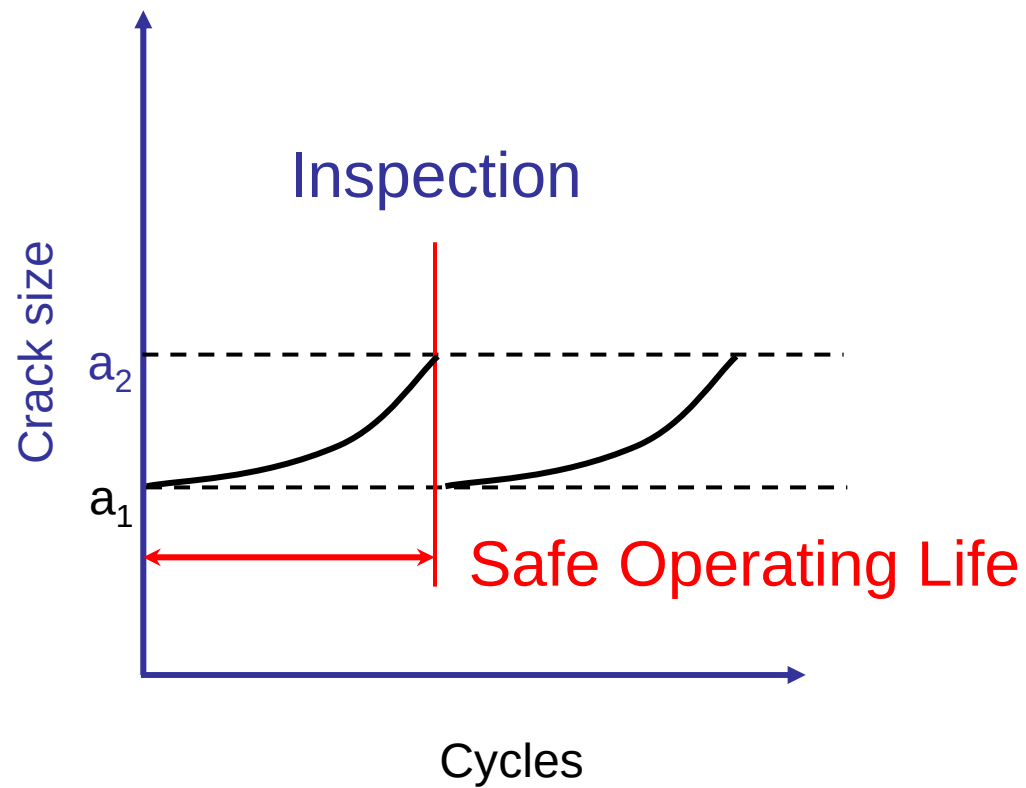
Safe Life



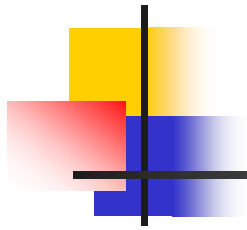
Choose an appropriate risk and replace critical parts after some specified interval



Damage Tolerant



Inspect for cracks larger than a_1 and repair

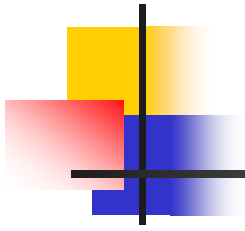


Inspection

A Boeing 777 costs \$250,000,000

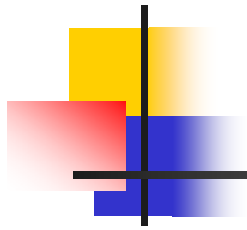
A new car costs \$25,000

For every \$1 spent inspecting and maintaining a B 777 you can spend only 0.01¢ on a car



Things Worth Remembering

- Questions to ask
 - Will a crack nucleate ?
 - Will a crack grow ?
 - How fast will it grow ?
- Similitude
 - Failure mechanism
 - Size Scale



Fatigue, How and Why

- Physics of Fatigue
- Material Properties
- Similitude
- **Fatigue Calculator**

www.FatigueCalculator.com

Fatigue Calculator - Microsoft Internet Explorer

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Fatigue Calculator

Fatigue Calculator: a useful set of web based fatigue analysis software tools for computing fatigue lives of metallic machine components and structures. No single method is best for all situations so several methods have been included: stress life, strain life, fracture mechanics crack growth and welds. Each of the methods have their own strengths and weaknesses. If you are not already, we suggest that you become familiar with constant amplitude fatigue analysis before using the other types of analysis.

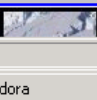
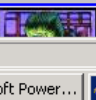
Founder / Moderator
[Darrell F. Socie](#)
Professor Emeritus of Mechanical Engineering
[University of Illinois at Urbana-Champaign](#)

Note: This site is best viewed using Internet Explorer with Javascript enabled. We are currently working to improve compatibility with other browsers.

[Constant Amplitude Fatigue Analysis](#) | [Probabilistic Fatigue Analysis](#)

☒ Experienced User Mode

Constant Amplitude Fatigue Calculators

Stress-Life  Use this method for long life situations where the strength of the material and the nominal stress control the fatigue life.	Strain-Life  This method is used for finite fatigue lives where plasticity around stress concentrations is important.
Crack Growth  Use this method to determine	Welds  Consider weld shapes and

Constant Amplitude Probabilistic

- Constant Amplitude Fatigue Calculators
 - [Stress-Life](#)
 - [Strain-Life](#)
 - [Crack Growth](#)
 - [Welds](#)
- Finders
 - [Stress Concentration Factors](#)

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Constant Amplitude Calculators

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Fatigue Calculator

Note: This site is best viewed using Internet Explorer with Javascript enabled. We are currently working to improve compatibility with other browsers.

Main Constant Amplitude Fatigue Analysis Probabilistic Fatigue Analysis

Founder / Moderator
Darrell F. Socie
Professor Emeritus, Mechanical Engineering
University of Illinois at Urbana-Champaign

Fatigue Technology

☐ Experienced User Mode

- **Constant Amplitude**
 - Fatigue Calculators
 - [Stress-Life](#)
 - [Strain-Life](#)
 - [Crack Growth](#)
 - [Welds](#)
 - Finders
 - Technical Background
 - Other
- **Probabilistic**
 - Fatigue Calculators
 - [Stress-Life](#)
 - [Strain-Life](#)
 - [Crack Growth](#)
 - [Welds](#)
 - Finders
 - Technical Background
 - Other

Constant Amplitude Fatigue Calculators

Stress-Life  Use this method for long life situations where the strength of the material and the nominal stress control the fatigue life.	Strain-Life  This method is used for finite fatigue lives where plasticity around stress concentrations is important.
Crack Growth  Use this method to determine how long it will take a crack to grow to a critical size.	Welds  Complex weld shapes and residual stresses require special fatigue considerations.

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Finders

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- Finders
 - Technical Background
 - Other
 - Probabilistic
 - Fatigue Calculators
 - Stress-Life
 - Strain-Life
 - Crack Growth
 - Welds
 - Finders
 - Technical Background
 - Other



the material and the nominal stress control the fatigue life.



around stress concentrations is important.

[Crack Growth](#)



Use this method to determine how long it will take a crack to grow to a critical size.

[Welds](#)



Complex weld shapes and residual stresses require special fatigue considerations.

Finders

[Stress Concentration Factors](#)



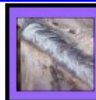
Compute the stress concentration factor for holes, grooves, notches, shoulders, and keyways.

[Stress Intensity Factors](#)



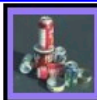
Compute the stress intensity factor for plates with various shaped cracks.

[Weld Classifications](#)



Select standard weld shapes.

[Material Properties](#)



Find material properties for fatigue analysis.

Technical Background

[Stress-Life](#)



[Material Properties](#)
[Modifying Factors](#)
[Stress Concentrations](#)
[Mean Stresses](#)

[Strain-Life](#)



[Material Properties](#)
[Stress Concentrations](#)
[Mean Stresses](#)

[Crack Growth](#)

[Welds](#)

Acknowledgements:

This site was written by Benjamin Socie with assistance from Bethany and Michael Socie.

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Deterministic Analysis

Stress Life Analysis - FatigueCalculator.com - Microsoft Internet Explorer

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Address <http://www.dev.fatiguecalculator.com/stresslifeexp.htm> Go

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Fatigue Calculator



Constant Amplitude Stress-Life Analysis

Enter as much data as you know. If it is not enough, you will be asked for more. Fields in red represent absolutely required data to begin calculations. Other data may become necessary as calculation proceeds.

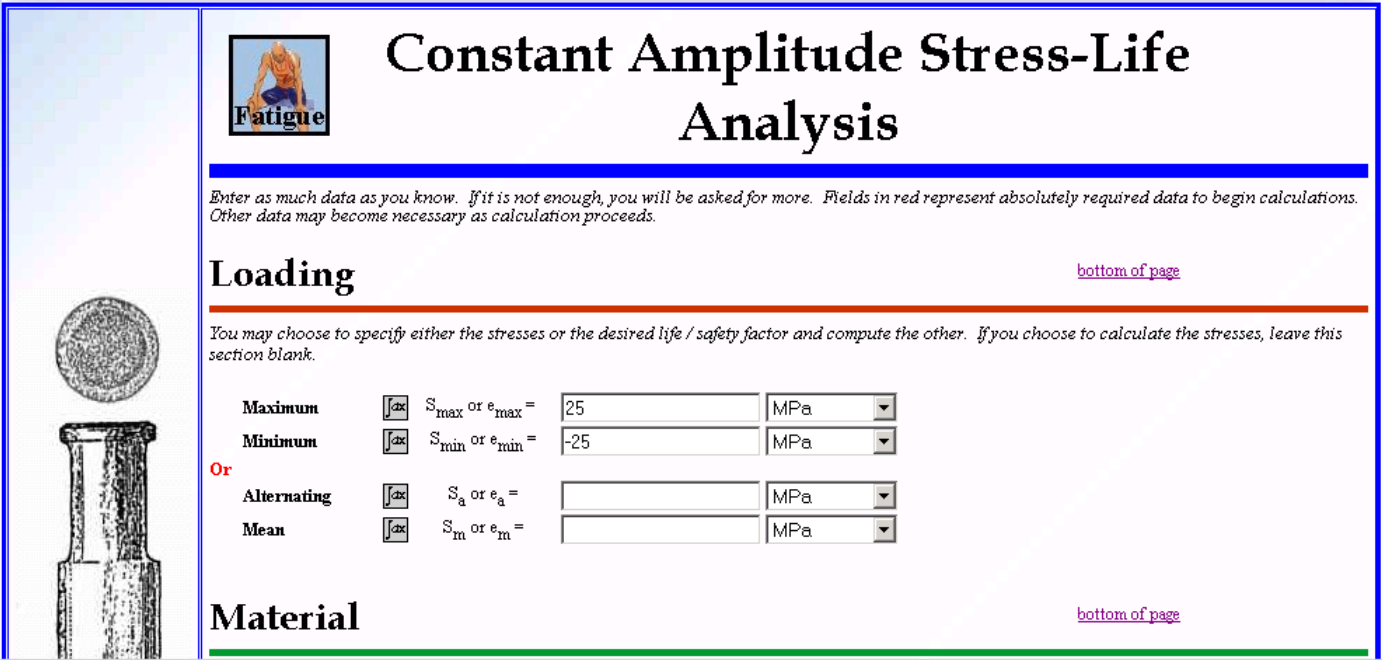
Loading

You may choose to specify either the stresses or the desired life / safety factor and compute the other. If you choose to calculate the stresses, leave this section blank.

Maximum	☐ S_{max} or e_{max} =	25	MPa
Minimum	☐ S_{min} or e_{min} =	-25	MPa
Or			
Alternating	☐ S_a or e_a =		MPa
Mean	☐ S_m or e_m =		MPa

Material

[bottom of page](#)



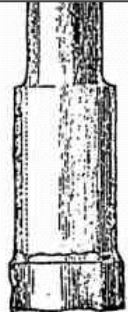
Deterministic Analysis (continued)

Stress Life Analysis - FatigueCalculator.com - Microsoft Internet Explorer

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Address http://www.dev.fatiguecalculator.com/stresslifeexp.htm Go



Typical broken axel
from the 1840s
From Glyon, 1844

Alternating ☐ S_a or ϵ_a = MPa

Mean ☐ S_m or ϵ_m = MPa

Material

[bottom of page](#)

You may use the Material Finder by clicking on the Material Property Finder Icon to look up the proper values for your material or specify values directly. If you do not enter a value for the Fatigue Limit, a default value of $1/2 S_u$ will be assumed.

Type

Ultimate Strength S_u = MPa

Fatigue Limit ☐ S_{FL} = MPa

 [Material Property Finder](#)

If this section is left blank, they will be estimated from the ultimate strength.

Intercept ☐ S_f = MPa

Slope ☐ b =

Modifying Factors

[bottom of page](#)

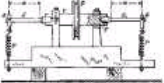
Either specify the modifying factor directly or choose a finish from the drop-down box. If you don't know, a default value of 1 will be used.

Surface Factor k_{SF} = or

Loading Factor k_L = or

Size Factor ☐ k_{size} =

Diameter d = mm



Wöhler's test machine
circa 1830

http://www.dev.fatiguecalculator.com/cgi-bin/StressShowMatProp.pl

Start Microsoft PowerPoint - ... Stress Life Analysis ... Corel PHOTO-PAINT 9 ... Results - FatigueCalcul... 1:34 PM

Deterministic Analysis (continued)

Stress Life Analysis - FatigueCalculator.com - Microsoft Internet Explorer

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Diameter d = mm

Stress Concentration

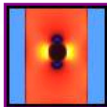
[bottom of page](#)

K_t

Use K_f in analysis?

K_f

Radius mm

 [Stress Concentration Factor Finder](#)

Safety Factor for Infinite Life

If you wish to calculate a safety factor, leave this section blank and click the Calculate Life / Safety Factor button below. If you wish to calculate the stresses, you must specify the desired safety factor and provide some additional information about the stresses.

Safety Factor

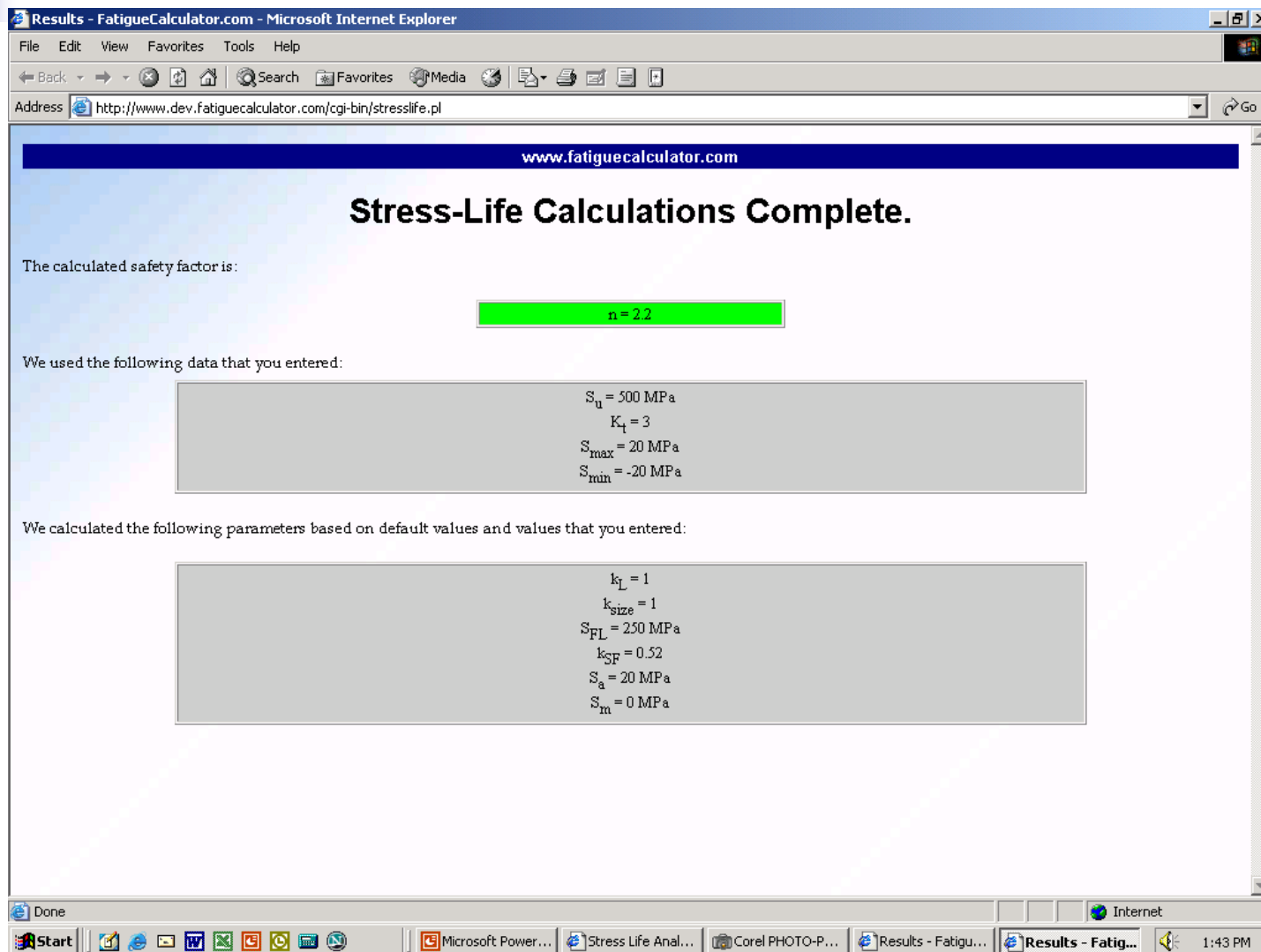
Mean Stress Definition =

 Data Acquisition circa 1910

 Early Strip Chart

Start | Internet | Microsoft PowerPoint - ... | Stress Life Analysis ... | Corel PHOTO-PAINT 9 ... | Results - FatigueCalcul... | 1:35 PM

Deterministic Analysis Results



The screenshot shows a Microsoft Internet Explorer window displaying the results of a stress-life calculation. The browser's address bar shows the URL <http://www.dev.fatiguecalculator.com/cgi-bin/stresslife.pl>. The page title is "Results - FatigueCalculator.com". The main heading on the page is "Stress-Life Calculations Complete." Below this, it states "The calculated safety factor is:" followed by a green box containing the value $n = 2.2$. The next section, "We used the following data that you entered:", lists the input parameters: $S_u = 500 \text{ MPa}$, $K_t = 3$, $S_{max} = 20 \text{ MPa}$, and $S_{min} = -20 \text{ MPa}$. The final section, "We calculated the following parameters based on default values and values that you entered:", lists the output parameters: $k_L = 1$, $k_{size} = 1$, $S_{FL} = 250 \text{ MPa}$, $k_{SF} = 0.52$, $S_a = 20 \text{ MPa}$, and $S_m = 0 \text{ MPa}$. The Windows taskbar at the bottom shows several open applications, including Microsoft PowerPoint, Stress Life Anal..., Corel PHOTO-P..., and Results - Fatigu... The system clock indicates 1:43 PM.

Results - FatigueCalculator.com - Microsoft Internet Explorer

File Edit View Favorites Tools Help

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Address <http://www.dev.fatiguecalculator.com/cgi-bin/stresslife.pl> Go

www.fatiguecalculator.com

Stress-Life Calculations Complete.

The calculated safety factor is:

$n = 2.2$

We used the following data that you entered:

$S_u = 500 \text{ MPa}$
 $K_t = 3$
 $S_{max} = 20 \text{ MPa}$
 $S_{min} = -20 \text{ MPa}$

We calculated the following parameters based on default values and values that you entered:

$k_L = 1$
 $k_{size} = 1$
 $S_{FL} = 250 \text{ MPa}$
 $k_{SF} = 0.52$
 $S_a = 20 \text{ MPa}$
 $S_m = 0 \text{ MPa}$

Done Internet

Start Microsoft PowerPoint Stress Life Anal... Corel PHOTO-P... Results - Fatigu... Results - Fatig... 1:43 PM

Probabilistic Analysis

Probabilistic Stress Life Analysis - FatigueCalculator.com - Microsoft Internet Explorer

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Address <http://www.fatiguecalculator.com/probabilistic/stresslife.htm> Go



Probabilistic Stress-Life Analysis

Enter as much data as you know. If it is not enough, you will be asked for more. Fields in red represent absolutely required data to begin calculations. Other data may become necessary as calculation proceeds. This is the probabilistic method, so select a distribution and scale parameter for each variable. The scale parameter has a different meaning for each distribution type, see [Description of Distribution Types](#) for more information. You may choose None if you wish to keep this variable constant throughout the probabilistic analysis.

[Description of Distribution Types](#)

Loading

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Loading typically follows a normal distribution. The coefficient of variation, COV, is typically around 0.2.

You may choose to specify either the stresses or the desired life / safety factor and compute the other. If you choose to calculate the stresses, leave this section blank.

		Value	Units	Distribution Type	Scale Parameter
Maximum	 $S_{\max}$ or $e_{\max}$ =		MPa	Normal	0.2
Minimum	 $S_{\min}$ or $e_{\min}$ =		MPa	Normal	0.2
Or					
Alternating	 S_a or e_a =		MPa	Normal	0.2
Mean	 S_m or e_m =		MPa	Normal	0.2

Material

[bottom of page](#)



Probabilistic Analysis (continued)

Probabilistic Stress Life Analysis - FatigueCalculator.com - Microsoft Internet Explorer

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Address http://www.dev.fatiguecalculator.com/probabilistic/stresslifeexp.htm Go

Mean ☐ S_m or e_m = MPa Normal 0.2

Material

[bottom of page](#)

You may use the Material Finder by clicking on the Material Property Finder Icon to look up the proper values for your material or specify values directly. If you do not enter a value for the Fatigue Limit, a default value of $1/2 S_u$ will be assumed.

 [Material Property Finder](#)

Material

	Value	Units	Distribution Type	Coefficient of Variation	Correlation Coefficient
Ultimate Strength	S_u = 500	MPa	Normal	0.1	
Fatigue Limit	☐ S_{FL} =	MPa	None		

If this section is left blank, values will be estimated.

Elastic Modulus	☐ E =	MPa	None		
Intercept	☐ S_f =	MPa	Log-Normal		
Slope	☐ b =		None		

Modifying Factors

[bottom of page](#)

Either specify the modifying factor directly or choose a finish from the drop-down box. If you don't know, a default value of 1 will be used.

Typical broken axel from the 1840s From Glyon, 1844

Wöhler's test machine circa 1850

Start Microsoft PowerPoint - ... Probabilistic Stress ... Corel PHOTO-PA Results - FatigueCalculator.com - Microsoft Internet Explorer

Probabilistic Analysis (continued)

Probabilistic Stress Life Analysis - FatigueCalculator.com - Microsoft Internet Explorer

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Address <http://www.dev.fatiguecalculator.com/probabilistic/stresslifeexp.htm> Go

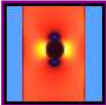
Modifying Factors [bottom of page](#)

Either specify the modifying factor directly or choose a finish from the drop-down box. If you don't know, a default value of 1 will be used.

		Value or Finish	Distribution Type	Coefficient of Variation
Surface Factor	$k_{SF} =$	or Machined	Normal	0.1
Loading Factor	$k_L =$	or None		
Size Factor	$k_{size} =$			
Diameter	$d =$	mm		

Stress Concentration [bottom of page](#)

Either specify K_f directly or enter K_t and the radius.



Stress Concentration Factor Finder

	Value	Distribution Type	Coefficient of Variation
K_t	3	Normal	0.05

Use K_f in analysis?

K_f

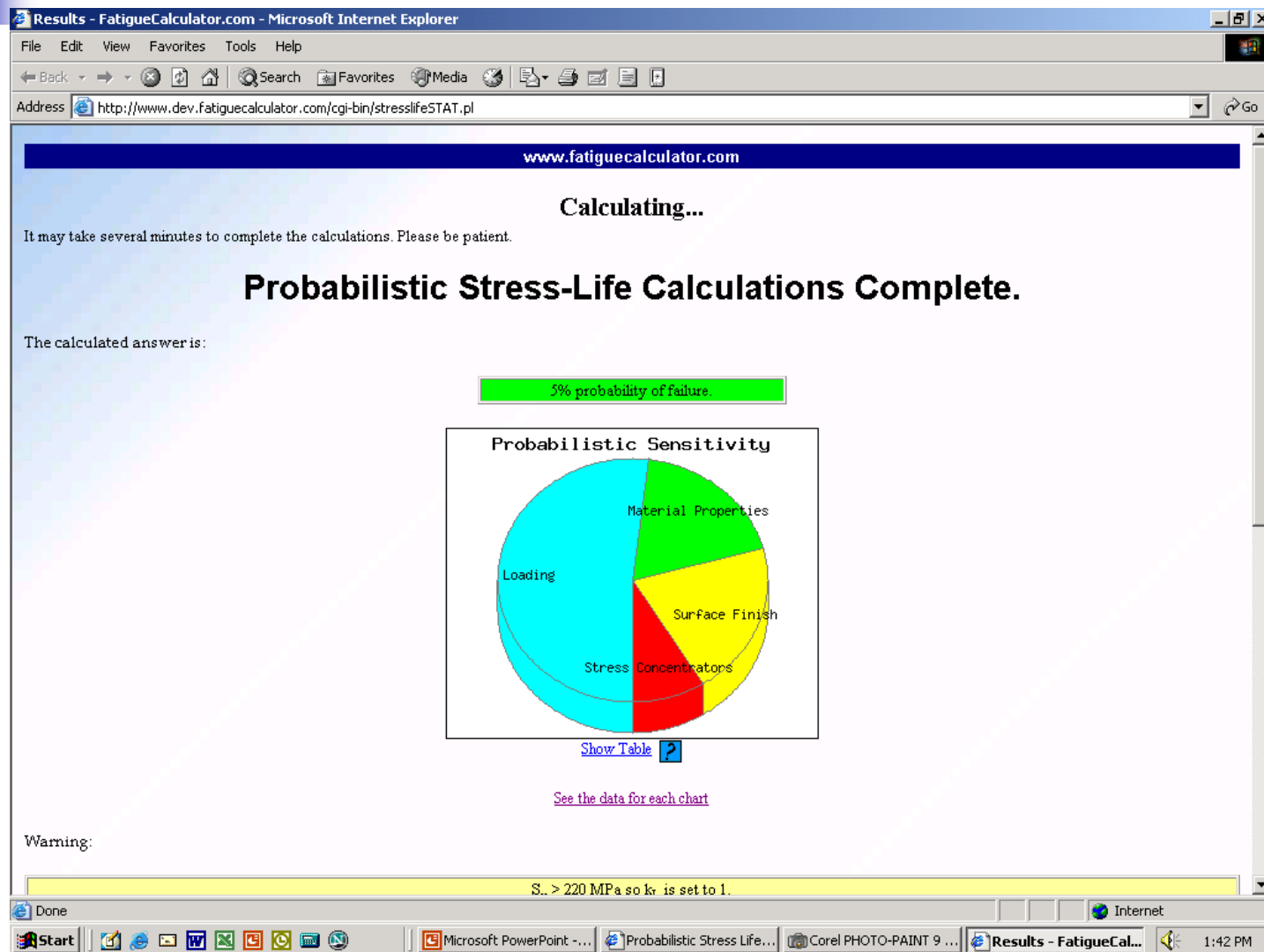
Radius mm

Calculate K_f

Calculate Life Clear Form

Start | Microsoft PowerPoint - ... | Probabilistic Stress ... | Corel PHOTO-PAINT 9 ... | Results - FatigueCalcul... | 1:39 PM

Probabilistic Analysis Results



Fatigue and Fracture (Basic Course)

